Two AI Agents, One Stellarator

Building an engineering simulation pipeline from an open dataset – including the dead ends

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Hobby project, driven by curiosity. Not affiliated with any research institute or company, not peer reviewed. All numbers are estimates from simplified models and may contain errors. The W7-X model is *inspired by* Wendelstein 7-X: coil paths come from public data [7]; casings, connectors, potting, supports and wall thicknesses are my own simplified assumptions and do not describe the real device. Code, computations and the draft text were produced with the help of AI agents; all figures are my own computations and renderings.

Introduction and motivation

I am not a fusion researcher. This is a private project, driven by curiosity: how does a stellarator work, and how far can one person get with open software and AI assistance on such a demanding part? Stellarator coils are a good example because design, manufacturing and strength meet here. Physics dictates the path, current and field create the forces, and the structure has to fit in between. Later the plasma vessel was added, with a cooling core of gyroid filaments.

A stellarator confines a fusion plasma with twisted 3D superconducting coils that carry large magnetic forces. In four days, a human and two AI agents built an end-to-end pipeline from a public coil dataset to a structural analysis of the coil casings: coil path → magnetic field → Lorentz force → casing geometry → stress. This report is mostly about *how* it was built: the division of labour between a local agent (Hermes) and a coding agent (Claude Code), a first model that was wrong while looking right, a fully parametric CAD model that broke the mesher, and a second, CAD-free code path that finally worked. For a W7-X-like system of 70 coils the result is 99 % of casing stresses below 43 MPa – under strongly simplified supports. A follow-up (Section 9) computes the flow through the gyroid cooling core of the 1 GW plasma vessel with FluidX3D and suggests improvements.

Contents

1 Setup **4**
- 1.1 Agents and responsibilities . . . 4
- 1.2 Tool chain . . . 4

2 Research phase **4**

3 First results – and why they were wrong **5**
- 3.1 Validating the solver . . . 5
- 3.2 The surrogate model (report v0.1) . . . 5

4 Restart: mesh directly from the data **7**

5 The parametric CAD branch **11**

6 The parallel branch: no CAD kernel **14**

7 Visualisation: from mesh to Blender **18**

8 Takeaways for working with coding agents **22**

9 Follow-up: cooling the plasma vessel **23**
- 9.1 Task and layout . . . 23
- 9.2 History . . . 23
- 9.3 An error in the lattice . . . 23
- 9.4 Is it enough? A rough heat estimate . . . 27
- 9.5 Suggested improvements . . . 28
- 9.6 Collaboration . . . 28

10 Follow-up: double filament, clearance and liquid lead **29**
- 10.1 Wall layout . . . 29
- 10.2 Water at 60 to 150 °C . . . 29
- 10.3 Clearance to the coils . . . 30
- 10.4 Liquid lead instead of water? . . . 31
- 10.5 Collaboration . . . 32

11 Coolant, strength and heat balance up to the turbine **33**
- 11.1 Variant f12: 5 mm inner skin . . . 33
- 11.2 Ranking by pairwise comparison . . . 33
- 11.3 Does the thin skin hold the pressure? . . . 34
- 11.4 Sourced reference values . . . 35
- 11.5 Decision: pressurised water as in a nuclear plant . . . 36
- 11.6 Derivation of the heat balance . . . 36
- 11.7 From plasma to grid: power flow and influences . . . 38
- 11.8 The components of the chain . . . 38
- 11.9 Do the pipes fit through the gaps between the coils? . . . 40
- 11.10 Blanket: comparison with the EU-DEMO WCLL concept . . . 41
- 11.11 Graded structure: flat contact at the skin, branching outward . . . 42
- 11.12 How does the heat spread sideways and outward? . . . 44
- 11.13 Divertor preheater: which design yields the most electricity? . . . 47
- 11.14 Exchanger modules built and simulated with FluidX3D . . . 50
- 11.15 Surrogate model from the flow simulations and the question of grading . . . 53
- 11.16 Collaboration . . . 56

12 Plasma: where does the heat come from, and how is it produced? **57**
- 12.1 The reaction: mass becomes energy . . . 57
- 12.2 Why it has to be so hot . . . 57
- 12.3 Where in the plasma the heat is produced . . . 57
- 12.4 Two routes out of the plasma . . . 58
- 12.5 What this means for the plant . . . 59

13 Liquid metal in a magnetic field: MHD in FluidX3D **60**

14 Alpha particles in the coil field: do they stay in the plasma long enough? **64**

15 Strength: where the force comes from and how a support structure takes it up **69**

1 Setup

Stellarator in five sentences. A fusion plasma (over 100 million degrees) is held by a magnetic field inside a ring-shaped vacuum vessel. To keep particles from drifting out, field lines must wind helically around the ring. A tokamak gets this twist from a large current in the plasma itself; a stellarator gets it entirely from external coils, which therefore have to be twisted in 3D. This makes steady-state operation easier but turns each coil into a demanding 3D part with tight tolerances. The reference device is Wendelstein 7-X (IPP Greifswald) with 50 non-planar and 20 planar superconducting coils [1].

1.1 Agents and responsibilities

Human	Goal and direction, reviews results *as images*, decides, supplies the key design ideas when stuck.
Hermes	Hermes Agent [15], run locally with open-weight models: literature research, project bookkeeping, writing reusable *skills*, long-running jobs overnight.
Claude Code	Coding agent [16]: simulation pipeline, debugging, post-processing, figures.

The agents never talk directly. They coordinate through files in the repository: an append-only findings log (`BEFUNDE.md`; every entry has observation, measurement, location, status), a handover file for decisions and running jobs (`ERINNERUNGEN.md`), and skills – short Markdown playbooks an agent loads before a task. Long simulations are started detached (`setsid nohup`) so they survive a session ending or a context compaction. The very first human intervention was a stop: Hermes started building tooling right away and was told to do the research first.

1.2 Tool chain

Physics data	simsopt [7] (coil Fourier coefficients), DESC [8] (plasma shape optimisation)
Geometry	Python + gmsh/OpenCASCADE [9] (direct meshing), FreeCAD 1.1 headless [11] (parametric CAD)
Solver	ElmerFEM 9.0 [10]: magnetostatics and linear elasticity in one coupled run
Own code	Biot-Savart (numba), FFT field solver, field-line tracer, collision checks
Visualisation	PyVista/VTK [12], ParaView [13], Blender 4.0 [14] via glTF
Hardware	one workstation with one GPU (reserved for the local agent)

2 Research phase

Hermes collected the basics: W7-X has a major radius of 5.5 m, five field periods and 2.5 T on axis [1]. Each coil is wound from a cable-in-conduit conductor: 243 NbTi strands of 0.57 mm, each with 144 filaments in copper, inside a co-extruded 16 mm × 16 mm aluminium jacket with 37 % void for helium cooling [2, 3]. From a set of 33 PDFs supplied by the user, four were relevant; the most useful one sizes coil structures for EU-DEMO with a 660 MPa stress limit and wedged inner legs [4]. A rate-limited arXiv search added work on HTS strain sensitivity [6] and on manufacturing tolerances (W7-X: 2 mm) [5].

Lesson – verify citations. The bibliography an agent produced during research contained entries without authors and with wrong venues or years. For this report every reference was checked individually against full text or publisher metadata (Crossref, arXiv API). Treat agent-written references as unverified until checked.

3 First results – and why they were wrong

3.1 Validating the solver

Before simulating coils, ElmerFEM was tested on a case with an exact answer: an infinite straight wire, $B = \mu_0 I / 2\pi r$. The first runs returned zero everywhere – silently. Root causes: a mesh format the converter misread, and wrong keywords in the solver input file, which Elmer ignores without an error. Starting from Elmer's own regression test instead of hand-written input fixed it: 0.09 % error inside the wire, 2 % outside (Fig. 4).

3.2 The surrogate model (report v0.1)

Hermes believed the real coil paths were unavailable (the URLs it tried returned 404) and generated a parametric surrogate: ellipses and bands, current calibrated to 2.5 T. Compared with W7-X the ampere-turns per coil matched to 0.90×. A report was written.

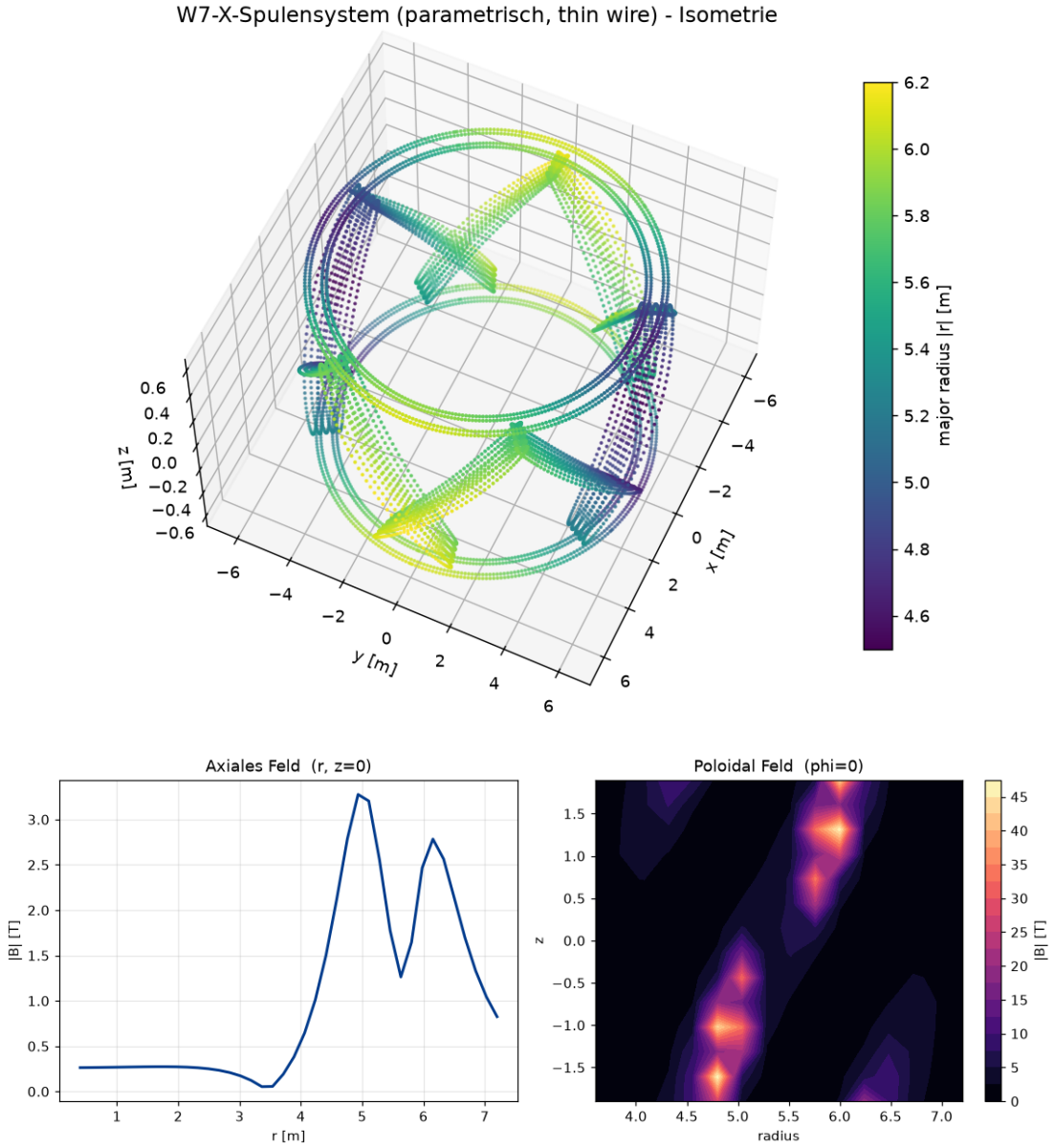

Figure 1: Report v0.1: the surrogate coil set and its field. Plausible numbers – but not a stellarator.

How to read the figure: Top: an early stand-in model of the coils made of thin wires; the colour only shows the distance from the centre of the ring. Bottom left: the magnetic field strength (in tesla, T; a fridge magnet has about 0.01 T) along a line across the machine. The peaks are where the line passes

close to wires. Bottom right: the same quantity as a colour map in a cross-section. The numbers look reasonable, but the coils are flat rings and slanted bundles. Such a field does not hold a plasma the way a stellarator does. That is why the project then restarted from real Wendelstein 7-X coil data.

A topological check by the coding agent – does each coil actually encircle the plasma axis? – failed for 20 of 50 coils; the "planar" coils were horizontal bands above and below the plasma. The good match was a coincidence. And the data had been available all along: simsopt ships the Fourier coefficients of the seven W7-X base coils in its package data [7]. Two more bugs surfaced later and changed published numbers: a sign error in the axis formula (calibration current 12.18 kA instead of 13.49 kA per turn), and Elmer choosing the current direction itself, so 25 mirrored coils ran backwards (1.38 T instead of 2.50 T on axis). The latter is now caught by an automated check.

Lesson – plausible is not correct. Agents are good at producing numbers that look right. Every geometry gets an independent invariant check (winding number around the axis, clearances), and every field or force is computed by a second, independent method before it goes into a report.

4 Restart: mesh directly from the data

With the real coil paths (7 base coils, 70 by symmetry) the expected physics appeared immediately: bean-shaped flux surfaces and the characteristic island chain (Fig. 2). A custom FFT voxel solver matched exact Biot-Savart to 1.3 % but destroyed the flux surfaces during field-line tracing, because a Cartesian grid breaks the device's symmetry – so field lines are always traced with Biot-Savart.

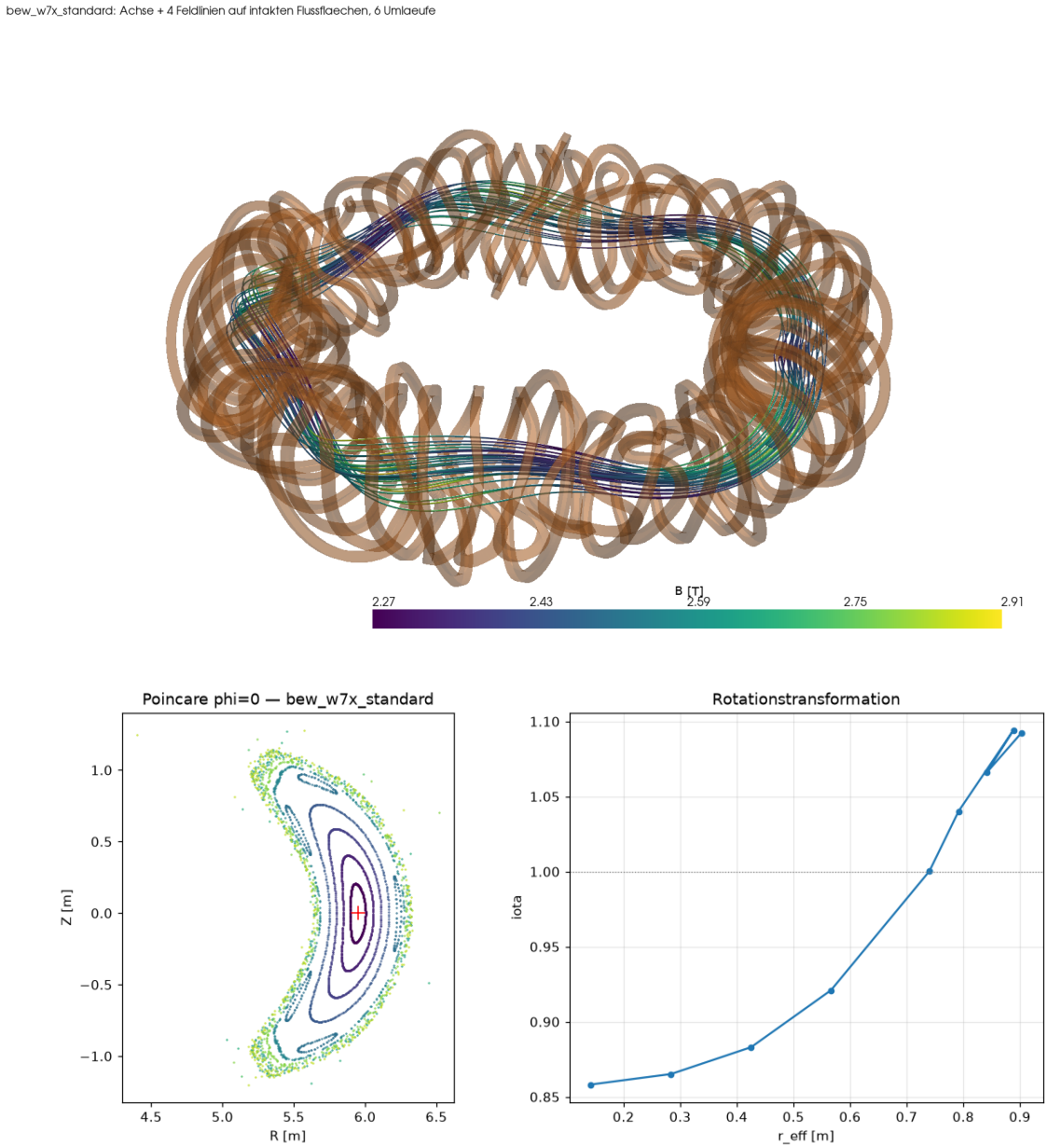

Figure 2: Field lines on flux surfaces (top), Poincaré section and rotational transform (bottom).

How to read the figure: Top: the 70 coils (brown) and a few magnetic field lines. They run around the ring many times, and their colour shows the field strength. Bottom left: a *Poincaré section*. Imagine cutting the ring at one place and putting a dot wherever a field line pierces the cut. If the dots form closed, nested curves like onion layers (*flux surfaces*), the plasma stays confined. The red cross is the centre (magnetic axis). At the very edge the curves break up into small islands and scattered dots; confinement ends there. Bottom right: the *twist*, i.e. how often a field line goes round the cross-section while it travels once along the ring. Where it reaches 1, the edge islands appear, as intended in W7-X. The model reproduces the known W7-X field.

The plan had been to model coils in CAD. Generating the mesh directly was faster and more robust: a rectangle at every path point, two ruled half-surfaces per coil, meshed by gmsh. Sweeps and smooth lofts failed with self-intersecting surfaces. A clearance check on centre lines replaced boolean intersection

tests (1.5 s instead of more than 45 min). Elmer with all 70 coils (3.3 M tetrahedra, 680 s, 10 GB) matched Biot-Savart on the axis to a median of 0.24 %.

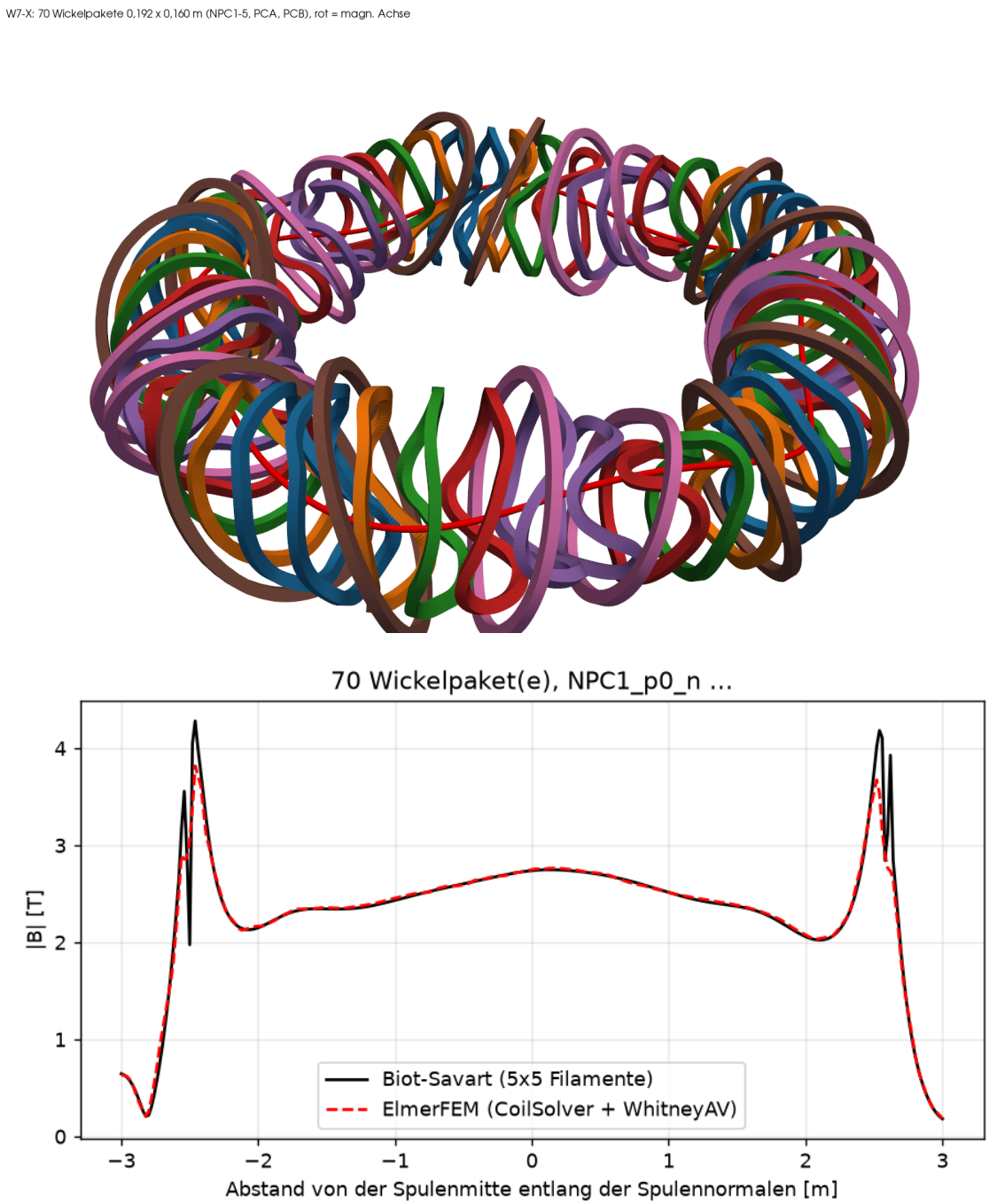

Figure 3: 70 winding packs meshed straight from the path data (top); FEM vs. Biot-Savart along a coil normal (bottom).

How to read the figure: Top: the 70 W7-X coils as winding packs with their real 19.2×16 cm cross-section, built from published path data. Colours mark the seven coil types; the red line is the centre of the plasma. Bottom: the field strength along a straight line through the centre of one coil, perpendicular to it (0 = coil centre). Two completely different methods give it. The black line comes directly from the formula for the field of a current (Biot–Savart). The red dashed line comes from a finite-element (FEM) calculation, which splits space into many small cells. The curves agree. Only at the peaks near ± 2.5 m, where the line passes through the conductor itself, is the FEM mesh coarser. So the FEM route computes correctly.

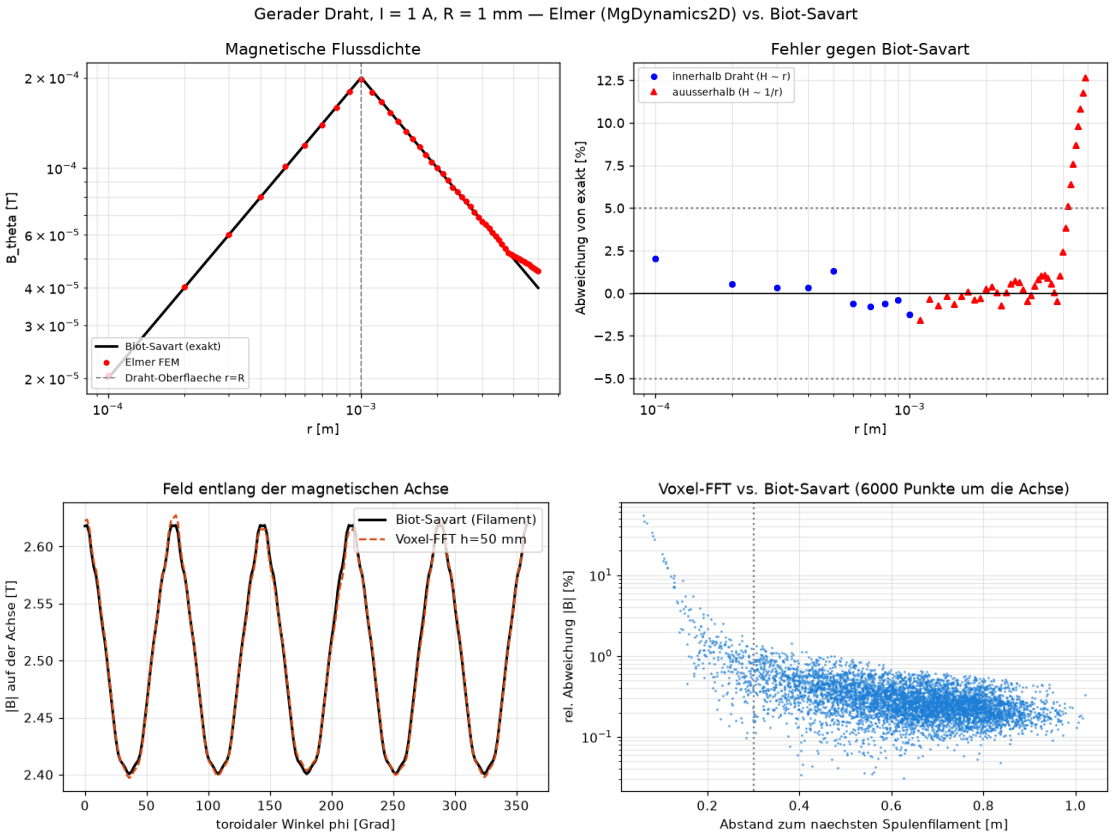

Figure 4: Validation: FEM on a straight wire (top), FFT voxel solver vs. Biot-Savart (bottom).

How to read the figure: Top: a check against a known answer, the magnetic field around a straight wire versus distance from its centre. Both axes are logarithmic (each major tick is a factor of ten). Inside the wire the field grows outwards, outside it falls; the kink is at the wire surface (dashed). Black is exact, red dots are Elmer. Right of it, the deviation: within ±2 %. Only at the far edge does it grow, because the artificial boundary of the computational domain interferes. Bottom left: the field strength along the centre of the plasma versus the angle around the ring; the five humps are the five-fold symmetry of W7-X. Black is the formula, orange the project's own fast solver (FFT), which divides space into 5 cm cubes. Bottom right: each dot is one test point, horizontally its distance to the nearest coil, vertically the deviation. More than 0.3 m from the coils, i.e. everywhere in the plasma region, it stays below 1 %. Both tools are reliable there.

A side study ran the pipeline backwards from a target of 1 GW net electric power: plasma size and field → own plasma shape (DESC) → coils (simsopt). With high-temperature superconductors the device ends up at about 10 m major radius with forces of roughly 1 GN per coil – about a thousand times W7-X. From here on, the support structure is the real problem.

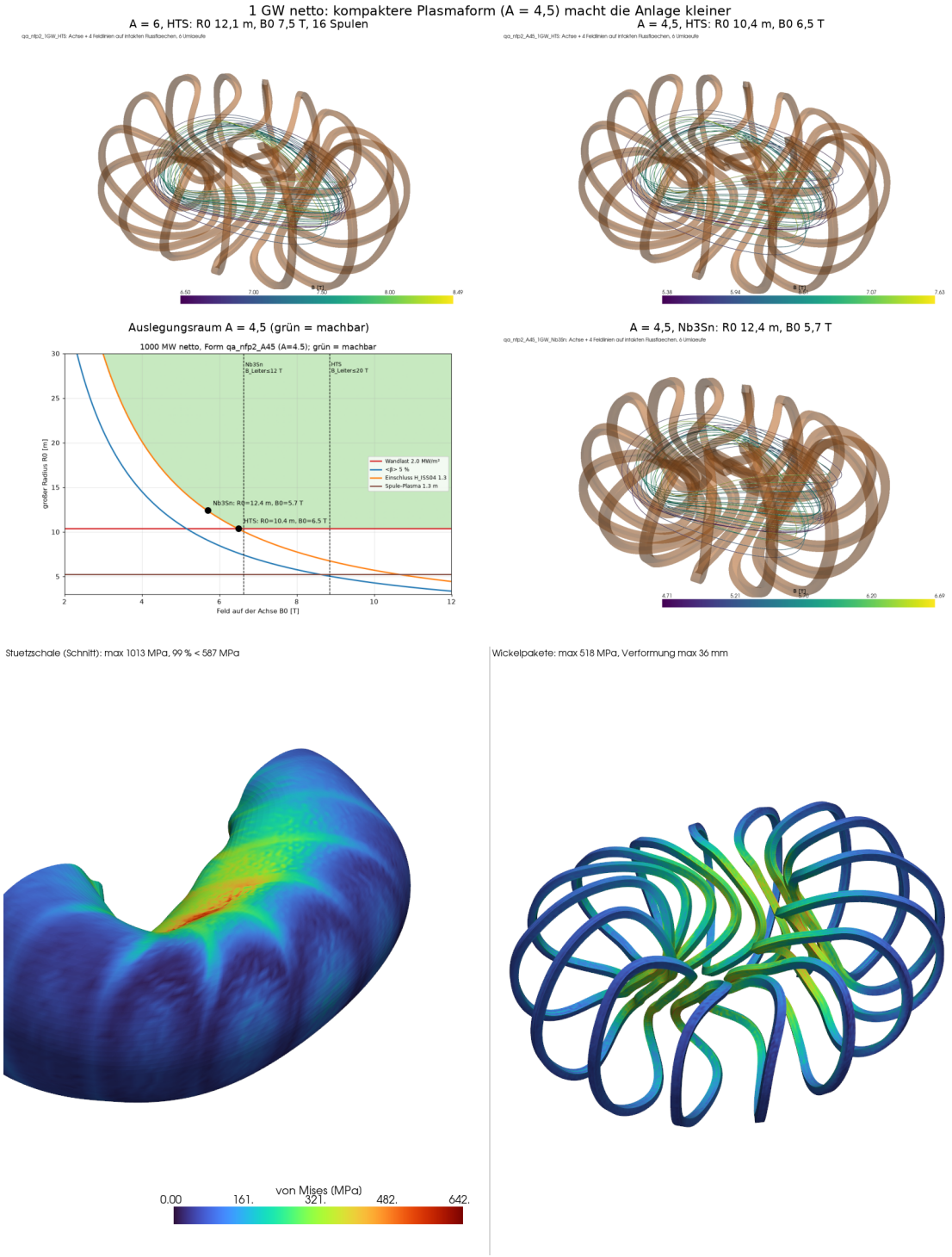

Figure 5: 1 GW sizing study (top) and a first support shell with coupled field → force → stress analysis (bottom).

How to read the figure: Top: three designs for a power plant with 1 GW delivered to the grid, each showing coils (brown) and field lines (colour = field strength). A is the *aspect ratio*, major over minor radius; a smaller A means a fatter, shorter ring. With $A = 4.5$ and high-temperature superconductor (HTS) the major radius shrinks from 12.1 m to 10.4 m. The chart at the lower left of the top figure is the *design space*: horizontally the field at the plasma centre, vertically the major radius. Each coloured line is a limit (heat load on the wall, plasma pressure, confinement, space between coil and plasma). Green is where all limits are met; the black dots are the chosen designs. The bottom figure shows the support shell and winding packs coloured by *equivalent stress*. This single number sums up how hard the material is loaded; steel withstands a few hundred MPa. Red means highly loaded. The chain field → force → stress works, but the shell is overloaded in places.

5 The parametric CAD branch

A continuous support shell could not be constructed cleanly. The human proposed a technique from commercial CAD (Creo's *Variable Section Sweep*): per coil a closed guide curve as the casing's inner contour, sketches normal to it, a loft through them; all walls grow outwards. The agent implemented it in headless FreeCAD as a fully parametric model: spreadsheet parameters, sketch dimensions as expressions, symmetric coils as clones, connectors between neighbours.

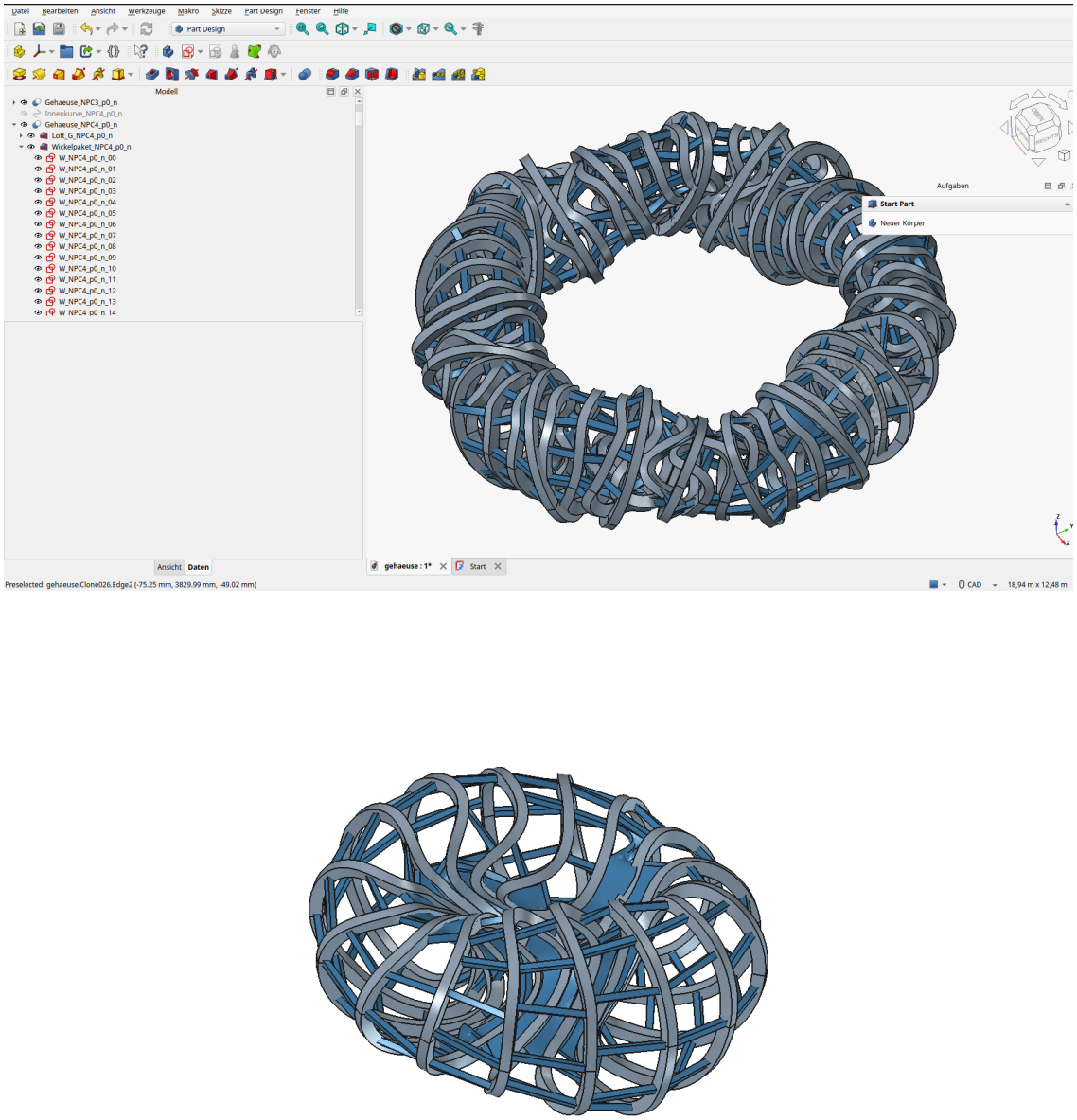

Figure 6: Parametric FreeCAD model: W7-X-like casings with connectors (top), compact 1 GW design (bottom).

How to read the figure: Top: a screenshot of the CAD program FreeCAD. On the left is the parts tree (winding pack and casing for each coil), on the right the 3D model: grey casings around all 70 coils and blue bars connecting them. *Parametric* means all dimensions are numbers in a table; change them and the model rebuilds itself. Bottom: the same procedure for the compact 1 GW shape with 16 coils. The geometry can thus be generated automatically for very different designs.

What went wrong:

1. **Estimated, not designed, walls.** The first coupled run worked but gave stresses up to eight times the allowable value.
2. **Wedges break the mesher.** Inner wedges (after [4]) produced a valid CAD solid, but gmsh failed with "overlapping facets": several boolean cuts on the same winding pack created coincident faces. Six attempts later (tolerances, fusing in FreeCAD, deliberate overlap; the last one hung gmsh for over an hour) the wedge was disabled.
3. **Collisions.** A loft bulges between sketch stations and corners of tilted neighbours overlap. An incremental assembly check (add coils one by one, count collisions, render a video) found about 100.
4. **Fix in CAD (human idea): subtract the neighbours.** Each casing minus its neighbours' envelopes plus a gap: 0 collisions in all 70 steps. Meshing the whole assembly stayed fragile.

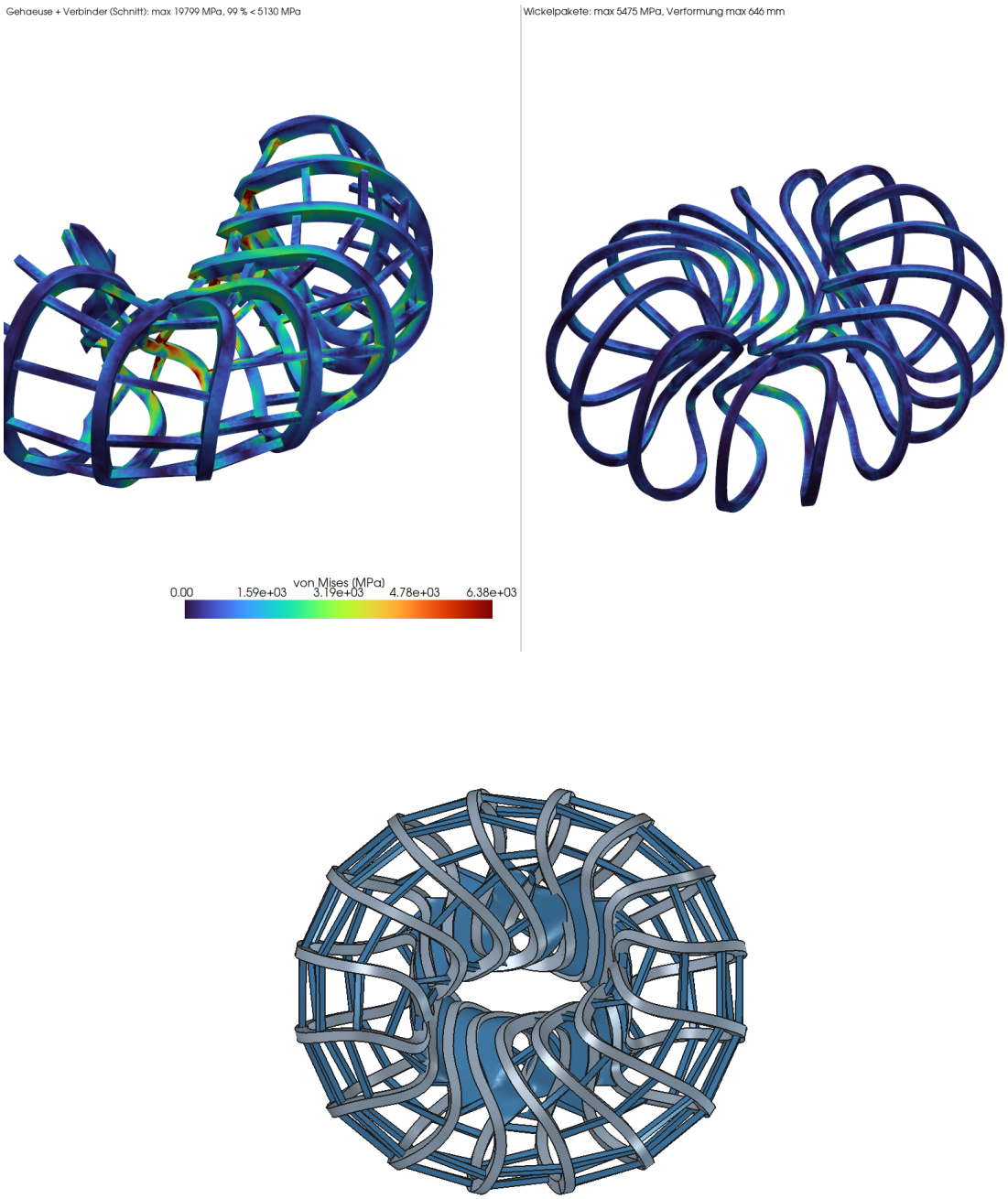

Figure 7: First stress run of the CAD casings (top); casings with inner wedges – valid in CAD, not meshable (bottom).

How to read the figure: Top left: casings and connectors (cut open); top right: the winding packs, coloured by equivalent stress. The scale goes above 6000 MPa, several times what steel can bear (roughly 500 MPa to 1000 MPa). Deformation reaches 646 mm. Bottom: a casing with a wedge on the inside as reinforcement, seen from above. It is valid in CAD, but the program that builds the computational mesh fails on it. So the walls were far too thin, and the obvious reinforcement could not be computed. That is why the second route without a CAD kernel was created.

Lesson – valid is not meshable. The more boolean operations meet on the same faces, the more sliver faces appear, and the mesher fails far away from where the problem was created. Keep the analysis model separate from the design model.

6 The parallel branch: no CAD kernel

The human's suggestion: *"keep this morning's working approach for the analysis and run the CAD path in parallel as an add-on."* Each casing became two rectangular tubes on the exact coil path (outer contour and winding-pack cavity) as separate surface meshes – no boolean operations at all. Wall thicknesses follow from load and available space; a disentangling step tests surface points of foreign casings (including intermediate rings between stations) against the local cross-section and shrinks the cheapest wall: 4980 → 40 → 0 contact points.

Model	steel casings ($E = 200\,\mathrm{GPa}$), homogenised winding packs ($E = 100\,\mathrm{GPa}$), linear elastic
Load	Lorentz force $\vec{J} \times \vec{B}$ from the magnetostatic solution in the same run
Supports	one clamped patch per casing (simplification; no connectors, no support ring)
Mesh / run time	3.02 M tetrahedra in 78 s; solve 55 min, about 14 GB RAM
Force per coil	FEM 2.42 MN vs. Biot-Savart 2.39 MN (+1 %)
Casing von Mises	mean 11 MPa, 99 % below 43 MPa, max 110 MPa; deformation below 1 mm

Because of the simplified supports and walls as thin as 3 mm in places this is *not* a strength proof – but it is a reproducible end-to-end pipeline.

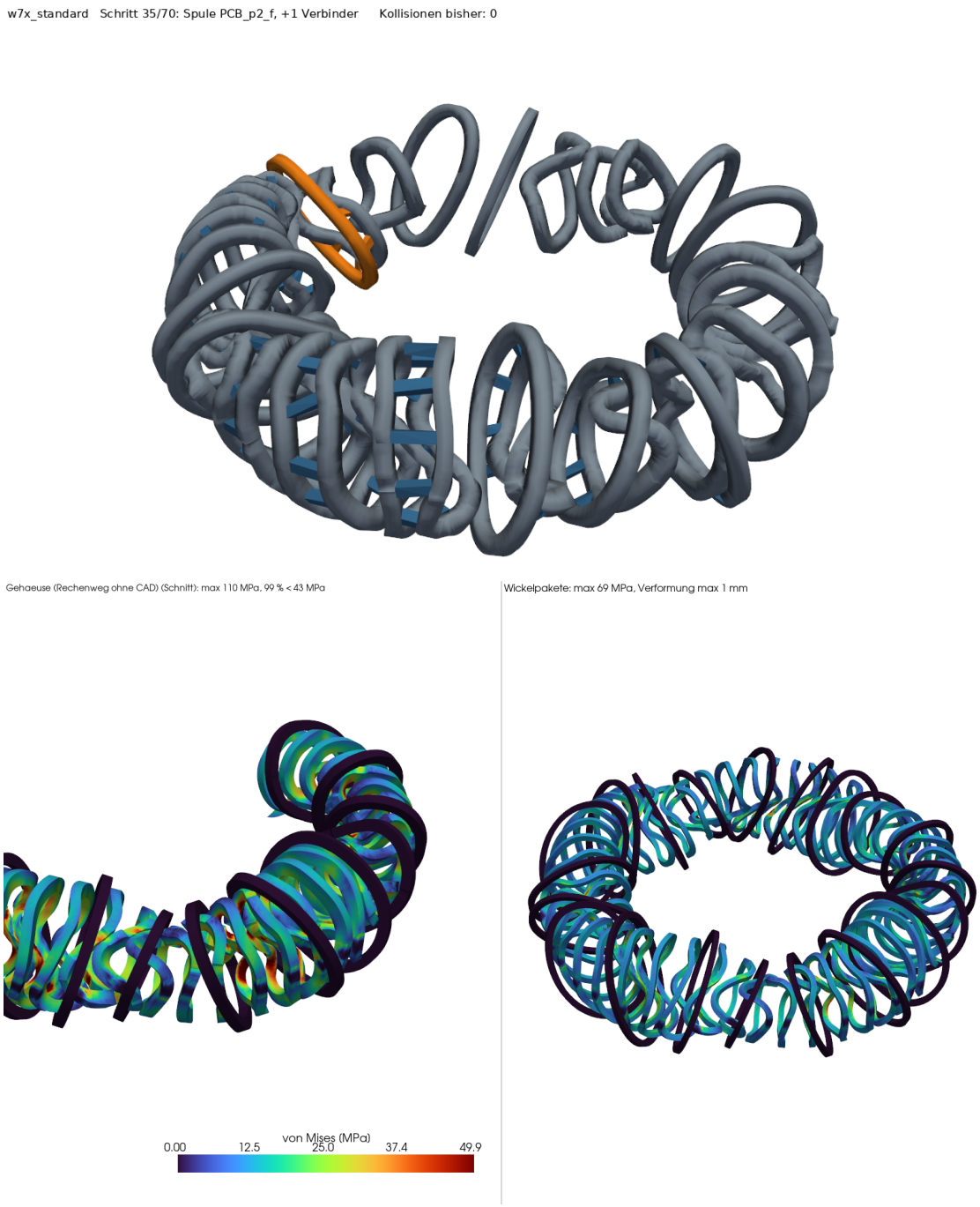

Figure 8: Incremental collision check in the CAD branch, step 35/70 (top); von Mises stress from the CAD-free branch (bottom).

How to read the figure: Top: the model is built coil by coil. After each step a program checks whether any parts touch. Orange is the coil just placed; at step 35 of 70 there is not a single contact. Bottom: the stress in the W7-X-like model from the second route. The colour scale only goes to 50 MPa. The casings reach at most 110 MPa, 99 % of the material stays below 43 MPa, and the winding packs deform by 1 mm. At the W7-X field strength (about 2.5 T) the load is small, far below the limit of steel.

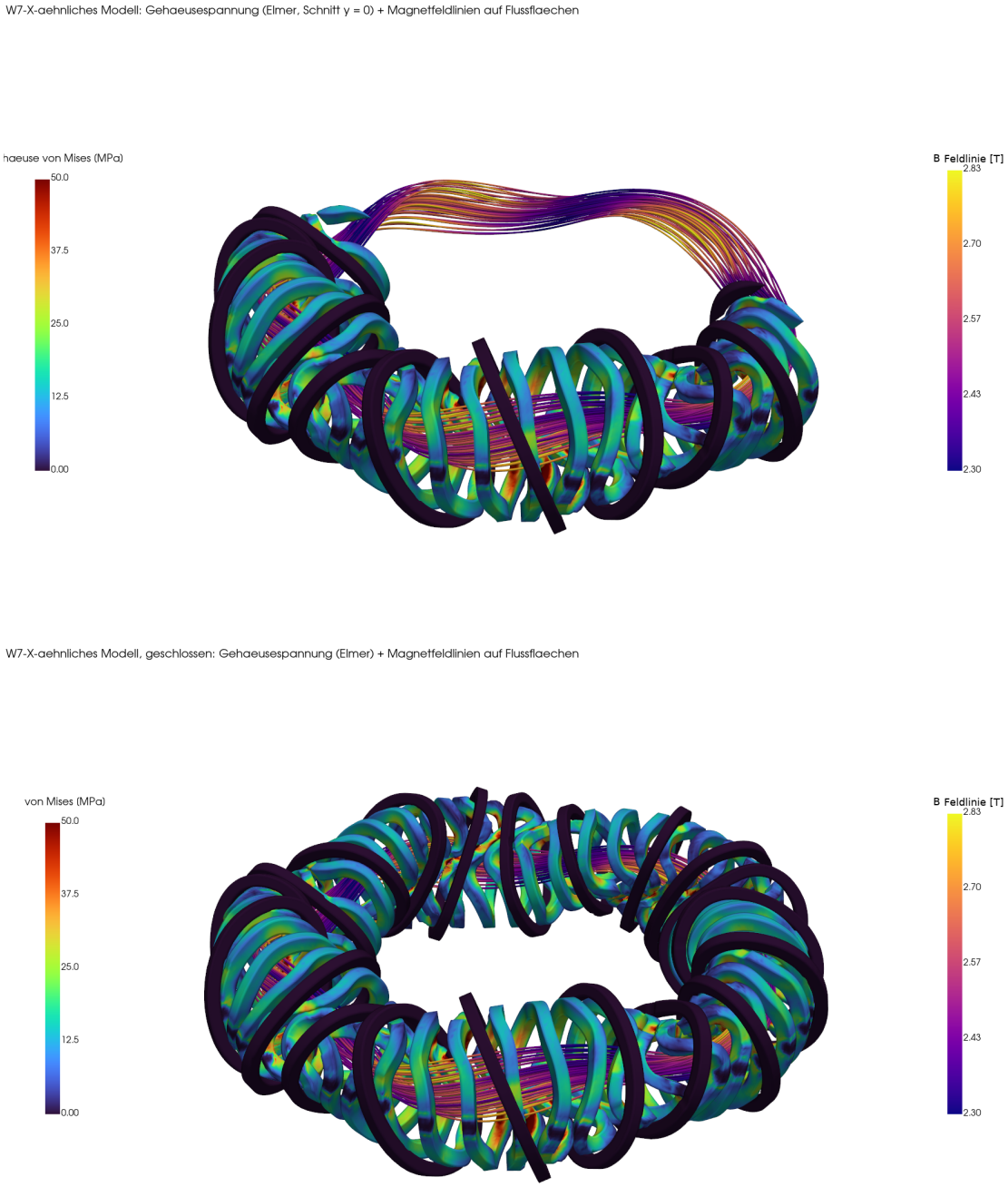

Figure 9: Casing stress with magnetic field lines on flux surfaces, cut open and closed.

How to read the figure: Both images show the same model with two colour scales. The casings are coloured by equivalent stress (left scale, 0 to 50 MPa). The thin lines are magnetic field lines, coloured by field strength (right scale). The top image is cut open so the field lines inside are visible; the bottom one is closed. You can see how the plasma runs through the cage of coils and where the casings are loaded most (yellow and red spots).

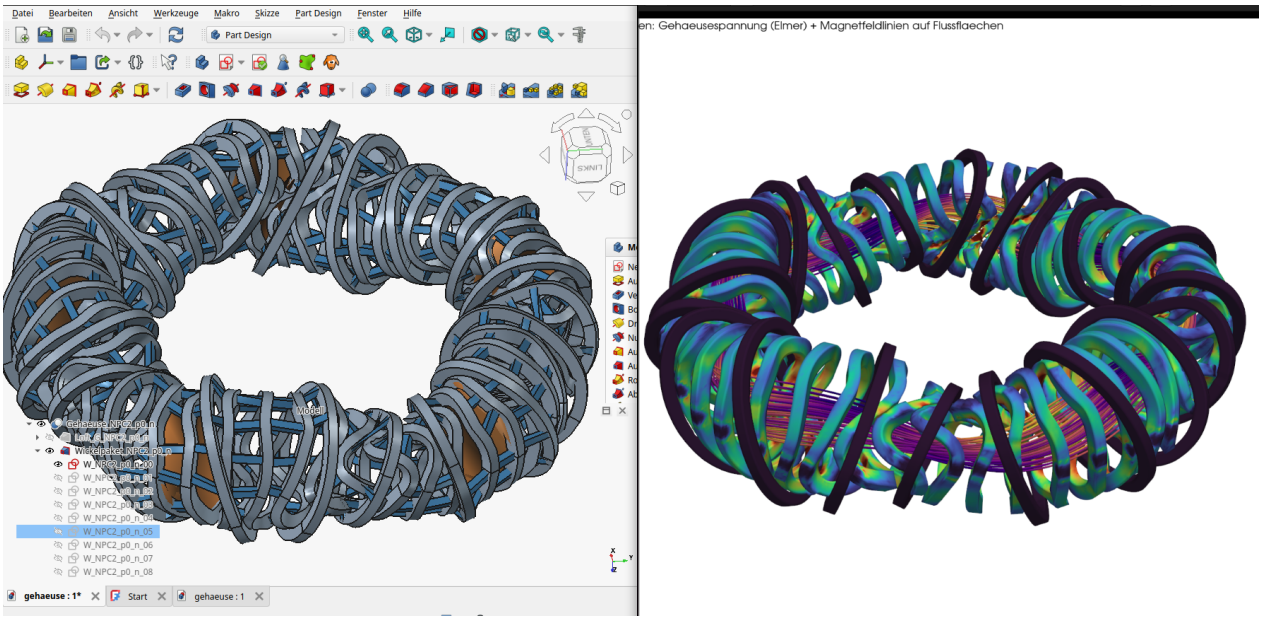

Figure 10: Both branches side by side: parametric CAD (left), stress analysis from the CAD-free pipeline (right).

How to read the figure: Left: the CAD model in FreeCAD; right: the computed stress. These are two separate routes. The CAD route gives exact, editable geometry for design and manufacture. The computational route builds the mesh directly from the path data and is therefore robust for the strength calculation. Both describe the same coils.

7 Visualisation: from mesh to Blender

A 3D cut-away model shows each coil's layers: cable (NbTi strands in copper, helium), aluminium jacket, turn insulation, ground insulation, potting (an assumption), steel casing; $12 \times 9 = 108$ conductors per coil. A separate conductor model answers the question "where does the current flow?": almost entirely in the superconducting NbTi; copper only takes over briefly during a quench; jacket and helium carry none.

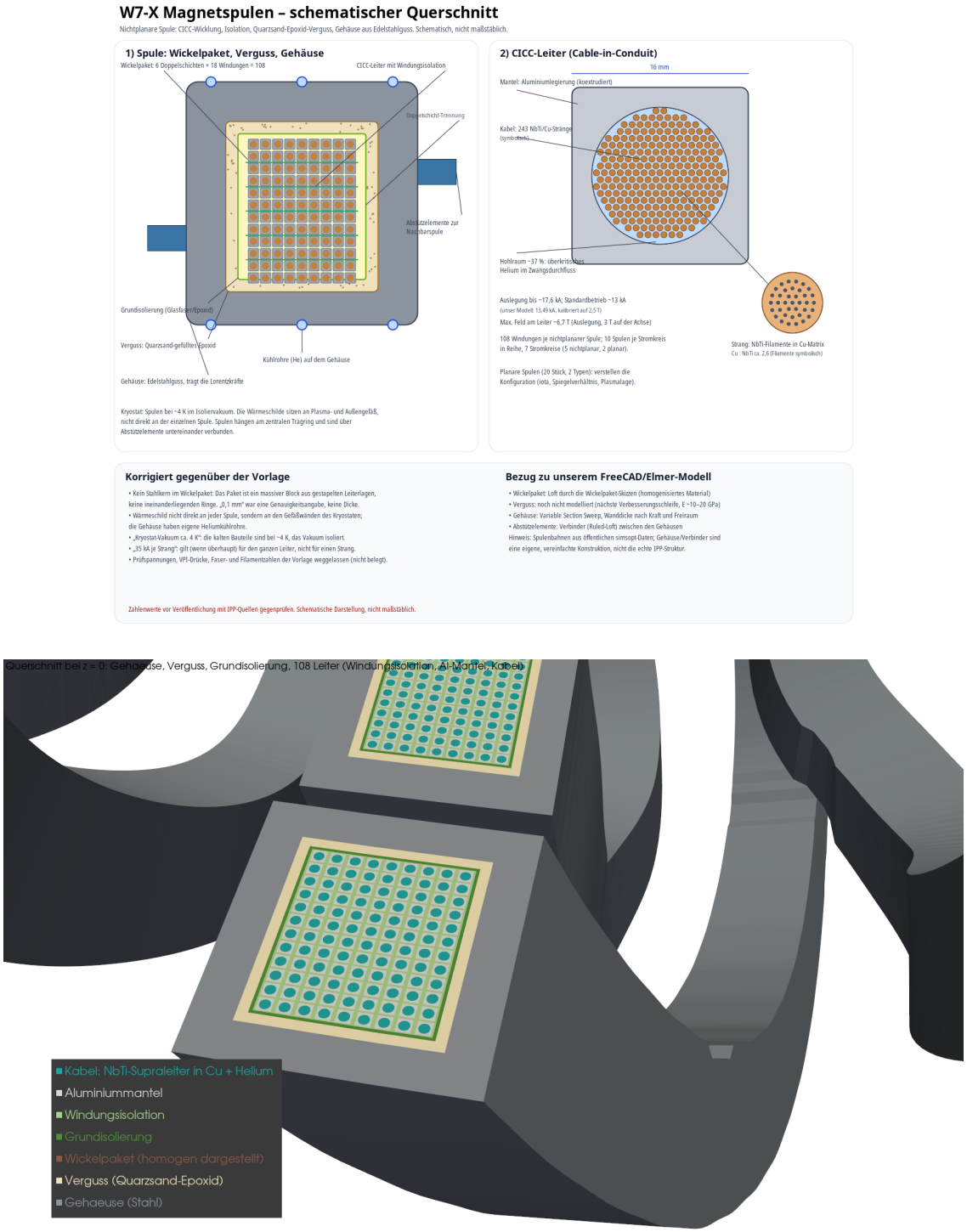

Figure 11: Coil layers (schematic, top) and in the 3D model (bottom).

How to read the figure: Top: a schematic cross-section of one coil (not to scale). Inside is the winding pack of 108 turns of conductor. Around it come insulation, a filling of quartz sand and epoxy, and an outer steel casing that carries the magnetic forces. On the right is the conductor itself: an aluminium jacket holding a cable of 243 strands. Liquid helium flows through the gaps and cools it to about 4 K, i.e.

−269 °C. Bottom: the same build-up as a 3D cut through two coils; the legend names the layers. The figure shows how many layers a superconducting coil needs.

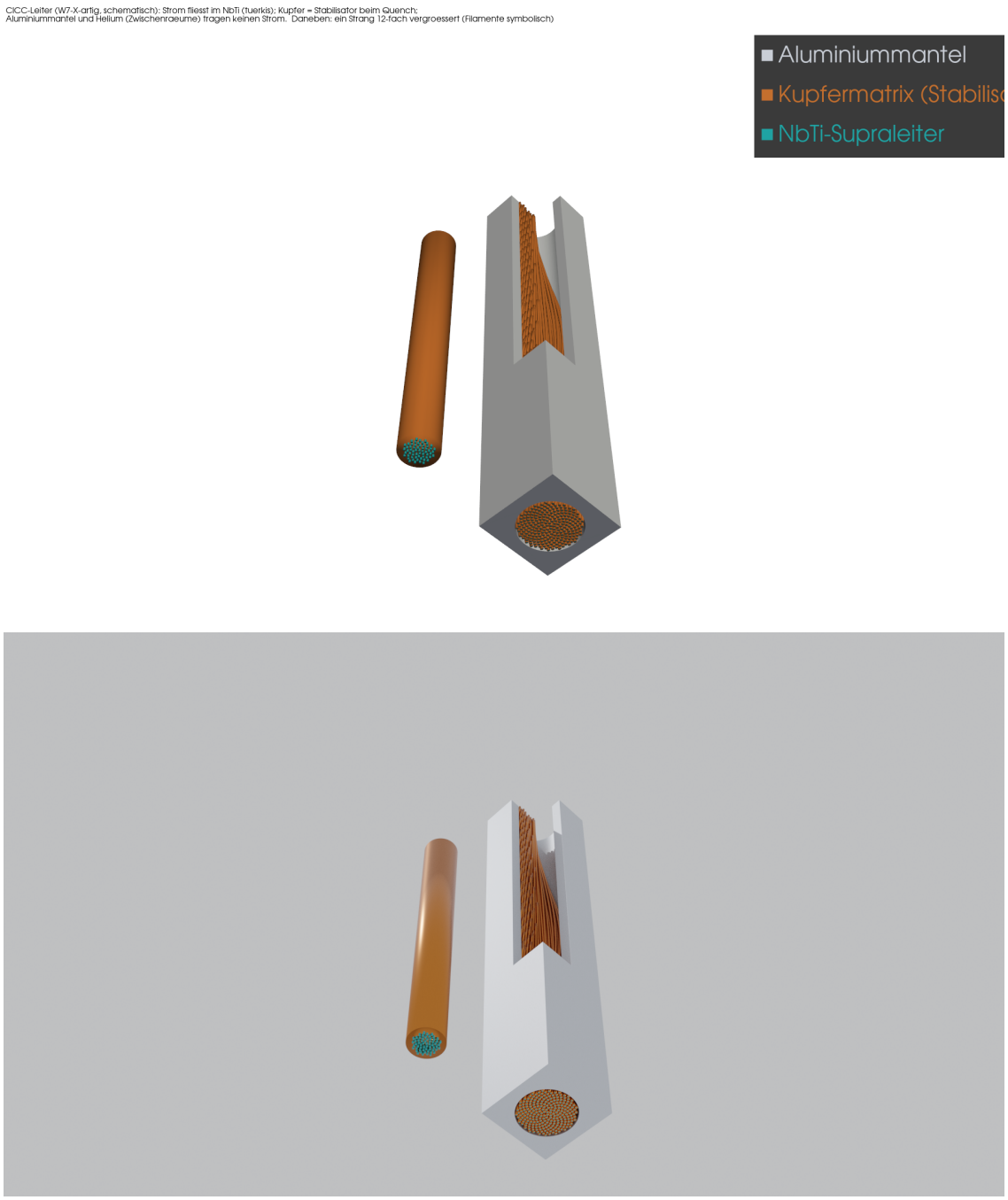

Figure 12: Conductor: opened jacket, twisted strands, one strand magnified with NbTi filaments (count symbolic). PyVista (top), Blender Cycles (bottom).

How to read the figure: A short piece of conductor with the aluminium jacket (grey) cut open. Inside are the twisted strands (copper coloured). Next to it one strand is magnified twelve times, showing the turquoise filaments of the superconductor niobium–titanium (NbTi). The current flows only in the superconductor, without resistance. The copper takes it over briefly if superconductivity breaks down (a *quench*). Bottom: the same rendered in Blender.

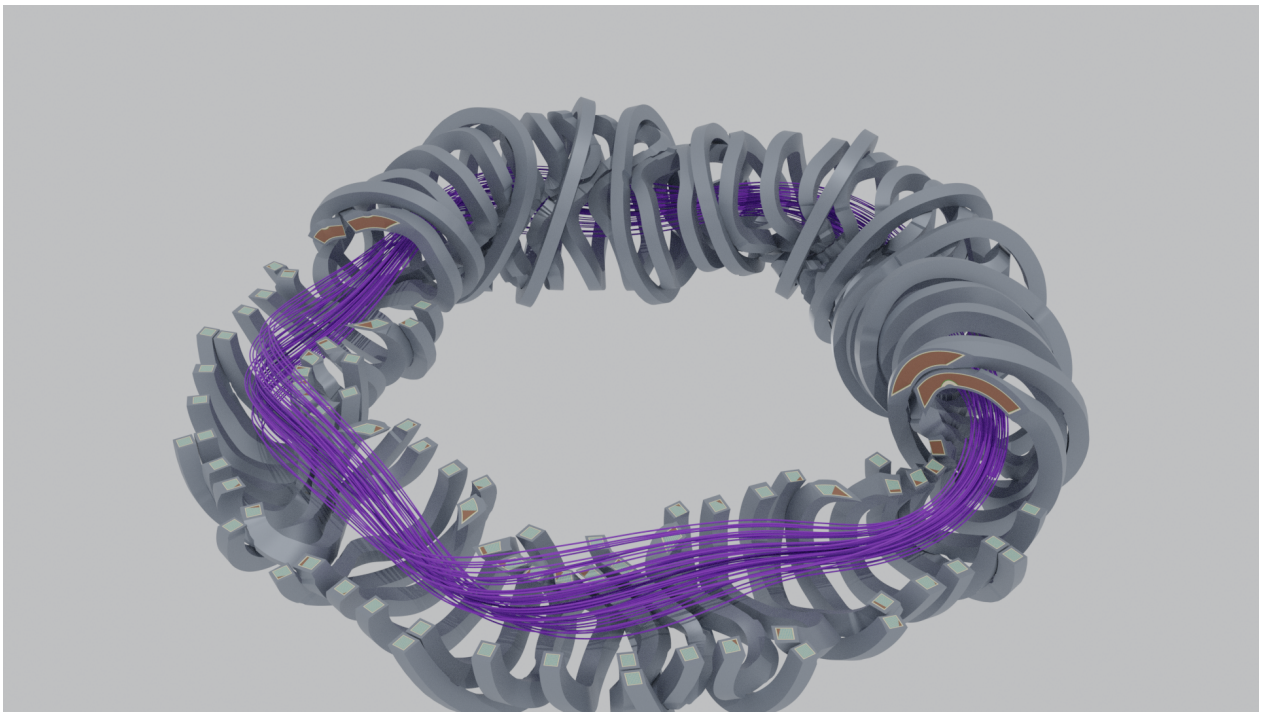

Figure 13: The full coil system with field lines, rendered in Blender.

How to read the figure: A rendered overall image for illustration only: grey coil casings, brown cut faces of the winding packs, and violet field lines showing where the plasma circulates.

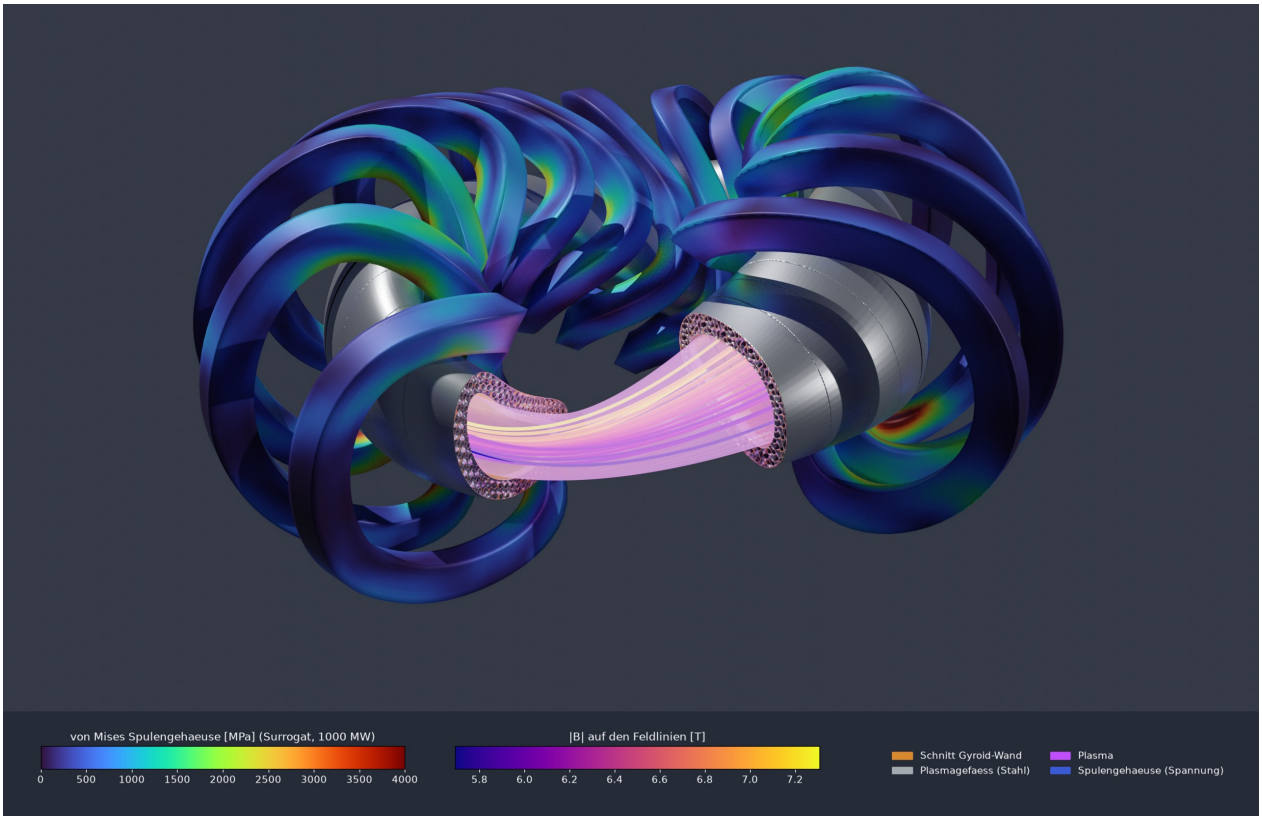

Figure 14: Later step: the 1 GW design with casing stress from the surrogate model (1000 MW), field lines, and a cut-open plasma vessel whose wall has a gyroid cooling core built with PicoGK. A rendering, not a cooling design.

How to read the figure: An outlook, not a finished design. The coil casings of the 1 GW plant are coloured

by stress (left scale; red spots would be too highly loaded for steel). The field lines are coloured by field strength (middle scale), and the plasma is pink. The plasma vessel (silver) is cut open. Its wall is a *gyroid*, a sponge-like, regularly repeating lattice. It is light and stiff and has channels for a coolant; its cut face is orange.

Getting PyVista's glTF export into Blender needed three fixes, each found by measuring instead of guessing: colours arrived as sRGB values in a linear slot (washed out; fixed by linearising), the scene was rotated by 90° about the vertical axis (fixed at the root node), and the distribution's Blender build lacks the denoiser (render fails; denoising disabled). Rendering runs on the CPU because the GPU is reserved for the local agent.

8 Takeaways for working with coding agents

- **Verification is the job.** The most expensive error looked plausible. Independent checks (invariants, second methods, regression tests from upstream projects) caught every major bug.
- **Silent failure is the default in scientific tools.** Wrong keywords ignored, current directions chosen by the solver, meshers hanging for an hour. Every pipeline stage now asserts on its output.
- **Prefer the boring path.** Direct mesh generation from data beat a CAD kernel with booleans. Keeping a working path alive while exploring the elegant one saved the project.
- **Humans supplied the decisive ideas** (research first, variable section sweep, neighbour subtraction, the parallel branch); agents supplied speed in implementing, testing and documenting them.
- **Files are the shared memory.** A findings log, a handover file and skills let two agents work on one project across sessions and context compactions.
- **References need checking** like any other agent output.

Open items

Potting layer between winding pack and casing; connectors and support ring instead of clamped patches; minimum wall thickness; wedge and casing in one body; wall sizing for the 1 GW design; HTS strain check; tolerance study (± 2 mm).

9 Follow-up: cooling the plasma vessel

Status 27 Oct 2026. Flow computed with cold water; heat only estimated roughly. The numbers show trends; they are not a design.

9.1 Task and layout

The plasma vessel wall of the 1 GW design is 0.9 m thick and has a core of gyroid filaments built with PicoGK [29]. A 10° segment (85°–95°) was studied. The assumed heat flux is 0.5 MW/m^2 on 26.7 m^2 of plasma-facing surface, i.e. 13.3 MW. With a 10 K temperature rise the segment needs 319 l/s of water.

The layout grew step by step from the user's instructions:

- Water flows *through the filaments*, not through holes in a perforated plate.
- The inlet is on the inboard side of the torus, the outlet outboard.
- The core sits between two inclined planes. In front of and behind it, the distribution and collection chambers taper off (Z-manifold). This shape came from a sketch the user drew on a screenshot.
- The gyroid is graded: a 130 mm layer with 100 mm cells on the plasma side, behind it a dense, coarse gyroid as load-bearing structure.

The flow was computed with FluidX3D [17], a lattice Boltzmann solver for GPUs [22, 23, 24, 25, 26, 27, 28]. The lattice has 8 mm spacing and 259 million cells. Each run covers 40 s of physical time with the LES turbulence model and takes about 15 min. FluidX3D has its own licence: free for non-commercial use only, source-available, not an open-source licence. The two modified files (`setup.cpp`, `defines.hpp`) are published as an "altered version" with the original licence in the repository folder `fluidx3d/`.

9.2 History

Variant	Change	Result
a, b	chamber with perforated plate; holes missed the plate (spotted by the user)	unstable, jets
f1	water through the filaments, open chambers	spread 141 %
f2	finer gyroid towards the plasma	spread 40 %
f4, f5	Z-manifold from the sketch, denser outer layer	spread 20 %, 7.9 kPa
f6–f9	thinner front layer, wall thickness by radius	dips at ±52°, cause searched
f10	lattice cells taken directly from PicoGK	dips gone, spread 36 %
f11	front wall 4.9 mm to 5.8 mm by radius	**spread 8 %**, 22 kPa, 288 l/s

Table 1: Variants of the cooling segment. Spread = standard deviation of the flow over 24 circumferential sectors divided by the mean.

9.3 An error in the lattice

In f6 to f9, two bands at ±52° of the circumference received almost no water. It looked like a design flaw. Three hypotheses were tested and rejected in turn: layer thickness, orientation of the layer relative to the gyroid, and inaccurate offset surfaces in PicoGK. Two intermediate findings were our own measurement errors.

The actual cause was the conversion of the water mesh into lattice cells. FluidX3D decides for each cell with a ray test whether it is inside. On nearly horizontal faces of the very fine gyroid mesh this failed: at $z = 1.906$ m a whole lattice layer became steel and blocked the band. It was found only by comparing the lattice with the mesh layer by layer. Since f10, PicoGK writes its voxel occupancy to a file that FluidX3D reads directly; it matches the mesh 100 %. The numbers of f6 to f9 are therefore distorted.

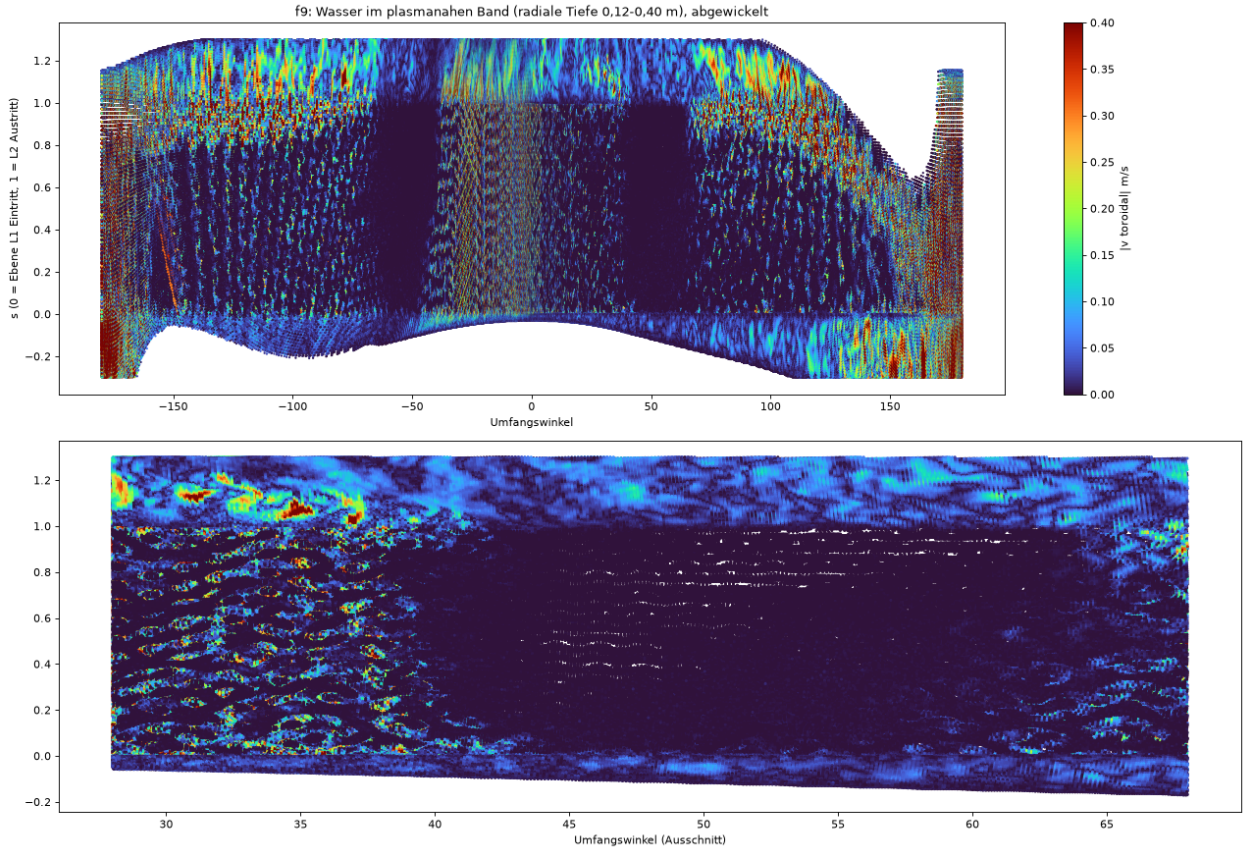

Figure 15: Lattice error (f9): water in the plasma-side layer, unrolled over the circumference (horizontal) and the path from inlet to outlet (vertical), colour = toroidal speed. The dark blocks at about ±50° carry no flow because one lattice layer there was wrongly set to steel. Zoom below. Labels in German.

> *How to read the figure:* Imagine the water-filled layer directly behind the plasma wall unrolled like a map. Horizontally it goes once around the cross-section (in degrees), vertically from inlet (0) to outlet (1). The colour shows how fast the water flows around the circumference; dark means it is standing still. Near ±50° there are two dark rectangles. Nothing flows there because the program had wrongly marked one layer of grid cells as steel. The bottom image is a close-up. Pictures like this reveal errors that averages hide. The bug was fixed afterwards.

After that, the distribution could be balanced in one step. The wall of the front filaments now varies with radius: 5.8 mm inboard, 4.9 mm top and bottom, 5.2 mm outboard. The spread dropped from 36 % to 8 %.

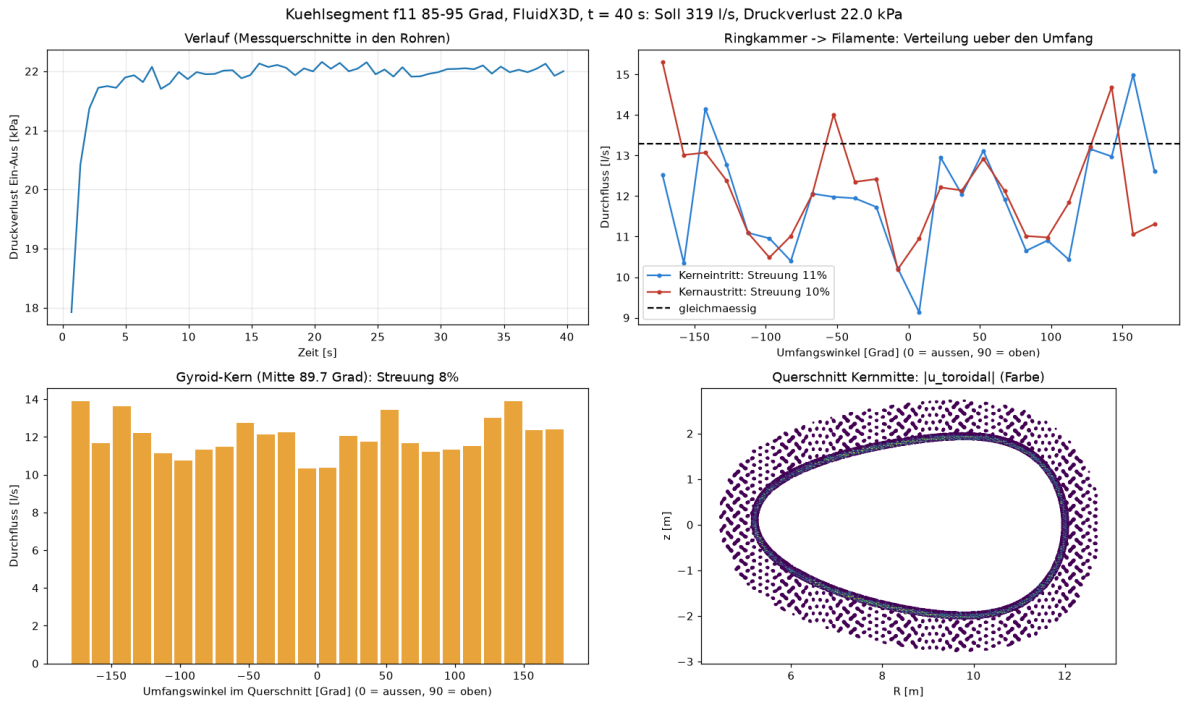

Figure 16: Evaluation of the FluidX3D run f11: pressure drop over time (steady after about 5 s), flow over the circumference at core inlet and outlet, flow through the gyroid core per sector (spread 8 %), and cross-section at mid core. Labels in German.

How to read the figure: Four panels. Top left: the *pressure drop*, i.e. how much pressure the pump must supply, over time. It rises during the first seconds and then stays flat at about 22 kPa (0.22 bar), so the flow has settled. Top right: the flow rate at each point around the circumference, entering (blue) and leaving (red) the core. The dashed line would be a perfectly even distribution; the deviation is about 10 %. Bottom left: the flow through the gyroid core per sector. The bars are similar in height (8 % scatter). Bottom right: a cross-section of the core centre. The water spreads fairly evenly, and the pumping effort is small.

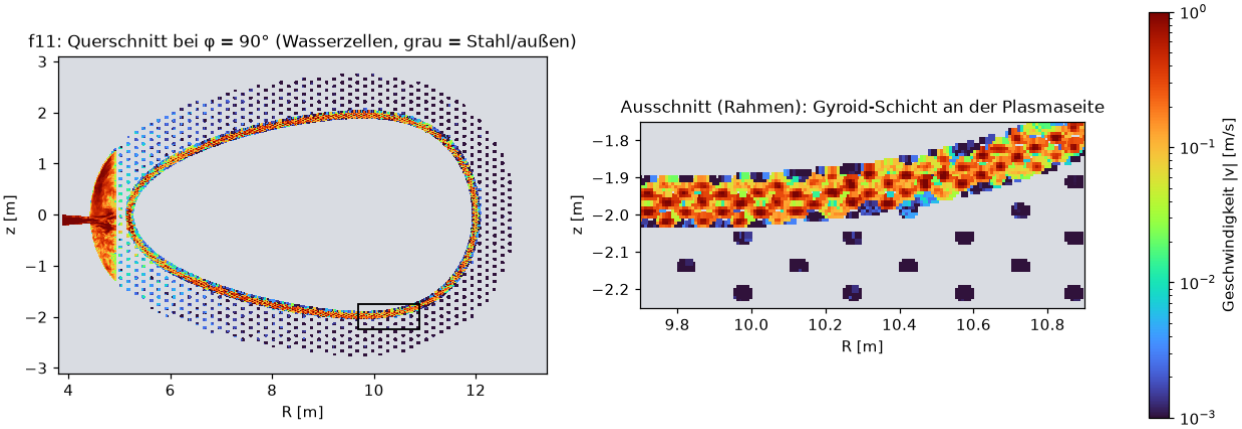

Figure 17: Cross-section through the f11 flow field at $\varphi = 90°$; every 8 mm lattice cell with water is a dot, colour = speed (log scale). Water enters on the left (inboard). Almost all the flow passes through the fine gyroid layer on the plasma side (zoom on the right, 0.1 m/s to 1 m/s). In the coarse dense gyroid behind it the water sits in closed pores; it only carries load. Labels in German.

How to read the figure: A cut across the cooling segment. Each cell with water is one dot, and the colour shows its speed on a logarithmic scale (red about 1 m/s, dark blue about 0.001 m/s). The water enters on the left at $R \approx 5$ m. The red ring on the plasma side shows that almost all the water flows through the fine layer right at the wall; the frame is magnified on the right. The dark dots behind it are water

standing still in the coarse gyroid. So cooling happens where the heat arrives, and the coarse gyroid only carries load.

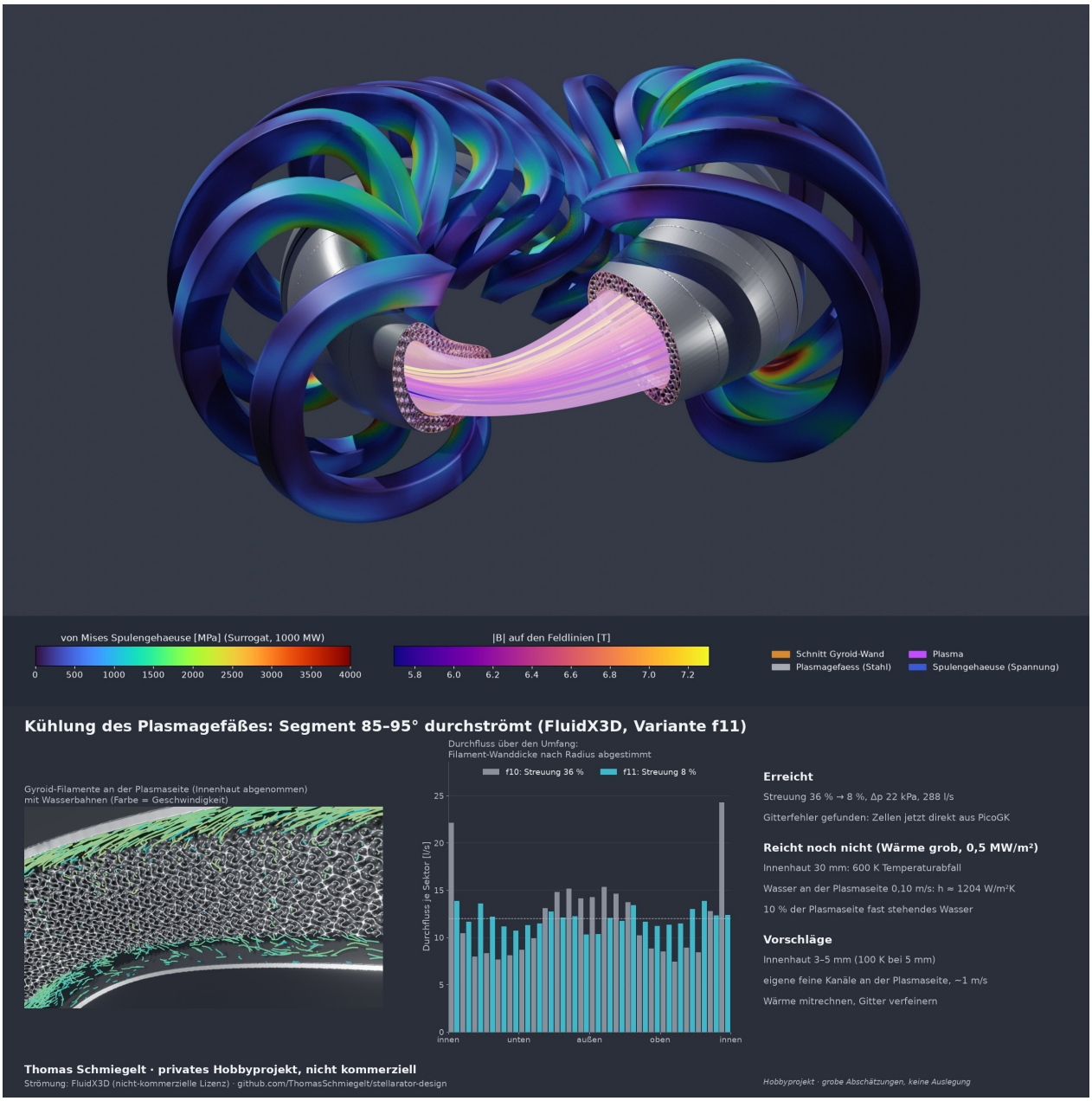

Figure 18: Overview of the 1 GW device extended by the cooling results: gyroid filaments on the plasma side with water paths (left), flow per circumferential sector before and after balancing (middle), findings (right). Labels in German.

How to read the figure: A summary on one board. Top: the plant. Bottom left: the gyroid filaments on the plasma side with the water paths. Centre: the flow per sector before balancing (grey, very uneven, 36 % scatter) and after (turquoise, 8 %). Right: the findings. Balancing greatly improved the distribution, but this layout does not yet reliably remove the plasma-side heat ($0.5\,\mathrm{MW/m^2}$).

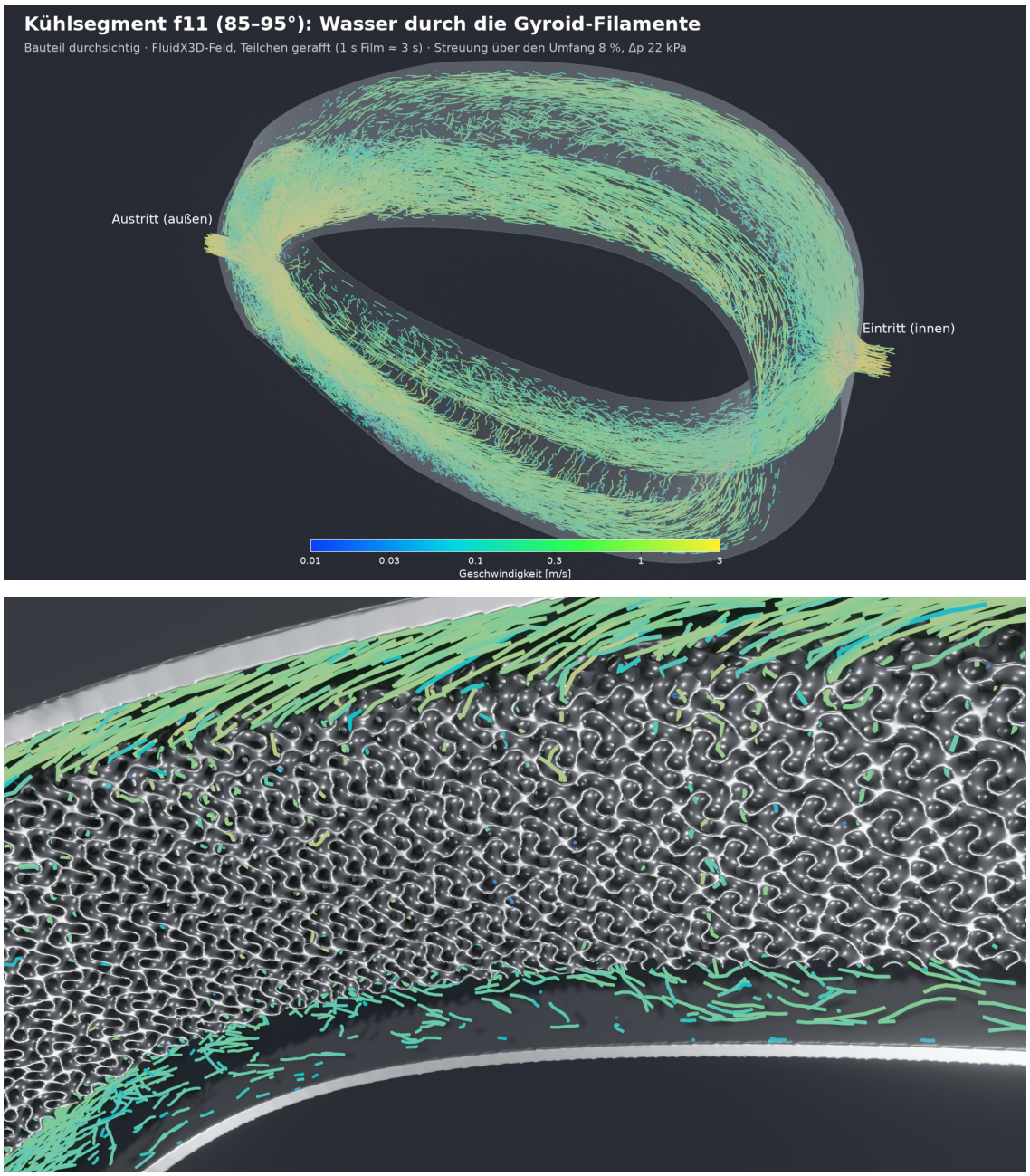

Figure 19: Frames from the flow films (Blender [14]): the segment rendered transparent, inlet inboard right, outlet outboard left (top); the lower half without the inner skin, looking at the plasma-side filaments (bottom). Colour = speed.

> *How to read the figure:* Stills from the flow videos. Each short streak is the track of an imagined water particle, and its colour shows its speed (blue slow, yellow fast). Top: the whole segment, transparent. The water enters at the inner right, spreads around the whole cross-section and leaves at the outer left. Bottom: a close-up of the gyroid filaments without the inner skin. Many fast tracks run in the gaps at the edge, fewer through the lattice itself. The water takes the easiest path.

9.4 Is it enough? A rough heat estimate

The flow computation contains no heat. A one-dimensional estimate for pressurised water at 300 °C (Dittus–Boelter, steel with $k = 25\,\mathrm{W/(m\,K)}$) shows two bottlenecks.

Quantity at 0.5 MW/m^2	Temperature drop
Conduction through the inner skin, 30 mm (as built)	600 K
Conduction through the inner skin, 5 mm	100 K
Transfer to the water, $v = 0.10$ m/s (f11), without / with fin effect	415 K / 210 K
Transfer to the water, fine 10 mm channels at 1 m/s	48 K

Table 2: Rough heat estimate (`stellarator/kuehlung_waerme.py`).

The distribution is solved, the cooling is not yet. The inner skin is far too thick and the water on the plasma side is too slow. In addition, on 10 % of the plasma side the water almost stands still (below 1 mm/s); hot spots would form there.

9.5 Suggested improvements

1. **Thinner inner skin:** 3 mm to 5 mm instead of 30 mm, possibly with tungsten armour. The gyroid core behind it carries the load.
2. **Faster flow on the plasma side:** either a thinner front layer (30 mm to 40 mm instead of 130 mm) with a finer cell, or a separate first-wall circuit with parallel channels at about 1 m/s. The gyroid behind then only carries load and removes volumetric heat. The pressure drop rises; at 22 kPa it is still small today.
3. **Remove stagnant water:** the near-zero speeds on the plasma side are probably at the end walls and chamber corners. They must be located in the field; guide vanes or a more open cell there are obvious options.
4. **Balance less sensitively:** one millimetre of wall thickness changes the flow by roughly a factor of 3.7. With typical manufacturing tolerances of ± 0.1 mm that is about ± 13 %. Defined orifices at the chambers, or a larger cell with thicker walls, are more robust.
5. **Compute the heat:** coupled flow and conduction (FluidX3D has a temperature extension, or Elmer for the steel). Only then can the fin effect of the filaments be judged.
6. **Check the lattice:** the front filament walls are about 10 mm thick and the lattice is 8 mm. A grid study at 4 mm to 6 mm is needed before the numbers are reliable.
7. **Reach the target flow:** 288 l/s instead of 319 l/s. The cause is the inlet boundary condition, not the geometry.

9.6 Collaboration

Here too, the decisive design ideas came from the user: water through the filaments, inlet inboard, inclined chamber boundaries. To agree on shape, a sketch on a screenshot worked best. This led to a small tool for drawing directly on the 3D model and handing the lines to the AI as 3D points [30]. The lattice error was found by the assistant, but only after several wrong guesses. The lesson: investigate an odd result before changing the design. The workflow is described in `docs/kuehlsegment_prozess.md` (German).

10 Follow-up: double filament, clearance and liquid lead

Of the improvements suggested in Section 9, the second one was taken up: the plasma side gets its own cooling circuit with fine channels. The requirements came from the user: the water should not get hotter than 150 °C, with an inlet of about 60 °C. On top of that came the idea of building the wall as a *double filament*: an inner part for cooling and, separate from it, an outer part for stiffness. All of it without colliding with the coils. All numbers in this section are rough estimates.

10.1 Wall layout

From the plasma side outwards: inner skin 5 mm, cooling channels, back wall 5 mm, dry gyroid for stiffness, outer skin 40 mm. The channels run toroidally across the segment from the inlet to the outlet plenum. The gyroid no longer carries water, so it may be more open and no longer has to be tuned to the cooling network.

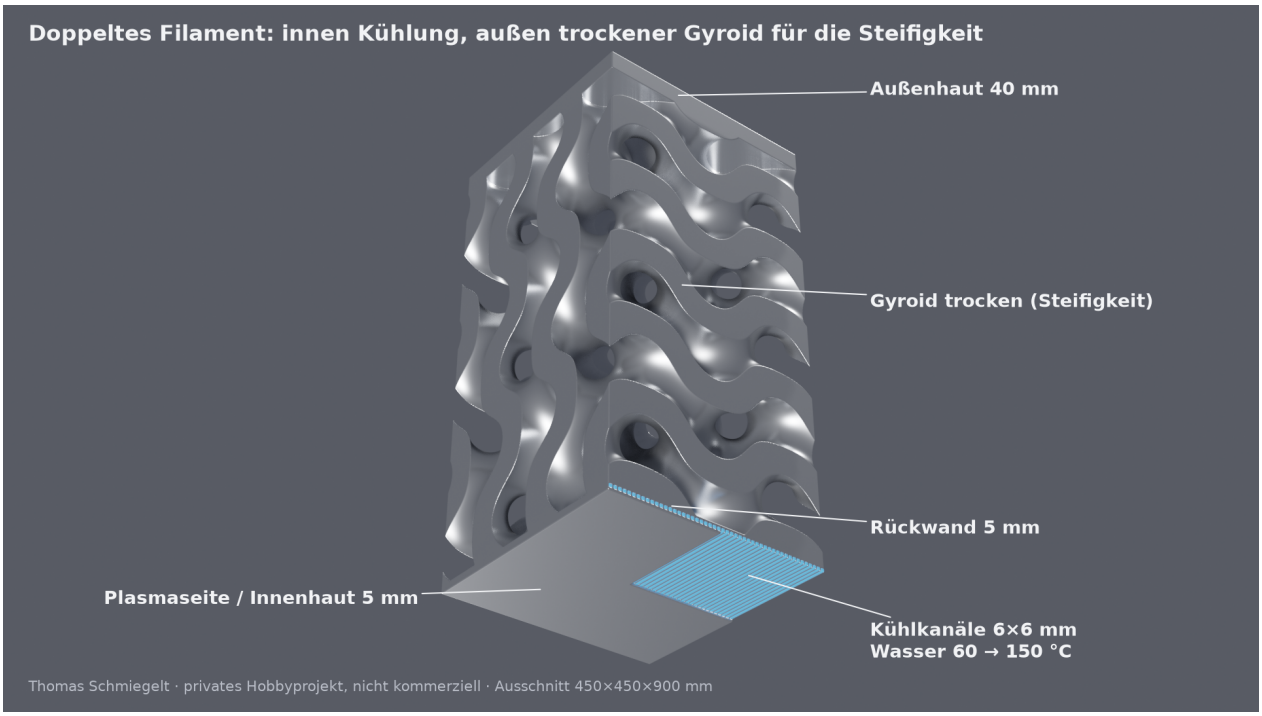

Figure 20: Wall piece with double filament (PicoGK, Blender), cut-out 450 × 450 × 900 mm. A window is cut into the inner skin; the cooling channels (blue) lie below. Labels in German.

> *How to read the figure:* A piece of the vessel wall, roughly the size of two shoe boxes. At the bottom is the plasma side (inner skin 5 mm). A window is cut out, showing the blue cooling channels beneath it, where water heats from 60 °C to 150 °C. Above them is a dry gyroid that only provides stiffness, and the outer skin is at the top. The jobs are separated: cooling close to the heat, load bearing behind it.

10.2 Water at 60 to 150 °C

At 0.5 MW/m² the segment (85°–95°) takes up about 12.6 MW on the plasma side. With a temperature rise of 90 K that is only 35 l/s. Spread over all channels in parallel the water would be too slow and would boil at the wall. Therefore three neighbouring channels are connected in series (U-turn at the end face). Method: heat transfer after Gnielinski, friction after Petukhov, temperature in the cross-section with Elmer for the worst channel (`stellarator/erstwand_kanal.py`).

Channels	6 × 6 mm, rib 3 mm, 616 strands with 3 passes each
Velocity	0.96 m/s to 2.15 m/s
Pressure drop	49 kPa
Steel maximum	300 °C
Channel wall maximum	202 °C (boiling from 212 °C at 2 MPa)

Table 3: First-wall channels with water 60 °C → 150 °C at 2 MPa (variant n5).

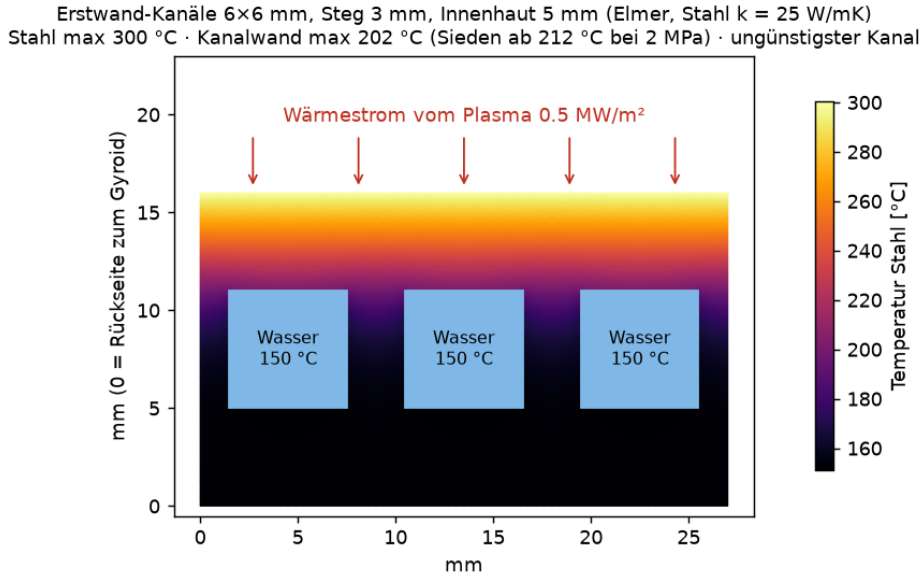

Figure 21: Steel temperature around the channels (Elmer, half pitch mirrored). Labels in German.

How to read the figure: A cut through the wall across three cooling channels (blue, water at 150 °C). The plasma's heat arrives from above (red arrows, 0.5 MW per square metre). The colour is the temperature in the steel: bright at the top, about 300 °C; dark at the bottom, about 160 °C. The steel stays within its allowed range. The channel wall reaches at most 202 °C, below the boiling point of 212 °C at this pressure, so the water does not boil.

So high pressures are not needed at 150 °C; 2 MPa leaves enough margin to boiling. The real drawback lies elsewhere: heat at 150 °C can hardly be turned into electricity.

10.3 Clearance to the coils

For every point of the plasma boundary the smallest distance to the coil casings and connectors of the surrogate model was computed (`stellarator/gefaess_abstand.py`). The minimum is 1.25 m to the casings and 1.70 m to the connectors; in the computed segment it is at least 1.88 m. Including the gap the wall needs 1.05 m; with 5 cm margin it fits everywhere.

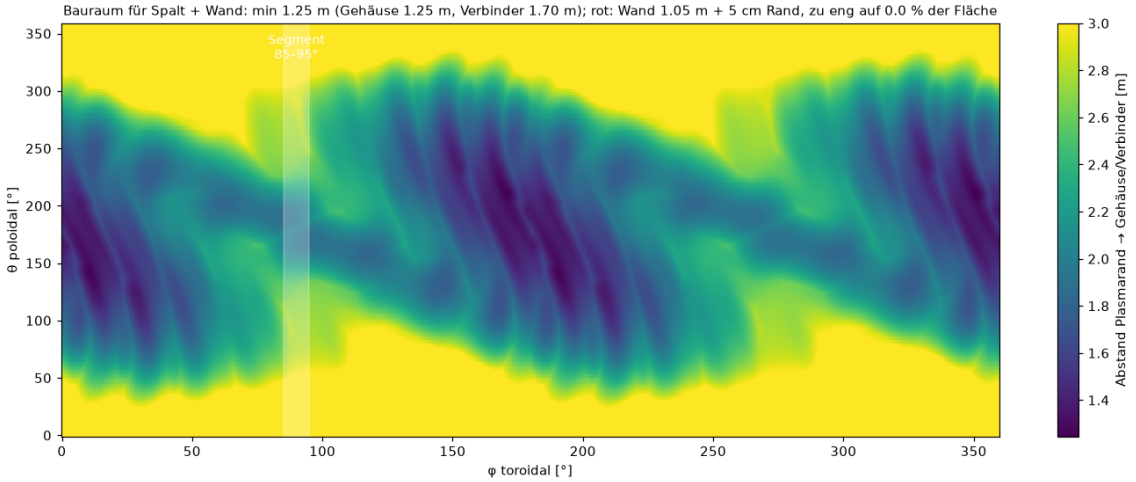

Figure 22: Distance from the plasma boundary to casings and connectors, unrolled over the torus. Red: limit at which the wall would no longer fit; it is reached nowhere. Labels in German.

How to read the figure: A map of the whole plasma surface, unrolled: horizontally once along the ring, vertically once around the cross-section. The colour is the free space between the plasma edge and the nearest casing or connector (yellow three metres or more, dark violet about 1.25 m). The wall needs

1.05 m plus a 5 cm margin, so the smallest gap of 1.25 m is enough everywhere. The slanted bands follow the coils. The light stripe marks the segment that was computed.

10.4 Liquid lead instead of water?

As a next step the user proposed a *gyroid within a gyroid*: cooled near the plasma with liquid lead, which passes its heat on to the water. A gyroid wall does split space into two separate, intertwined networks and would make a compact heat exchanger. Lead needs no pressure, and lead-lithium would also breed tritium. Three points speak against putting the exchanger into the wall, however:

1. Lead melts at 327 °C, lead-lithium at about 235 °C. Next to water at 60 °C to 150 °C it would freeze.
2. A leak between hot lead and water right at the plasma is dangerous and cannot be repaired.
3. In the magnetic field, induced currents brake the metal as soon as it flows across the field.

For the third point the pressure drop was roughly estimated (`stellarator/blei_kanal.py`): $\mathrm{d}p/\mathrm{d}x = \sigma\, v\, B_\perp^2\, k_p$ with $k_p = c/(1+c)$ for conducting steel walls (c wall conductance ratio) and $k_p \approx 1/\mathrm{Ha}$ for an insulating layer. The field of a stellarator is mainly toroidal, so toroidal channels run almost along the field; the pitch of the field lines was estimated at about 6°.

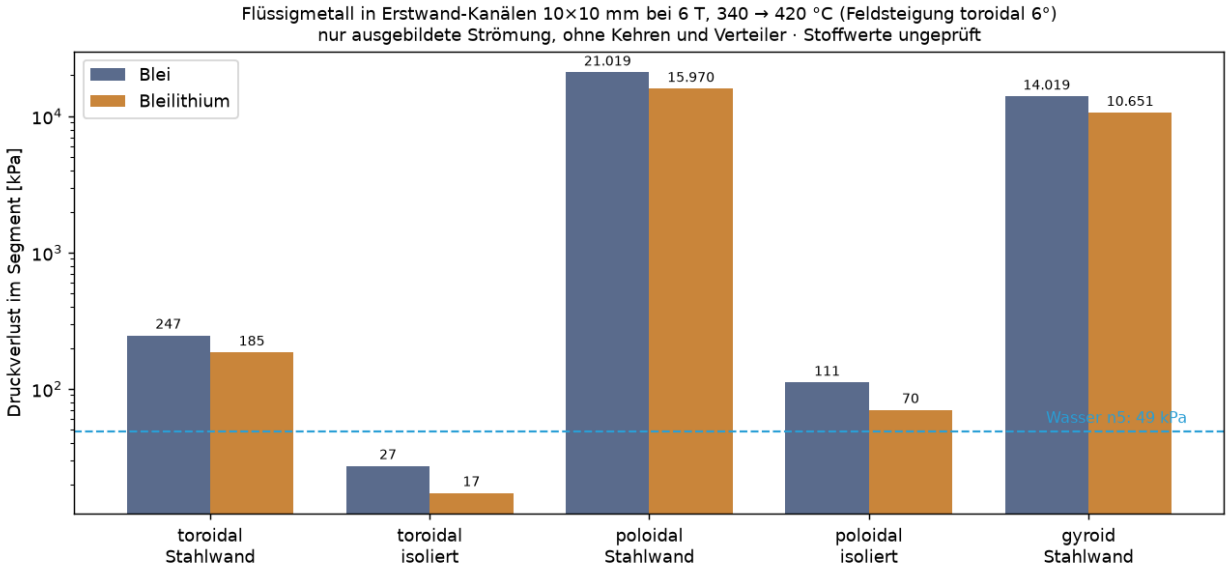

Figure 23: Pressure drop in the segment for lead and lead-lithium per channel direction at 6 T, logarithmic. Water (n5) for comparison. Labels in German.

How to read the figure: Each bar is the pressure needed to pump liquid metal through the channels of one segment (blue lead, orange lead–lithium). The scale is logarithmic; each major tick means ten times more. The dashed line is water for comparison. When a conducting metal moves through a magnetic field, electric currents arise in it that brake it (the *MHD effect*). If it flows along the field (toroidal) or the walls are electrically insulated, the pressure stays small. Across the field (poloidal) or in the gyroid it becomes a thousand times larger, above 10 MPa. Liquid metal is therefore only an option for channels along the field.

Channel direction	Pressure drop	Pumping power / heat	
toroidal, steel wall	247 kPa	0.2 %	suitable
toroidal, insulated	27 kPa	0.02 %	suitable
poloidal, steel wall	21 MPa	17 %	unsuitable
gyroid, steel wall	14 MPa	11 %	unsuitable

Table 4: Lead in 10 × 10 mm channels, 340 °C → 420 °C, 0.87 m/s, 103 l/s per segment (variant b5).

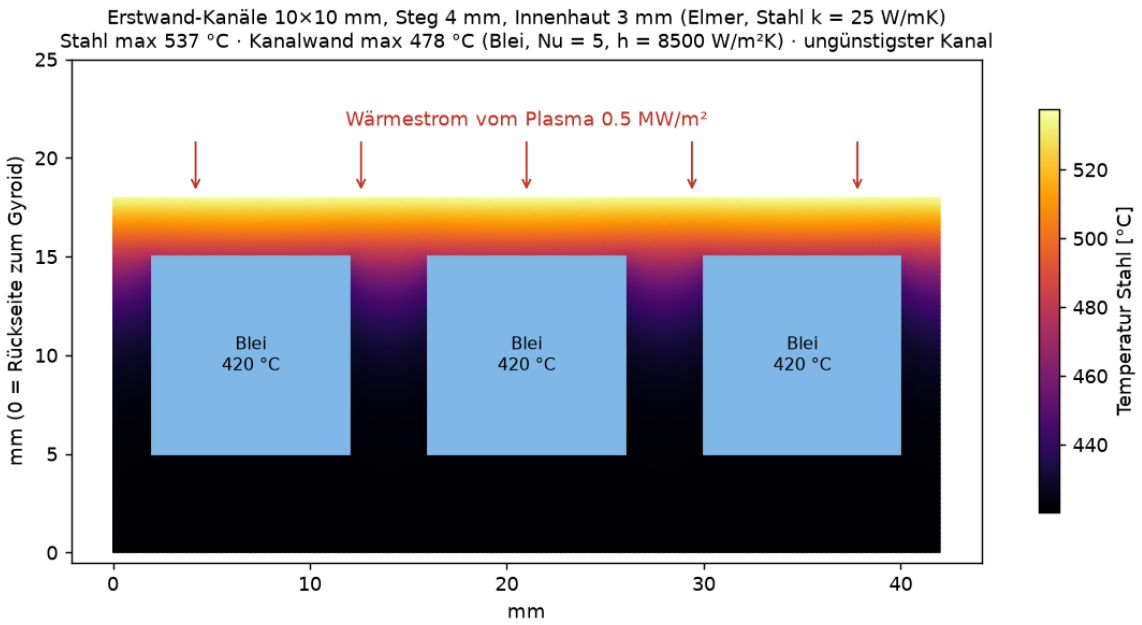

Figure 24: Steel temperature with lead (b5, inner skin 3 mm): at most 537 °C. With a 500 °C outlet and 20 mm channels it would be 711 °C. Labels in German.

> *How to read the figure:* Like the water calculation above, but with liquid lead at 420 °C in larger channels and a thinner 3 mm inner skin. The colour is the steel temperature, at most 537 °C on the plasma side. That works, but it is close to the limit of the steel. With hotter lead at the outlet it would get too hot.

Result: **the gyroid is not suitable for liquid metal**, because it keeps turning the flow across the field. Toroidal channels, on the other hand, work well. A sensible layout: lead or lead-lithium in toroidal channels on the plasma side, the dry gyroid behind it for stiffness, and a steam generator outside the coils that passes the heat to the water. At about 420 °C the heat can also be converted to electricity much better.

Not included are the additional losses in bends, manifolds and changes of cross-section (often the largest share), corrosion of steel in lead, volumetric heat from neutrons, and a more accurate heat transfer (conservatively Nu = 5 here). The material properties have not been checked against sources.

10.5 Collaboration

The ideas in this section came from the user: the temperature limit, the double filament, the check against the coils and the gyroid within a gyroid with lead. The assistant calculated them, named the limits and proposed variants. Along the way a small browser tool was built for looking at flow sections: one specifies section planes and hands them to the AI with “send and next section” or “send and finish” (`stellarator/schnitt_werkzeug.py`).

11 Coolant, strength and heat balance up to the turbine

This section follows the cooling from a single segment to the grid. Three questions started it: what does the thinner inner skin gain? Which variant is best when the figures of merit contradict each other? And which pressure and temperature are realistic? The user settled the last one: *reuse an existing turbine from a nuclear power plant and keep everything else as simple as possible.*

11.1 Variant f12: 5 mm inner skin

f12 is f11 with a 5 mm instead of 30 mm inner skin and a 500 mm outlet pipe (rule: outlet always larger than inlet). Flow with FluidX3D as in Section 9.

	f11	f12
Spread around the circumference (gyroid core)	8 %	11 %
Pressure drop	22.0 kPa	19.4 kPa
Flow rate	288 l/s	289 l/s
Velocity on the plasma side	0.10 m/s	0.10 m/s
Temperature drop across the inner skin at 0.5 MW/m^2	600 K	100 K

Table 5: f11 and f12 compared.

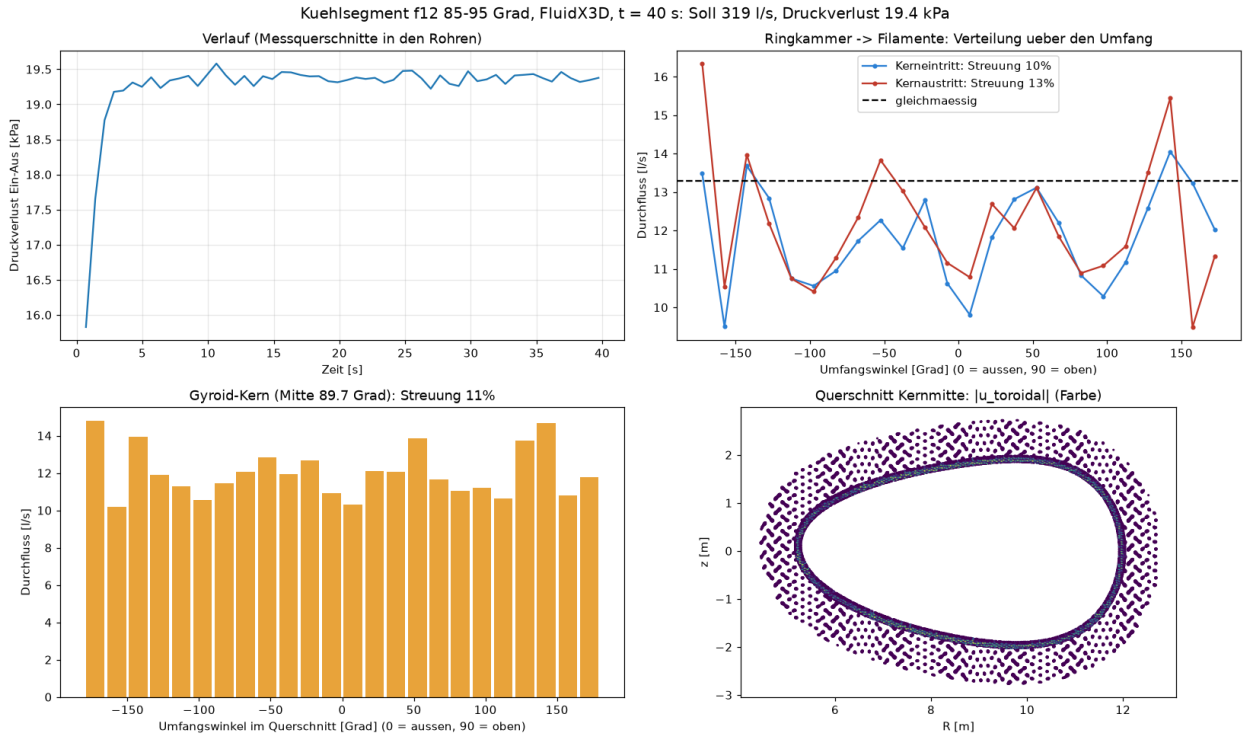

Figure 25: Evaluation of f12: pressure drop over time, distribution around the circumference and flow through the gyroid core.

> *How to read the figure:* Same layout as the f11 evaluation: top left the pressure drop over time (settled at 19.4 kPa), top right the distribution around the circumference, bottom left the flow through the gyroid core per sector (11 % scatter). Variant f12 needs slightly less pumping pressure with a similar distribution.

The wall is no longer the bottleneck. What remains is the transfer into the slow water on the plasma side, about 410 K.

11.2 Ranking by pairwise comparison

Each variant yields several figures of merit that often conflict: f12 beats f11 on temperature and pressure drop but loses on spread. The user suggested deciding how to proceed by pairwise comparison so that resources

are used well. Implemented in `stellarator/varianten_vergleich.py`:

1. **Pareto front:** a variant drops out if another one is worse in no criterion and better in at least one. No weights are needed.
2. **Weights by AHP** (Analytic Hierarchy Process): the user compares the criteria in pairs on a scale of 1 to 9. From the comparison matrix A with $a_{ji} = 1/a_{ij}$ the weight vector w is the eigenvector of the largest eigenvalue $\lambda_{\max}$. Consistency is checked by $\mathrm{CR} = \frac{(\lambda_{\max}-n)/(n-1)}{\mathrm{RI}_n}$ with the random index $\mathrm{RI}_4 = 0.90$; below 0.1 the judgements count as consistent.
3. **Rank:** weighted sum of the figures of merit normalised to 0 to 1.

The user's judgements: temperature is five times as important as spread, pressure drop and stagnant water each; pressure drop is three times as important as spread and five times as important as stagnant water. This gives weights of 0.61 for temperature, 0.23 for pressure drop, 0.08 for spread and 0.08 for stagnant water, with $\mathrm{CR} = 0.10$, right at the limit.

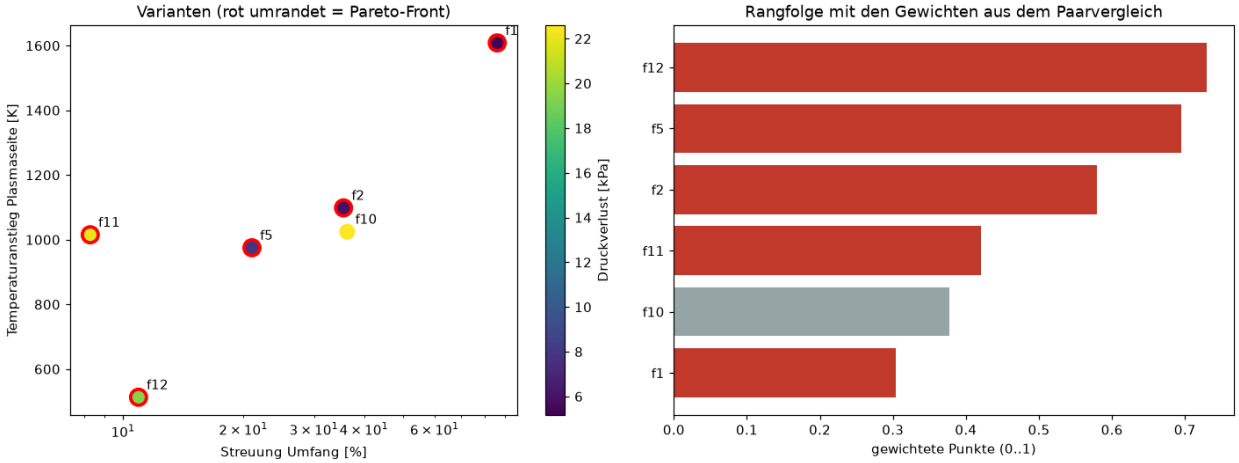

Figure 26: Left: variants by temperature rise and spread, colour pressure drop, red outline the Pareto front. Right: ranking with the user's weights.

> *How to read the figure:* Left: each dot is one computed variant (f1 to f12). Horizontally is how unevenly the water is spread, vertically how much the plasma side heats up. Both should be small, so lower left is better. The colour shows the pressure drop. Circled in red is the *Pareto front*: no other variant is better in every respect at once. Right: the ranking by points. How much each criterion counts was set by the user in a pairwise comparison. f12 comes first.

Ranking: f12 (0.73) ahead of f5 (0.69), f2 (0.58), f11 (0.42), f10 (0.38) and f1 (0.30). Notably, f11 had been optimised for an even distribution, which the user in the end weighted low. The pairwise comparison thus showed that effort had gone into a secondary goal.

11.3 Does the thin skin hold the pressure?

A thin skin is good thermally but has to carry the coolant pressure. `stellarator/haut_festigkeit.py` computes with Elmer a steel plate of thickness t over 2×2 gyroid cells ($a = 100\,\mathrm{mm}$). It is clamped where the 10 mm gyroid walls touch the skin. In the plane of the skin these are the points with

$$\frac{|f|}{|\nabla f|}\,\frac{a}{2\pi} < \frac{w}{2}, \qquad f = \sin X\cos Y + \sin Y\cos Z + \sin Z\cos X, \quad X = 2\pi x/a.$$

In between, the coolant pushes towards the plasma. As a check, a strip with straight supports at a spacing of $b = a/2 - w = 40\,\mathrm{mm}$ was computed. With both edges clamped, $\sigma = p\,b^2/(2t^2)$: at $t = 5\,\mathrm{mm}$ Elmer gives 34.3 MPa per MPa of pressure, the formula 32.0 MPa.

Skin thickness	Stress per MPa of pressure	Allowable pressure at 150 MPa
5 mm	39 MPa	3.8 MPa
10 mm	14 MPa	10.5 MPa
15 mm	9 MPa	16 MPa

Table 6: Inner skin over the gyroid (Elmer, linear elastic, no thermal stresses).

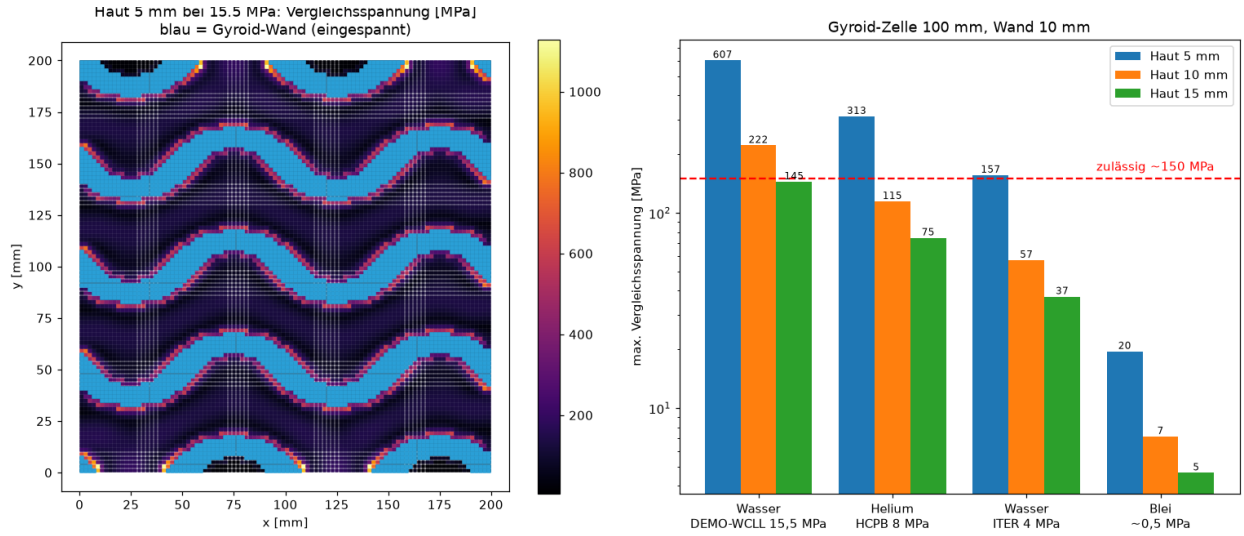

Figure 27: Left: von Mises stress in the 5 mm skin at 15.5 MPa, gyroid walls in blue. Right: peak stress per coolant and skin thickness.

> *How to read the figure:* Left: a top view of a 20×20 cm piece of skin pressed by the cooling-water pressure of 15.5 MPa (155 bar). It is supported by gyroid walls (blue). The colour is the stress; bright yellow means high, mainly between the supports. Right: the highest stress for each coolant and skin thickness on a logarithmic scale. The red line is the allowed value of about 150 MPa. With high-pressure water the skin must be about 15 mm, with helium 10 mm. For lead, which flows almost without pressure, the thickness hardly matters.

For pressurised water as in a nuclear plant (≈ 16 MPa) a water-filled gyroid is therefore unsuitable. High-pressure water belongs in small channels: a 6 mm channel under a 5 mm skin reaches only about 11 MPa by the same formula. That is the double filament of Section 10.

User's remark: the gyroid only rests on lines. The gyroid walls touch the skin only along narrow lines (26 % of the area in the model). Forces collect there; in the figure above the stress peaks sit exactly at the edges of these joints. In the double filament the water pressure is balanced inside the channel plate (skin and back wall pull on each other through the webs), so the gyroid does not carry it. What remains are thermal expansion between the hot channel plate and the gyroid, eddy-current forces and fatigue under cyclic operation, all acting on these lines. Possible remedies: fillets at the joints, a gyroid that thickens towards the skin, or a layer of ribs as a transition. Not yet calculated.

11.4 Sourced reference values

Coolant data of real plants were researched with the local agent PI. A first attempt with another local agent (Hermes) produced invented values and links. For PI a skill `web-recherche` was therefore written: it actually fetches pages and records each value only with a verbatim quote and its source. A checker confirms that each quote appears in the fetched text. Because the local model has little context, PI researches each plant in a separate call; the user suggested this after a long run had produced nothing. All 29 quotes were confirmed. PI still misread two rows (inlet and outlet swapped at the steam generator); they were corrected on review.

Plant	Coolant	In / out	Pressure	Source
EPR	water	295 / 320 °C	15.5 MPa	Wikipedia
VVER-1000	water	289 / 319 °C	16.0 MPa	Wikipedia
BREST-OD-300	lead	340 / 540 °C	–	Wikipedia
ALFRED	lead	390 / 520 °C	unpressurised	Wikipedia
HTR-PM	helium	250 / 750 °C	7.0 MPa	OSTI report
ITER blanket	water	70 °C / –	4 MPa	iter.org
EU-DEMO HCPB	helium	300 / 500 °C	8 MPa	KIT

Table 7: Sourced coolant data; full quotes in `docs/recherche/pi/ergebnis.md`.

11.5 Decision: pressurised water as in a nuclear plant

With an existing turbine from a pressurised water reactor the steam conditions are fixed, and with them the primary circuit: pressurised water at about 289 °C → 319 °C and 16 MPa as in the VVER-1000. Steam generators, main coolant pumps and pressuriser are off-the-shelf. Lead (freezing at 327 °C, braking in the magnetic field) and helium (8 MPa, blowers) are dropped. The first-wall channels (6×6 mm, three passes) were recomputed for these conditions: steel at most 441 °C, channel wall 340 °C (boiling at 347 °C), pressure drop 0.24 MPa. The steel thus sits in the working range of EUROFER (about 300 °C to 550 °C); with water at 60 °C to 150 °C it was below it. The margin to boiling, 7 K, is tight (first stated as 5 K with a wrong saturation temperature).

11.6 Derivation of the heat balance

The chain runs backwards from the grid to the plasma, with the same assumptions as the power balance of stage 0 (`invers/stufe0_bilanz.py`), here split by component (`stellarator/waermebilanz.py`).

1. Thermal power from grid power. P_{grid} goes to the grid; the turbine converts the heat P_{th} into electricity with efficiency η. Pumps ($f_p P_{\text{th}}$) and the remaining house load P_{house} (coil refrigeration, tritium plant, buildings) are subtracted:

$$P_{\text{grid}} = \eta P_{\text{th}} - f_p P_{\text{th}} - P_{\text{house}} \quad \Rightarrow \quad P_{\text{th}} = \frac{P_{\text{grid}} + P_{\text{house}}}{\eta - f_p} = \frac{1000 + 60}{0.35 - 0.04}\,\text{MW} = 3419\,\text{MW}.$$

2. Fusion power. One D-T fusion releases 17.59 MeV; the alpha particle carries 3.52 MeV, the neutron 14.07 MeV. The alpha fraction is $f_\alpha = 3.52/17.59 = 0.200$. Nuclear reactions in the blanket multiply the neutron energy by $M = 1.15$:

$$P_{\text{th}} = P_{\text{fus}}\left[(1 - f_\alpha)M + f_\alpha\right] = 1.120\,P_{\text{fus}} \quad \Rightarrow \quad P_{\text{fus}} = 3053\,\text{MW}.$$

3. Split by component. The neutrons pass through the first wall and deposit their energy in the blanket. The alpha power stays in the plasma and leaves it as radiation (onto the whole first wall) or as particle flow along the field lines (onto the divertor). Half each way is assumed ($f_{\text{div}} = 0.5$):

$$P_{\text{blanket}} = (1 - f_\alpha)\,M\,P_{\text{fus}} = 2808\,\text{MW},$$
$$P_{\text{first wall}} = (1 - f_{\text{div}})\,f_\alpha\,P_{\text{fus}} = 305\,\text{MW}, \qquad q_{\text{first wall}} = P_{\text{first wall}}/S = 305\,\text{MW}/1230\,\text{m}^2 = 0.25\,\text{MW/m}^2,$$
$$P_{\text{divertor}} = f_{\text{div}}\,f_\alpha\,P_{\text{fus}} = 305\,\text{MW}, \qquad A_{\text{divertor}} \geq P_{\text{divertor}}/q_{\max} = 305\,\text{MW}/10\,\text{MW/m}^2 = 31\,\text{m}^2.$$

S is the plasma surface of the HTS design (from stage 0: neutron wall load 1.99 MW/m²). The first-wall channels are designed for 0.5 MW/m², a factor of 2 margin for local peaks. 10 MW/m² is the design load of the water-cooled divertor elements of Wendelstein 7-X (Wikipedia).

4. Cooling water. With $\Delta T = 30$ K and $c_p = 5380$ J/(kg K), $\rho = 732$ kg/m³ at about 300 °C:

$$\dot{m} = \frac{P}{c_p\,\Delta T}: \qquad \dot{m}_{\text{blanket}} = 17\,400\,\text{kg/s}\ (23.8\,\text{m}^3/\text{s}), \qquad \dot{m}_{\text{first wall}} = 1890\,\text{kg/s}\ (2.6\,\text{m}^3/\text{s}).$$

The pumping power for the first-wall channels alone is small: $\dot{V}\,\Delta p/\eta_{\text{pump}} = 26.4\,\text{m}^3/\text{s} \cdot 0.24\,\text{MPa}/0.8 \approx 8$ MW, even if all the water passed through such channels. The assumed $f_p P_{\text{th}} = 137$ MW leaves room for steam generators, piping and blanket.

5. Turbine. $P_{\text{gross}} = \eta\, P_{\text{th}} = 1197\,\text{MW}$, waste heat at the condenser $(1-\eta)P_{\text{th}} = 2223\,\text{MW}$. Comparison with sourced plants:

- **Isar 2 (Konvoi):** 3950 MW thermal, 1485 MW gross, 1410 MW net, i.e. $\eta_{\text{gross}} = 0.376$; shut down since 15 April 2023. The fusion plant would need 87 % of its heat and 81 % of its turbine output: **one turbine of this class is enough**.
- **VVER-1000:** 3000 MW thermal, net efficiency 31.7 %, about 1 GW electric. The fusion plant delivers 114 % of this heat; a single turbine of this size would be too small.

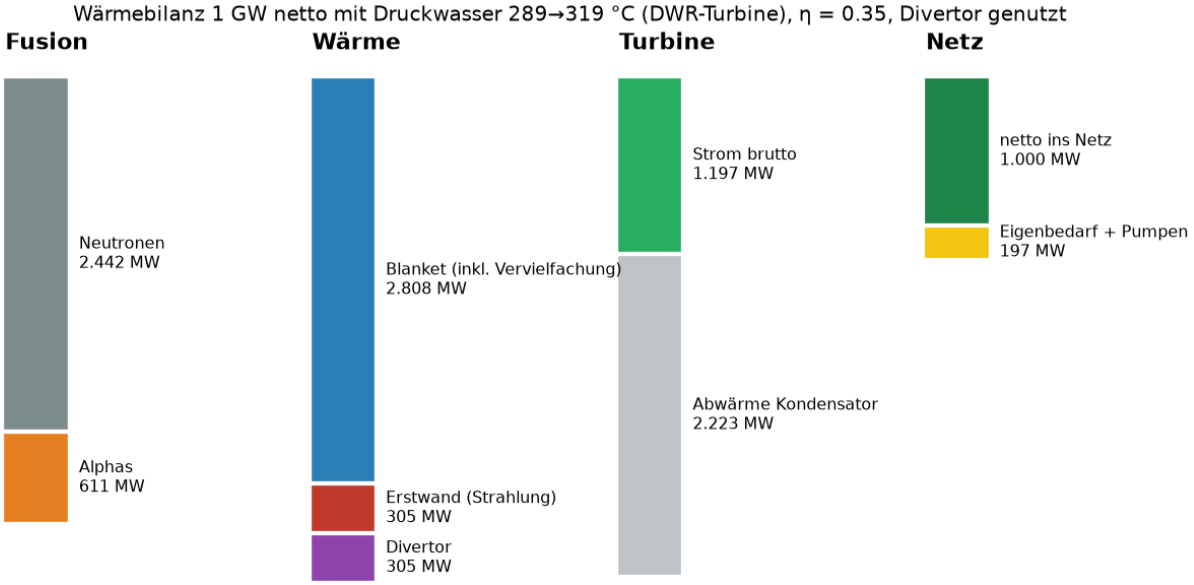

Figure 28: Heat balance for 1 GW net with pressurised water and a turbine from a pressurised water reactor (labels in German).

> *How to read the figure:* Read the columns from left to right; their height is the power. Fusion delivers neutrons and alpha particles (helium nuclei). These become heat in the blanket (increased by nuclear reactions in the blanket), in the first wall and in the divertor. The turbine turns this into electricity (gross), and the larger part goes to the condenser as waste heat. After subtracting pumps and auxiliary loads, 1000 MW reach the grid. Only about a third of the heat becomes electricity, as in every steam power plant.

6. Sensitivity. Efficiency has the largest effect on the required fusion power:

Case	P_{fus}	P_{th}	P_{gross}
$\eta = 0.35$, divertor heat used	3053 MW	3419 MW	1197 MW
$\eta = 0.376$ (as Isar 2)	2817 MW	3155 MW	1186 MW
$\eta = 0.317$ (VVER-1000, net)	3417 MW	3827 MW	1213 MW
$\eta = 0.35$, divertor heat lost	3353 MW	3419 MW	1197 MW

Table 8: Fusion power required for 1 GW net.

The divertor is cooled at low temperature (copper alloy). Its heat can preheat the feedwater. If it is lost, the required fusion power rises by 10 %.

Limits. f_{div} and M are assumptions. Part of the neutron heat is already deposited in the first wall and is counted here with the blanket. The blanket itself is open: it must remove 2.8 GW at pressurised-water conditions and breed tritium at the same time. The EU-DEMO WCLL concept (water as coolant, lead-lithium as breeder) aims at exactly this; sourced values are still missing here because the publishers' sites block automated access. Whether a turbine from a decommissioned plant is actually available was not checked; the reference plants only show the matching size class.

11.7 From plasma to grid: power flow and influences

The heat balance above is derived as a chain of formulas. Figure 29 shows the same numbers as a flow: the width of each band is proportional to the power. Below the nodes are the parameters that set each value. An interactive version, in which every node and every parameter can be clicked, exists as a web page; the figure is produced by `stellarator/einflussgraph.py`.

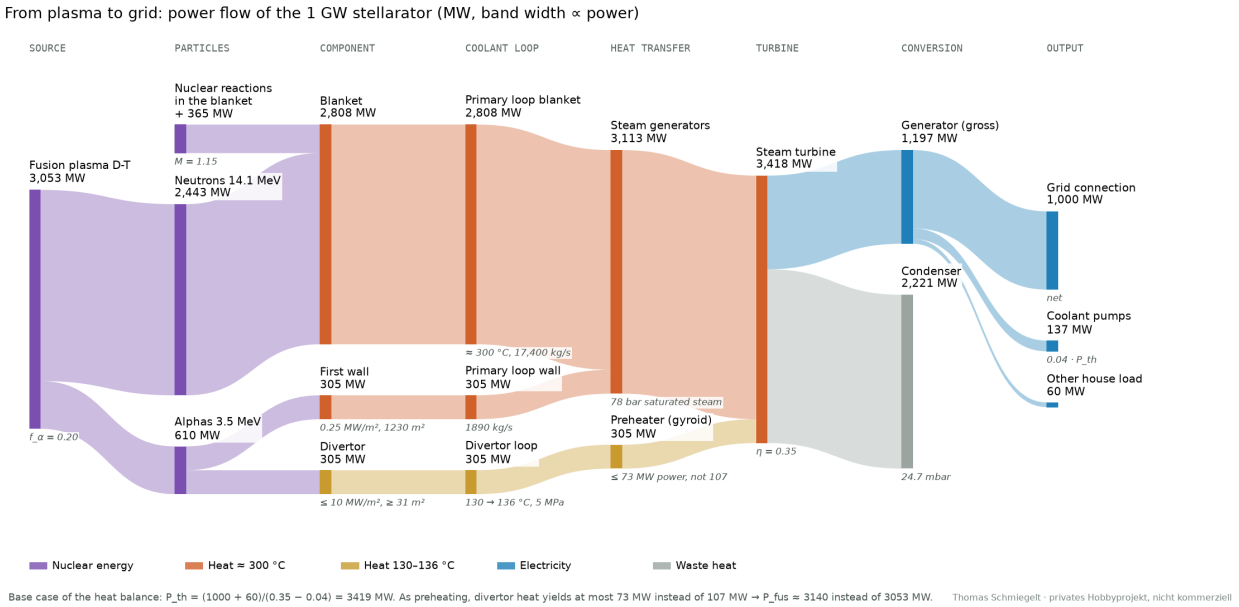

Figure 29: Power flow from the fusion plasma to the grid connection (base case of the heat balance, MW).

> *How to read the figure:* The bands flow from left (plasma) to right (grid), and each band's width is proportional to the power. The colours show the form of energy: violet nuclear, orange heat around 300 °C, ochre heat around 130 °C, blue electricity, grey waste heat. Follow one band, for example neutrons → blanket → cooling loop → steam generator → turbine → generator → grid. The small italic notes under the nodes are the quantities that set each value. The wide grey band to the condenser is the largest loss.

How to read it. Of 3053 MW fusion power, 1000 MW reach the grid, just under one third. The largest loss is not in the plasma and magnet part of the plant but in the turbine: the condenser rejects 2223 MW as waste heat. This is true of every steam plant and follows from the live-steam temperature. The chain has three places where the design directly affects the efficiency:

- the **coolant temperature** of blanket and wall (about 300 °C). It sets the live-steam pressure of 78 bar and thus $\eta = 0.35$;
- the **divertor temperature** (136 °C). It turns 305 MW of heat into preheating only, which yields at most 73 MW of electricity (Section 11.13);
- the **pressure losses** in channels and exchangers. Through the pumps (137 MW) they come directly off the net output.

The physics parameters (f_α, M) determine how much fusion power is needed for 1 GW net, not the efficiency downstream.

11.8 The components of the chain

This section explains the components from the plasma to the grid in plain words. General background is referenced; numbers marked "project" are assumptions or results of this design.

1. Fusion plasma and magnet coils. In the plasma, deuterium and tritium (D-T) fuse at about 150 million °C [45]. Gas this hot touches no wall. Magnetic fields hold it in suspension. A stellarator shapes this field with twisted external coils alone and drives no current through the plasma. It therefore runs steadily

by nature, but needs stronger magnets than a tokamak [42]. Large machines such as Wendelstein 7-X use superconducting coils cooled to about 4 K with liquid helium [43]. This needs a cryogenic plant. Project: the plasma delivers $P_{\text{fus}} = 3053\,\text{MW}$ (about 3140 MW with the preheater correction). Open: the forces between the coils are about 1000 times those of W7-X, and the support structure is not solved yet.

2. Neutrons and alpha particles. Each D-T fusion releases 17.6 MeV. The neutron carries about 80 %; the helium nucleus (alpha particle) carries the rest [44, 46]. The neutron has no charge. It leaves the magnetic field and deposits its energy in the wall and the blanket. The alpha particle stays in the plasma and heats it [47]. Project: $f_\alpha = 0.2$ (exactly 3.52/17.59). Assumption: the alpha power leaves the plasma half as radiation onto the first wall and half as a particle stream onto the divertor ($f_{\text{div}} = 0.5$). This split is open; it depends on the real plasma.

3. First wall. The first wall is the lining that faces the plasma. It protects the coils and catches heat and radiation [48]. At ITER it sits on detachable panels, and the armour changes from beryllium to tungsten [50]. Tungsten has the highest melting point of all metals, but it is brittle [48]. Project: the wall takes 305 MW of radiation. That is $0.25\,\text{MW/m}^2$ on $1230\,\text{m}^2$. The cooling channels are designed for $0.5\,\text{MW/m}^2$, a factor of 2 in reserve. The cooling water (1890 kg/s) warms by 30 K. Open: neutron damage to the wall material, which is not studied here.

4. Blanket. The blanket sits behind the first wall. It has three jobs: it slows the neutrons and turns their energy into heat; it breeds tritium when neutrons hit lithium; and it shields the coils [49, 50]. Tritium hardly exists in nature, and only the blanket can supply it [49]. Lead or beryllium multiply the neutrons (multiplication) [49]. Project: $M = 1.15$ (assumption). The blanket removes 2808 MW with 17 400 kg/s of pressurised water at about 300 °C and $\Delta T = 30\,\text{K}$. The EU-DEMO concept WCLL (water, lead-lithium) aims at these conditions [31]. Open: can the blanket remove 2.8 GW and breed enough tritium at the same time?

5. Divertor. The divertor guides particles and heat that escape the plasma along open field lines onto dedicated plates [51]. It removes ash and impurities and keeps the plasma clean [52]. The ash consists mainly of helium, which D-T fusion produces [44]; vacuum pumps remove it. At the ITER divertor the heat load is steadily about $10\,\text{MW/m}^2$, with slow peaks up to $20\,\text{MW/m}^2$ [52]. Project: 305 MW on at least $31\,\text{m}^2$ at $10\,\text{MW/m}^2$. It is cooled cold (water 130 °C in, 136 °C out, 5 MPa), as in EU-DEMO [33]. The project assumes a copper alloy for this. Open: the lifetime under this load.

6. Primary cooling circuits. The cooling circuits of blanket and wall carry the heat out. They use pressurised water, as in a pressurised-water reactor. The high pressure keeps the water liquid although it is far above 100 °C [53]. Project: about 300 °C, $\Delta T = 30\,\text{K}$, 17 400 kg/s through the blanket and 1890 kg/s through the wall. The pressure of about 16 MPa is an assumption. Choosing these conditions is also a project decision: it allows a turbine from a pressurised-water reactor. Open: the pipes must pass through the narrow coil gaps. The calculation shows ample room (more than a factor of 10 in reserve).

7. Steam generator. The steam generator is a heat exchanger. Hot primary water flows through many thin tubes; on the outside, secondary-loop water boils into steam [54]. The tube walls separate the loops, so primary water does not reach the turbine [54, 53]. The project assumes that the primary water may carry tritium or activation products. Project: live steam at 78 bar, saturated, 2443 kg/s. These are the EPR values [39]. The steam is saturated, as in a pressurised-water reactor; the project assumes $\eta = 0.35$ for it. Open: tritium may migrate through the tube wall; this is not studied.

8. Divertor preheater. The divertor heat arrives at only about 130 °C. That is too cold for steam, but fine for preheating feedwater. This saves extraction steam, which then does work in the turbine. Project: the gyroid exchanger, a printed wall with winding channels, is to transfer the 305 MW. Result of the calculation: the preheater yields at most 73 MW of electricity, and only without temperature approach and without pumps. For comparison, 107 MW would be 35 % of 305 MW. If the heat is lost entirely, the required fusion power rises by 10 %. Open: the correlations come from small-scale experiments.

9. Steam turbine. The turbine lets the steam expand in many stages and turns pressure into rotation [55]. With reheat, the steam returns to a heater after the high-pressure section and then flows on. This yields more work and fewer droplets on the last blades [55]. Project: heat $P_{\text{th}} = 3419\,\text{MW}$, efficiency $\eta = 0.35$, so

1197 MW gross. For comparison, Isar 2 had 3950 MW thermal and 1485 MW gross; a turbine of this class suffices. Assumption: such a turbine is available. Open: whether a decommissioned turbine can be had.

10. Condenser and cooling tower. The condenser cools the exhaust steam back into water. It works below atmospheric pressure so that the turbine extracts more energy [56]. The heat released must go somewhere: a wet cooling tower dumps it by evaporating part of the water [57]. Project: 2223 MW of waste heat at 24.7 mbar (20.9 °C). That is almost twice the electrical output, as in every steam power plant. Open: site and water demand of the cooling tower; the project has not determined them.

11. Generator. The generator turns the rotation of the turbine shaft into electricity. It uses electromagnetic induction: a moving magnetic field induces a voltage in windings [58]. Power plants use synchronous generators, which must run in step with the grid [58, 55]. Project: it produces 1197 MW gross of alternating current. Of this, 197 MW go to the plant's own demand. The balance uses only the overall efficiency η; there is no separate generator efficiency in it. Open: selection and size.

12. Plant's own demand. The plant uses part of its electricity itself. Project: the coolant pumps need 137 MW ($0.04\,P_{\mathrm{th}}$, assumption). A further 60 MW covers the rest: the coil cryogenic plant, the tritium plant and buildings. The cryogenic plant keeps the coils at a few kelvin [43]. The tritium plant is to recover tritium from the blanket and exhaust gas and feed it back to the plasma (project description, not referenced). The value for the rest is an assumption; for the first-wall channels alone the project finds only about 8 MW of pumping power. Open: the total of 197 MW is not built up from single items.

13. Transformer and grid connection. A transformer raises the generator voltage. At high voltage the same power needs less current, and the lines lose less heat [59]. The high-voltage grid then takes the power. Project: 1000 MW net go to the grid. This is the target from which the whole chain was computed backwards. Transformer losses are not included in the balance. Open: grid connection point and design.

11.9 Do the pipes fit through the gaps between the coils?

All the cooling water has to go from the vessel through the coils to the steam generators and back. On the inboard side the winding packs of the HTS design leave only 0.58 m (stage 3). This question could rule out the pressurised-water solution, so it was checked before anything else (`stellarator/rohr_durchfuehrung.py`).

Demand. From the heat balance: $\dot V = (\dot m_{\mathrm{blanket}} + \dot m_{\mathrm{first\ wall}})/\rho = 19\,290\,\mathrm{kg/s} \,/\, 732\,\mathrm{kg/m^3} = 26.4\,\mathrm{m^3/s}$ in each direction, so twice that pipe area for supply and return. With velocity v and inner diameter d_i one needs

$$n = \frac{2\,\dot V}{v\,\pi d_i^2/4} \quad \text{pipes, e.g. } n = \frac{2 \cdot 26.4}{10 \cdot \pi \cdot 0.8^2/4} = 11 \text{ at } d_i = 0.8\,\mathrm{m},\ v = 10\,\mathrm{m/s}.$$

The pipe wall follows from the boiler formula $s = p\,d_i/(2\sigma)$: at 16 MPa and 150 MPa, $s = 0.053\,d_i$, so the outer diameter is $D = d_i(1 + p/\sigma) = 1.11\,d_i$.

Space. From each point of the vessel's outer surface (plasma boundary plus 1.05 m) a straight pipe runs 4 m outward along the surface normal. The smallest distance $a_{\min}$ of this path to winding pack, potting, casing and connectors of the surrogate model ($\mathrm{R_0} = 10.4$ m) limits the pipe diameter:

$$D_{\max} = 2\,(a_{\min} - a_{\mathrm{shield}}), \qquad a_{\mathrm{shield}} = 0.15\,\mathrm{m}.$$

a_{shield} is room for a gap and a thermal shield: the pipe is at 300 °C, the superconducting coils are cryogenic. Pipes must be at least $D + 0.2$ m apart; counting was greedy, freest spots first.

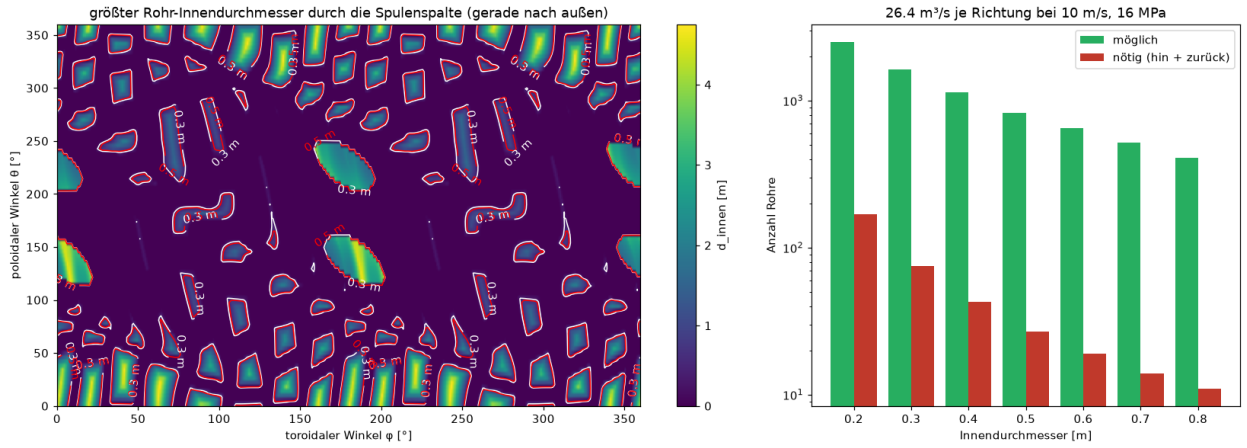

Figure 30: Left: largest inner diameter of a straight pipe over the unrolled vessel surface ($\theta = 0$ outboard, 180° inboard); the windows between coils and connectors are clearly visible. Right: possible and required pipes per size (labels in German).

How to read the figure: Left: again the unrolled vessel surface. The colour is the largest pipe diameter that fits straight outwards between the coils from that spot. Dark means a coil is in the way. The bright islands are the windows; the red line outlines spots of at least 0.3 m. Right: green bars show how many pipes of a given size fit, red bars how many are needed for the coolant out and back (logarithmic). Green is far above red everywhere, so there is plenty of room.

d_i	required ($v = 10$ m/s)	required ($v = 5$ m/s, shield 0.3 m)	possible (default / cautious)
0.3 m	75	150	1634 / 1386
0.5 m	27	54	824 / 732
0.8 m	11	22	410 / 340

Table 9: Pipes through the gaps between the coils.

Result. **The pipes fit with plenty of room**, by more than a factor of 10 even in the cautious case. On 19 % of the vessel surface a straight path for 0.8 m pipes is free. The windows lie mainly outboard (θ near 0) and at top and bottom. Inboard (θ around 180°) almost everything is covered by coils: the inboard blanket has to be fed through headers from above and below. The initial concern about 0.58 m applied only to the narrowest inboard gap.

Limits. The support structure between the coils is still open (forces about 1000 times those of W7-X). It will occupy part of the windows, as will access for heating, diagnostics, vacuum pumps and maintenance. Only straight pipes along the normal were considered; with bends more room would be found.

11.10 Blanket: comparison with the EU-DEMO WCLL concept

The blanket takes 82 % of the heat and has to breed tritium at the same time. The reference is the water-cooled lead-lithium blanket (WCLL) of the European DEMO [31]. The user downloaded the open-access article; all values below are quoted verbatim from it (`docs/recherche/wcll/wcll_werte.md`).

	EU-DEMO WCLL [31]	this plant
Cooling water	295 → 328 °C, 15.5 MPa	289 → 319 °C, 16 MPa
Circuits	first wall and breeding zone separate	first wall and blanket separate
EUROFER steel	not above 550 °C	first wall at most 441 °C
First-wall heat flux	0.32 MW/m^2	0.25 MW/m^2 (mean), channels for 0.5
Fusion power / net	1998 MW / 500 MW	3053 MW / 1000 MW
Neutron wall load	1.04 MW/m^2	1.99 MW/m^2
Breeder	PbLi, Li enriched to 90 % ^{6}Li	open
Tritium breeding ratio	1.14 (target 1.15)	open

Table 10: WCLL and the pressurised-water design of this plant.

Consequences.

1. **The choice of pressurised water is confirmed.** DEMO uses practically the same conditions for WCLL as a pressurised water reactor, with two separate circuits for first wall and breeding zone. The heat balance here is split the same way.
2. **Lead returns, but as breeder, not as coolant.** In WCLL the PbLi flows only slowly through the blanket and carries the bred tritium out. The heat is taken up by water in double-walled tubes. The objections to lead as a coolant (fast flow across the field) hardly apply.
3. **The neutron wall load is twice that of DEMO.** In stage 0 the HTS design was shrunk up to the limit of $2\,\mathrm{MW/m^2}$. That means roughly twice the heat density in the blanket and twice the radiation damage to the steel, hence a shorter life of the blanket modules, and more shielding in front of the coils. The Nb_3Sn design would be at $0.59\,\mathrm{MW/m^2}$, but is almost twice as large.
4. **The tritium breeding ratio is tight.** Even DEMO misses the target of 1.15 slightly. A stellarator has more openings and a more complex shape, losing blanket area. This is a risk of its own and would need a neutronics calculation.

11.11 Graded structure: flat contact at the skin, branching outward

The user suggested solving the line-contact problem differently: "On the blanket side one would need a flat support, then it becomes a balancing act. On the one hand the metal strands must not be so dense that they obstruct the water too much, on the other hand they must be thick enough for heat conduction. Further out the material can branch heavily and become thin." Three design rules follow (pressurised water $p = 16\,\mathrm{MPa}$, skin $t = 5\,\mathrm{mm}$, allowable $\sigma = 150\,\mathrm{MPa}$):

1. Support spacing at the skin. With the factor 1.22 from the Elmer calculation over the gyroid (39.2 instead of 32.0 MPa per MPa):

$$\sigma = 1.22\,\frac{p\,b^2}{2t^2} \le \sigma_{\mathrm{allow}} \quad \Rightarrow \quad b \le t\sqrt{\frac{2\sigma_{\mathrm{allow}}}{1.22\,p}} = 5\,\mathrm{mm}\cdot\sqrt{\frac{300}{19.5}} \approx 20\,\mathrm{mm}.$$

At 4 MPa it would be 39 mm. With supports every 20 mm the thin skin holds pressurised water as in a nuclear plant.

2. How deep the strands conduct heat. A strand of thickness d in water acts as a cooling fin with $m = \sqrt{4h/(k\,d)}$; roughly the length $2/m$ is effective (steel $k = 25\,\mathrm{W/(m\,K)}$):

d	$h = 1200\,\mathrm{W/(m^2K)}$ (slow)	$h = 10\,000\,\mathrm{W/(m^2K)}$ (fast)
3 mm	8 mm	3 mm
10 mm	14 mm	5 mm

Table 11: Effective fin depth $2/m$.

The heat is in the water within 3 mm to 15 mm. Only this first layer needs thick strands; behind it the strands only carry load.

3. How thin it may get further out. The pressure pushes inner and outer skin apart; the strands hold them together in tension. Their area fraction φ must everywhere satisfy

$$\varphi \ge \frac{p}{\sigma_{\mathrm{allow}}} = \frac{16}{150} \approx 11\,\%.$$

Branching is therefore allowed as long as the sum of the cross-sections remains. For a gyroid of cell a and wall thickness t_g, $\varphi \approx 3.09\,t_g/a$.

Implementation in PicoGK. `picogk/GradientWand`: 5 mm skin, then four layers whose cell doubles from layer to layer. Between two layers the distance fields of the gyroids with cell a and $2a$ are blended smoothly, so the walls branch. The joints are filleted with 1.5 mm.

Layer	Cell	Wall	to depth	Steel fraction
0	20 mm	2.5 mm	30 mm	39 %
1	40 mm	3 mm	80 mm	23 %
2	80 mm	4.5 mm	180 mm	17 %
3	160 mm	8 mm	570 mm	15 %

Table 12: Layers of wall piece g1 ($240 \times 240 \times 600$ mm, voxel 1 mm); 84 % water overall.

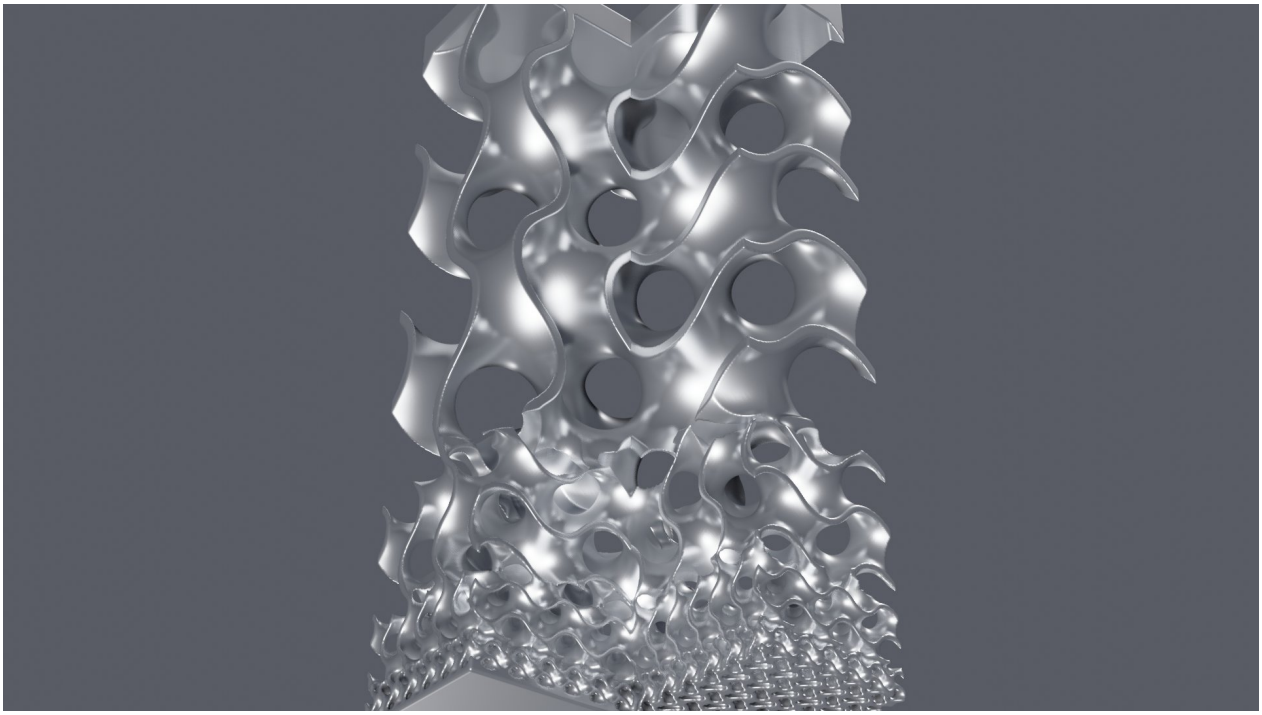

Figure 31: Graded wall g1 seen from the plasma side, one quarter cut away: small cells at the skin at the bottom, the cell doubles with each layer further out.

How to read the figure: A 3D image of the graded wall with a quarter cut away. At the bottom is the smooth skin facing the plasma, above it a lattice. With each layer outwards the cell size doubles: fine at the skin, where heat and pressure arrive, coarse further out. A bone is built in a similar way.

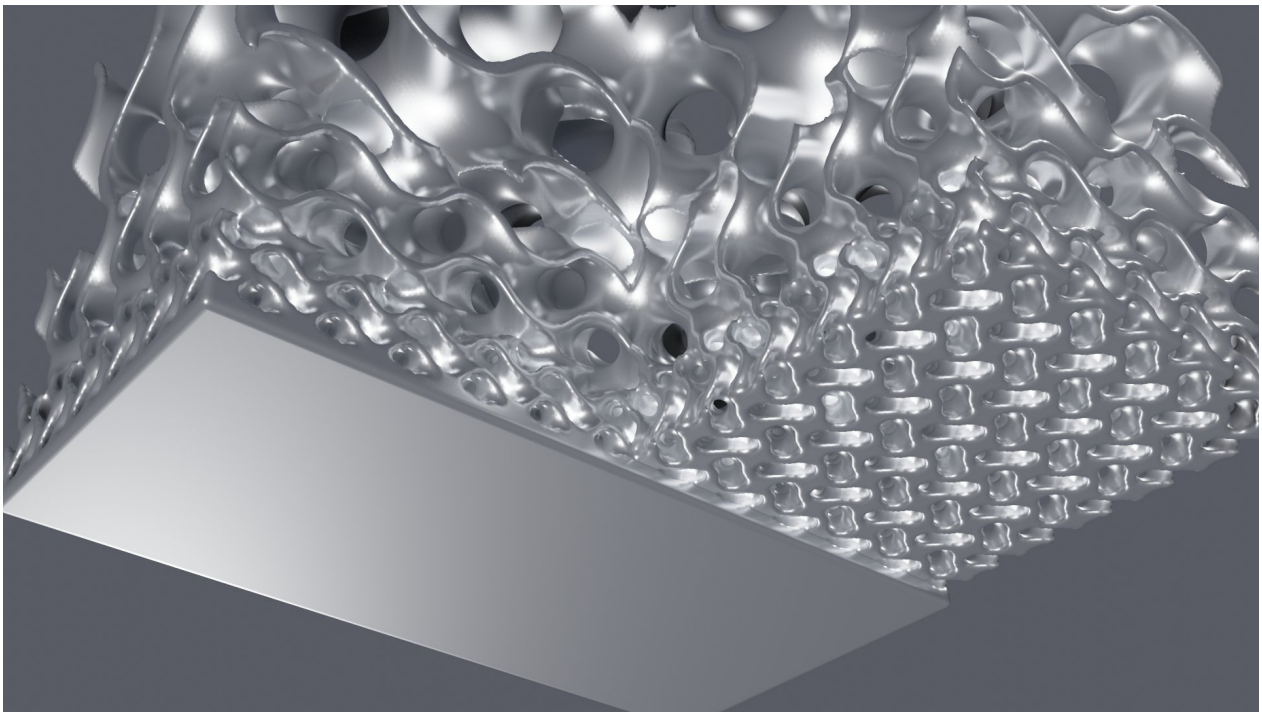

Figure 32: Close-up of the skin with a window: the fine layer supports the skin almost over its full area, support spacing about 10 mm.

> *How to read the figure:* A close-up of the skin with a window cut into it. Beneath it is the finest lattice layer, supporting the skin about every 10 mm. The skin rests on an almost continuous support and can stay thin.

Still open are the flow test (FluidX3D: does the water pass through the dense skin layer or bypass it further out?) and the strength test (Elmer: skin and branches at 16 MPa). The wall piece also writes out the water volume for this.

11.12 How does the heat spread sideways and outward?

The user's question: how do we spread the heat sideways and so lead the heat flow outward?

Estimate. A plate of thickness t and conductivity k, cooled on one side with heat transfer coefficient h, spreads a hot spot sideways only over the spreading length

$$\lambda = \sqrt{k\,t/h}. \tag{1}$$

For 5 mm steel ($k \approx 25\,\mathrm{W/(m\,K)}$) and $h = 10^4\,\mathrm{W/(m^2\,K)}$ that is 3.5 mm. A gyroid wall of thickness t acts as a fin cooled on both sides. Its effective depth is

$$1/m = \sqrt{k\,t/(2h)}, \tag{2}$$

which is 1.8 mm for 2.5 mm steel. Both values are small compared with the 20 mm cell. Expectation: in steel the heat stays where it arrives and enters the water directly behind the skin.

Calculation. Elmer HeatSolve (steady state) on a cut-out of wall g1:

- Cut-out 80 × 80 × 100 mm, one full period of the first three layers. Same layers and blending as `GradientWand`; fillets from morphological closing.
- Voxel hexahedra of 0.5 mm, 1.4 to 1.5 million elements.
- Plasma side: $q = 0.5\,\mathrm{MW/m^2}$.
- Wetted surfaces: $h = 10^4\,\mathrm{W/(m^2\,K)}$ to water at 304 °C. The voxel staircase area exceeds the smooth area by the factor $|n_x| + |n_y| + |n_z|$, so h is divided by this factor on each face.

- Sides and back: adiabatic.
- Energy balance: input and output agree to 0.1 %.

Five cases were computed:

- uniform load;
- hot spot (radius 10 mm, $2q$);
- the same spot with 3 mm tungsten in front of the skin ($k = 130$);
- skin and structure of copper alloy ($k = 320$, like CuCrZr in the divertor);
- doubled h (faster water).

The k values are rough assumptions, not sourced.

Case	plasma side max	wet wall max	removed within 10 mm	removed under spot
steel, uniform	469 °C	364 °C	95 %	–
steel, hot spot	588 °C	399 °C	95 %	6.9 %
tungsten + steel, spot	535 °C	377 °C	95 %	5.6 %
copper alloy, spot	351 °C	340 °C	62 %	5.8 %
steel, spot, h doubled	559 °C	363 °C	98 %	7.0 %

Table 13: Heat test on wall g1. Saturation temperature at 16 MPa: 347.4 °C. "Removed within 10 mm": share of heat entering the water within 10 mm behind the plasma side of the steel skin. "Removed under spot": share leaving within the spot radius. The spot receives 9.4 % of the heat; any shortfall was spread sideways.

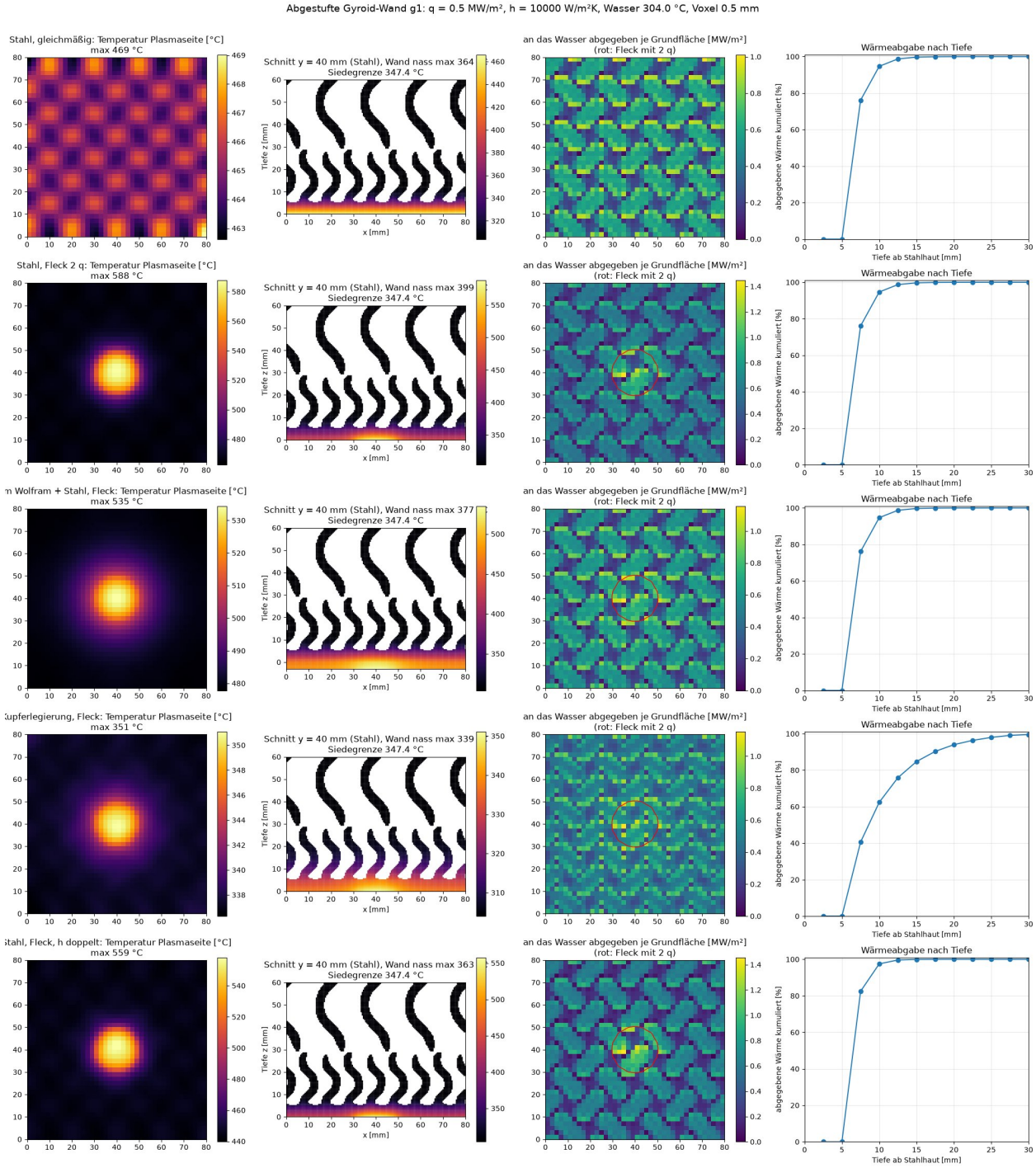

Figure 33: Heat test of wall g1. Columns from left: plasma-side temperature; section through the middle; heat given to the water per footprint area (red: spot); cumulative heat removal over depth.

> *How to read the figure:* Five rows for five cases (from the top): steel with even heating, steel with a hot spot, steel with a tungsten layer, a copper alloy, and steel with twice as good water cooling. The first column shows the temperature of the plasma side seen from above (the hot spot glows); the second a cut through the wall (black is water). The third column shows where heat passes into the water, and the fourth what share of the heat has been released by which depth. For steel the curve rises almost to 100 % within the first 5 mm, so only the front, finest layer does any work. Only with copper (fourth row) does it rise more slowly, letting heat reach deeper layers.

Result.

- **In steel the heat does not travel into the depth.** 95 % leaves the steel within the first 5 mm behind the skin, through the back of the skin and the roots of the struts. The struts themselves stay

almost at water temperature. The wetted area is about 15 times the footprint but is hardly used. In steel the graded structure is a load-bearing structure, not a heat exchanger. This confirms the estimate.

- **The saturation temperature is exceeded even under uniform load.** 3 % of the wet wall is above 347.4 °C, where the struts join the skin and less area cools locally. With the hot spot the wall reaches 399 °C.
- **Tungsten spreads the hot spot sideways**: 53 K less at the hottest point. Because the extra 3 mm of tungsten must also conduct the heat, the plasma side is 12 K warmer on average.
- **Faster water** lowers the wall temperature most for the same material, but barely helps the spot.
- **Copper alloy is the only case that carries heat into the depth.** Only 62 % leaves within the first 5 mm; the rest leaves further back through the struts. The plasma side stays below 351 °C and the wet wall below saturation. The fins now work because $1/m = \sqrt{kt/(2h)}$ grows to 6 mm with $k = 320$.

Conclusions.

- Only a well-conducting material spreads heat sideways and into the depth. Options are a thin tungsten or copper layer at the skin, or copper-alloy struts in the first 10 mm to 20 mm. Copper is problematic under neutrons (embrittlement, activation) but common in divertors. The structure further back can stay steel; it then only carries load.
- With steel alone the water must flow fast directly at the skin. The skin layer therefore needs its own supply, as the rules above already require.
- In the blanket the heat is generated throughout the volume by neutrons. There, branching well-conducting paths to the water make sense, as constructal theory describes for cooling a heat-generating volume [32]. That is the next step, together with flow (FluidX3D) and strength.

Limits of the calculation:

- h is uniform because no flow calculation is coupled.
- There is no volumetric neutron heating.
- k is constant.
- The sides are adiabatic instead of periodic.

Script: `stellarator/gradient_waerme.py`.

11.13 Divertor preheater: which design yields the most electricity?

The divertor heat arrives at low temperature. In EU-DEMO the target coolant enters at 130 °C and 5 MPa and heats up by 6 K. The authors plan to use this heat for preheating [33]. In this design it amounts to 305 MW (see heat balance).

How the preheater works (without formulas). The divertor is the *exhaust* of the plasma: about 10 % of the power lands there on a small area, up to 10 MW/m^2. For copper and tungsten to survive this, it is cooled with fairly cold water, 130 °C in and 136 °C out. That is too cold for the steam generator; the turbine needs steam of almost 300 °C, which the blanket supplies. Without further measures this heat would have to be discarded.

But the steam cycle has a place with even colder water: the *condensate*, which forms from the steam in the condenser behind the turbine. It is only about 21 °C and has to be heated again anyway on its way back to the steam generator. In every steam plant today this is done with *bleed steam*: steam is tapped from the turbine and does no more work afterwards. If the divertor heat takes over this preheating, the steam stays in the turbine and keeps driving the generator down to the condenser. This extra work is the gain. An everyday comparison: if you preheat bath water with waste heat, the water heater burns less gas; the waste heat does not turn into gas, it only saves some.

The preheater is a *heat exchanger*: two separate channel systems, one for the warm divertor water and one for the cold condensate, with a steel wall between them. The waters never mix; only heat passes through the wall. Because the divertor heat only replaces the coldest, almost fully expanded bleed steam, it is worth less than its amount suggests: at most 73 MW of electricity, not $0.35 \times 305\,\text{MW} = 107\,\text{MW}$. The calculation below derives this number and compares the designs.

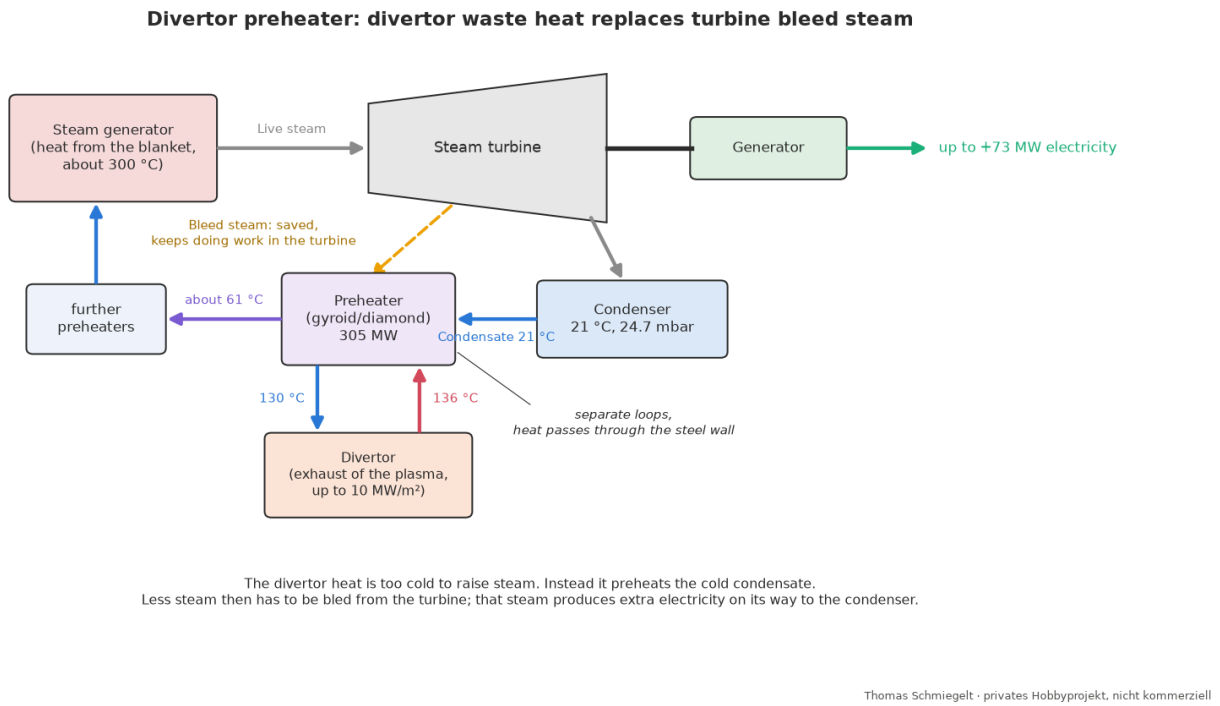

Figure 34: Principle of the divertor preheater in the steam cycle.

How to read the figure: The arrows show where water and steam flow. At the top the main path: the steam generator supplies steam, the turbine drives the generator, and the steam turns back into water in the condenser. Below, the water's way back: the cold condensate (blue) runs through the preheater and leaves about 40 K warmer (violet). Underneath, the divertor's own loop: its water arrives at 136 °C (red), gives off heat and returns at 130 °C. The dashed yellow arrow is the bleed steam that would otherwise have done this preheating. It is saved and instead produces electricity in the turbine, up to 73 MW.

Question and hypothesis. Which preheater design extracts the most net electricity from these 305 MW? The candidates are a tube bundle, a plate exchanger, gyroid (uniform and graded) and diamond.

$$P_{\text{net}} = \Delta P_{\text{turbine}} - P_{\text{pump, divertor side}} - P_{\text{pump, condensate side}}.$$

Hypothesis H1: at equal pumping effort, a TPMS structure reaches a smaller approach temperature and therefore more net electricity.

1. Plant side. The divertor water preheats condensate, which saves bleed steam. The saved steam still does work on its way to the condenser. For heating by $\mathrm{d}T$:

$$\mathrm{d}W = \dot{m}_k \, c_p \, \mathrm{d}T \, \frac{h_A - h_{\text{exh}}}{h_A - h'_A}, \qquad p_A = p_s(T + \Delta T_{\text{TTD}}).$$

Here h_A is the bleed steam enthalpy, h'_A the enthalpy of saturated condensate at bleed pressure and h_{exh} the enthalpy at the turbine exhaust. ΔT_{TTD} is the terminal temperature difference of the heater. All states come from CoolProp [37].

The expansion line starts with saturated steam at 78 bar, as in the EPR [38]; English Wikipedia gives 77.2 bar at the turbine inlet [39]. It ends at a condenser at 24.7 mbar (20.9 °C). The condensate flow is the EPR steam flow (2443 kg/s) scaled to our 3419 MW thermal, i.e. 1820 kg/s. The condensate then heats up by $Q/(\dot{m}_k c_p) = 39.8$ K. It can reach at most 136 °C.

Without approach temperature and without pumping this gives an **upper limit of** 73.3 MW, i.e. 24 % of the divertor heat.

2. Heat exchanger side. Counterflow with $UA = Q/\Delta T_{\text{lm}}$. The approach at the hot end is $\Delta T_1 = 136\,^\circ\text{C} - T_{k,\text{out}}$. At the cold end $\Delta T_2 = 130\,^\circ\text{C} - T_{k,\text{in}} = \Delta T_1 + 33.8$ K. The divertor water acts almost like a source at fixed temperature. ΔT_{lm} therefore stays large even for a small ΔT_1, e.g. 7 K at $\Delta T_1 = 0.3$ K.

Both sides share the wall area A. With $d_h = 4V_f/A$ the fluid volume per side is $V_f = A\,d_h/4$. For equal path length this gives

$$\frac{Re_{\text{hot}}}{Re_{\text{cold}}} = \frac{\dot{m}_{\text{hot}}\,\mu_{\text{cold}}}{\dot{m}_{\text{cold}}\,\mu_{\text{hot}}} \approx 7,$$

independent of geometry. The reason is the unequal mass flows: 12 000 kg/s of divertor water against 1820 kg/s of condensate. A symmetric TPMS structure therefore cannot keep both sides inside the valid range of the correlation ($2961 < Re < 18254$). Several passes on the condensate side (n) solve this, because the condensate path becomes n times longer:

$$A_{f,\text{hot}} = n\,A_{f,\text{cold}}\,\frac{d_{h,\text{hot}}}{d_{h,\text{cold}}}, \qquad \Delta p = 4f\,\frac{L}{d_h}\,\frac{\rho v^2}{2}, \qquad P_{\text{pump}} = \frac{\Delta p\,\dot{V}}{\eta_P}.$$

Core volume: $V = A\,(d_{h,\text{hot}} + d_{h,\text{cold}})/4 + A\,t$, without headers and manifolds.

For each design the scan covers 40 approaches (0.3 K to 25 K), 12 hydraulic diameters, 13 ratios of the two diameters, 24 velocities and $n = 1$ to 8. For each volume the best valid design counts (Pareto front).

Design	Heat transfer and friction (Fanning)	Band
Diamond	$Nu = 0.2644\,Re^{0.69}\,Pr^{1/3}$, $f = 1.850\,Re^{-0.17}$, CFD [34]	Li 2020, Li 2022 [35]
Gyroid uniform	as diamond, $f \times 1.145$ (12.3–16.7 % more friction [34])	Li 2020, Li 2022
Gyroid graded	$j/f^{1/3}$ 28 % better (measured [36]), here $Nu\times$ 1.28	as above
Plate (stopgap)	as uniform gyroid; measured gyroid is “slightly lower” [36]	up to +10 %
Tube bundle	Gnielinski and Petukhov, shell side as tube side (weakest assumption)	–

Table 14: Correlations; verbatim quotes in `docs/recherche/vorwaermer/`.

Hydraulic diameters: tube 10 mm to 25 mm, plate 3 mm to 10 mm, TPMS 2 mm to 15 mm. Assumptions: wall 0.5 mm stainless steel (printed as in [36]), terminal temperature difference 3 K, isentropic turbine efficiency 0.85 (HP) and 0.88 (LP), reheat at 11 bar to 270 °C, pumps 0.8, condensate pressure 2 MPa.

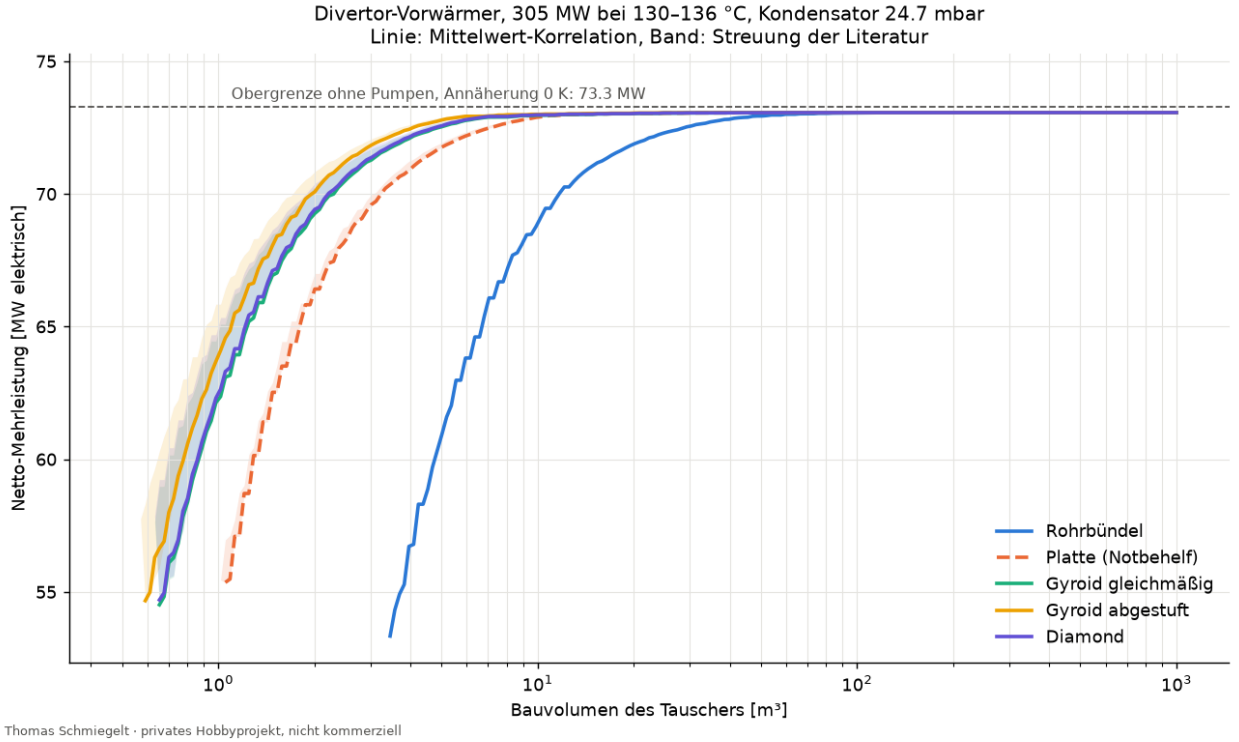

Figure 35: Net extra power versus core volume (labels in German). The line is the central correlation, the band the literature spread. Dashed: plate (stopgap).

How to read the figure: Horizontally: the volume of a heat exchanger in cubic metres (logarithmic, 1 to 1000). Vertically: the extra electrical power gained by using divertor heat to preheat the turbine feedwater, with pumping power already subtracted. Each line is one exchanger type. All approach the upper limit of 73.3 MW. The further left a curve rises, the smaller the exchanger can be. Gyroid and diamond reach nearly the maximum with a few cubic metres; the tube bundle needs about ten times as much. The band shows the scatter in the literature.

Design	Volume for 95 %	Volume for 99 %	Net up to 50 m^3
Tube bundle	11 m^3	30 m^3	72.9 MW
Plate (stopgap)	2.8 m^3 to 3.0 m^3	6.7 m^3 to 7.3 m^3	73.1 MW
Gyroid uniform	1.8 m^3 to 2.1 m^3	4.5 m^3 to 5.0 m^3	73.1 MW
Gyroid graded	1.6 m^3 to 1.8 m^3	3.8 m^3 to 4.2 m^3	73.1 MW
Diamond	1.7 m^3 to 2.1 m^3	4.4 m^3 to 4.9 m^3	73.1 MW

Table 15: Core volume for 95 and 99 % of the upper limit. The range comes from the correlations.

3. Findings.

- **Efficiency does not decide the design.** From about 30 m^3 on, every design reaches 72.9 MW to 73.1 MW. Pumping power is then 0.01 MW to 0.14 MW. H1 holds only at small volume. Given enough size, all designs exhaust the upper limit.
- **The designs differ in volume.** For 99 % a TPMS exchanger needs 4 m^3 to 5 m^3, a plate about 7 m^3 and a tube bundle about 30 m^3. The graded gyroid is smallest, diamond close behind. The choice must therefore be made on volume, cost, manufacturing, fouling and strength, not on efficiency.
- **Correction of the heat balance.** The balance above counted the divertor heat at the full efficiency of 0.35, i.e. 107 MW of electricity. As preheating it yields at most 73 MW. For 1 GW net the required fusion power thus rises by about 3 %, from 3053 MW to roughly 3140 MW.
- **Mass flows.** Divertor water and condensate flow at a ratio of 7:1. A symmetric exchanger therefore needs several passes on the condensate side. The best designs use $n = 8$ and sit at the lower limit $Re \approx 3000$. Slower flow would be even better, but verified correlations for it are missing.
- **Bigger levers than the design.** A condenser at 50 mbar (river cooling) lowers the upper limit to 63.5 MW. A condensate flow of 75 % lowers it to 68.7 MW. In EU-DEMO the cassette body also removes heat at 180 °C to 210 °C (4.2 MW per cassette) [33]. Heat at higher temperature is worth more. That is the next calculation step.

Limits of the calculation:

- The correlations scatter widely, and the plate curve is not a model of its own.
- The expansion line is simplified: one efficiency per section, bleeds without feedback on the steam flow.
- Headers, manifolds and flow distribution are missing. The best TPMS designs are only about 1.3 m long and need a large frontal area for that.
- Whether a 0.5 mm printed wall holds 5 MPa against 2 MPa has not been checked. That is a question for strength experts.

Script: `stellarator/vorwaermer.py`, results in `stellarator/daten/vorwaermer.json`.

11.14 Exchanger modules built and simulated with FluidX3D

The choice of preheater design so far rests on literature correlations whose range of validity (Pr = 3–5) does not cover divertor water with Pr ≈ 1.3–1.8. The modules were therefore built and the flow computed directly. The available computer cannot handle OpenFOAM; the lattice Boltzmann solver FluidX3D runs on the graphics card [17, 24].

1. Modules. PicoGK builds one graded two-fluid block each ($48 \times 48 \times 96$ mm, wall 0.5 mm, shell 1.5 mm): gyroid with cell size 4.33 mm $\to$ 5.8 mm and diamond with 5.37 mm $\to$ 7.2 mm. Both give the same hydraulic diameter $d_h = 2 \to 2.9$ mm. The cell size grows logarithmically along the block, with phase $p_z = 2\pi L/(a_1 - a_0)\ \ln(a/a_0)$. End caps close the cold side, side windows the warm side. At 0.2 mm voxel size the block is 39.7 % steel and 30 % each warm and cold side. The plates (flat and chevron, $\beta = \pm 60°$) are built parametrically in FreeCAD.

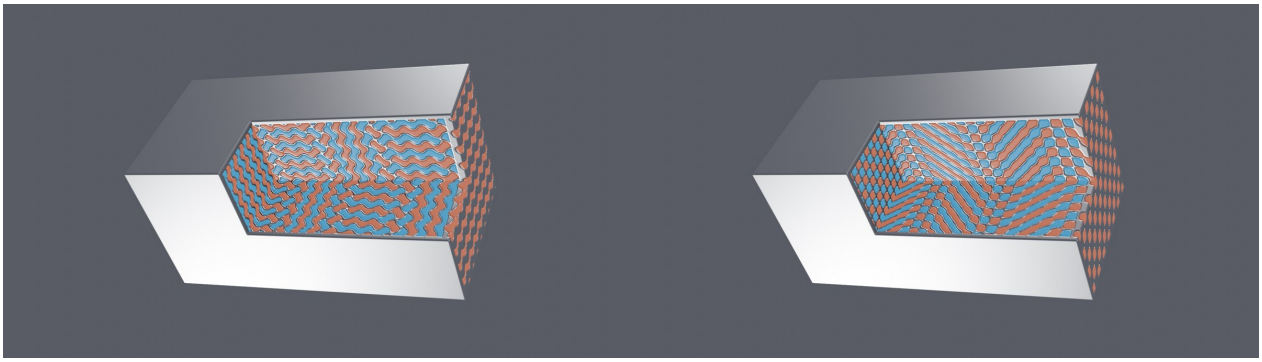

Figure 36: Graded two-fluid modules, cut open. Left gyroid, right diamond. Red is the warm (divertor) side, blue the cold (condensate) side, grey the steel wall.

> *How to read the figure:* Two exchanger blocks cut open, gyroid lattice on the left, diamond lattice on the right (two different lattice shapes). Red is the space for the warm water from the divertor, blue for the cold condensate, grey the steel wall between them. These are two separate, intertwined channel systems; the two waters never touch, only the heat passes through the wall.

2. Domain. A domain with inlet and outlet fails at Re ≈ 3000: the required pressure drop needs more than 25 % density difference in the lattice Boltzmann method, far outside its validity. The domain is therefore a section of one side, periodic in all directions. A body force f_z drives the flow, a uniform heat source q heats the fluid, and the wall holds $T_w = 0$. Then

$$f = \frac{f_z\, d_h}{2\, u^2}, \qquad h = \frac{q\, V_f}{A\, T_m}, \qquad \mathrm{Nu} = \frac{h\, d_h}{D}, \qquad T_m = \frac{\sum u_z T}{\sum u_z}.$$

f_z is iterated until Re matches within 15 %.

3. Changes to FluidX3D. Five points had to be changed (sources with licence note in `fluidx3d/`):

- The temperature model (D3Q7) has $c_s^2 = 1/4$. The actual diffusivity is $D = \alpha/2$; a pure diffusion test confirmed this (measured $D/\alpha = 0.44$–0.53). The code therefore passes $\alpha = 2D$.
- The wall sat one cell too far into the fluid when the wall temperature was set in fluid cells. This gave a Nu about three times too high. Solid cells now use anti-bounce-back: $g_{\mathrm{in}} = -g_{\mathrm{out}} + \frac{1}{4}(T_w - 1)$.
- At Re = 3000 the cell Péclet number is about 185. A turbulent diffusivity $D_t = \nu_t/0.85$ from the solver's Smagorinsky model is therefore added.
- The simple BGK collision for temperature became abruptly unstable at 96 cells per d_h after 172 000 steps (temperature to 10^{20} within 500 steps). The relaxation time $\tau \approx 0.504$ is close to the stability limit 0.5. It was replaced by a TRT collision after Ginzburg [41]: the antisymmetric part relaxes at the rate that sets D, the symmetric part with the "magic" combination $\Lambda = (1/\omega^+ - \frac{1}{2})(1/\omega^- - \frac{1}{2}) = 1/4$.
- The heat source is either uniform or proportional to u_z. The second form is the usual formulation for fully developed periodic flow with axially uniform heat flux (Patankar, Liu and Sparrow 1977 [40]). A uniform source raises Nu even in the laminar gap from 8.235 to 10.

4. Verification. For the laminar plate channel the values are known exactly: $f\,\mathrm{Re} = 24$, and Nu = 10 (uniform source) or 8.235 (source $\propto u$). The simulation gives $f\,\mathrm{Re} = 29.7 \to 25.7 \to 24.7$ at 12, 24, 48 cells per channel height. For Nu it gives $10.6 \to 10.3$ (BGK, uniform) and $8.9 \to 8.5$ (TRT, $\propto u$). The error falls with order $p \approx 1.3$. The choice of Λ (1/4 or 3/16) did not change the test case.

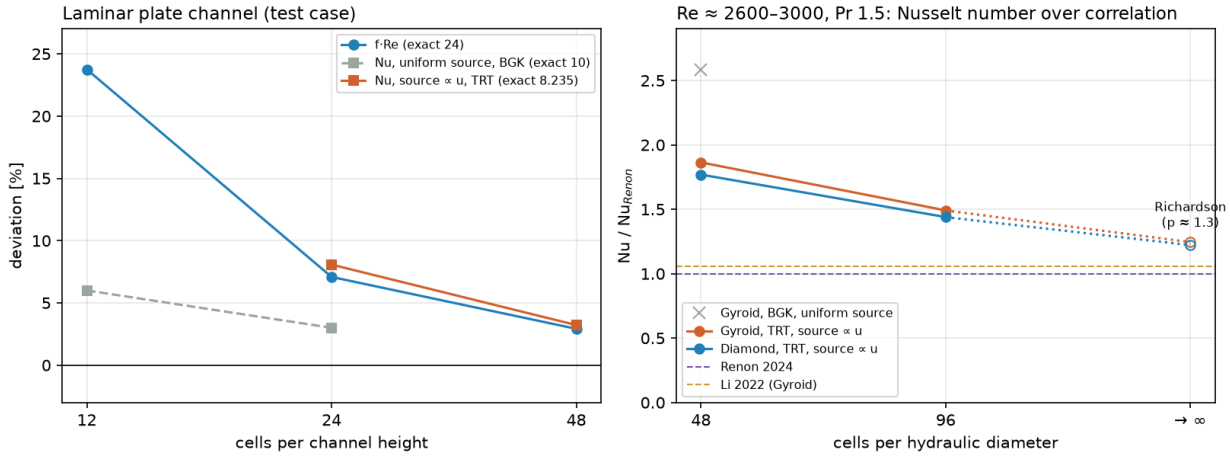

Figure 37: Left: test case laminar plate channel, deviation from the exact value. Right: Nusselt number relative to Renon at the same Re, over grid resolution; open circles: Richardson estimate.

> *How to read the figure:* Left: a check of the flow program on a case with an exactly known answer, flow between two plates. Vertically is the deviation in percent, horizontally how fine the computational grid is. The finer, the more accurate; at 48 cells it is about 3 %. Right: the *Nusselt number*, a measure of how well heat passes from the wall into the water, divided by the literature value (Renon, = 1). The open circles extrapolate to an infinitely fine grid (Richardson) and lie about 25 % above the literature. The grey cross comes from an older, too inaccurate method.

5. Comparison of designs. All runs with TRT and source $\propto u_z$, Pr = 1.5. The driving force is adjusted until Re matches within 15 %; at 96 cells gyroid and diamond stayed somewhat below. $j/f^{1/3}$ with $j = \mathrm{Nu}/(\mathrm{Re}\,\mathrm{Pr}^{1/3})$ is the area goodness: heat transfer per pumping effort.

Design	Grid	Re	f	Nu	$j/f^{1/3}$	$\mathrm{Nu}/\mathrm{Nu}_{\mathrm{Renon}}$
Gyroid	48	2952	0.805	139.8	0.0445	1.86
Gyroid	96	2583	0.732	102.0	0.0383	1.49
Diamond	48	2974	0.668	133.4	0.0448	1.77
Diamond	96	2676	0.596	100.9	0.0391	1.44
Chevron ±60°	48	2908	1.006	121.5	0.0364	–
Chevron ±60°	96	2921	0.863	103.5	0.0325	–

Table 16: FluidX3D results per side (periodic section). Renon applies to TPMS only.

Finding.

- **Hydraulics:** f lies within the band of the correlations (Renon 0.48–0.56, Li 2022 0.81–0.83 [34, 35]). At equal d_h diamond has about 20 % less friction than gyroid; Renon gives 12–17 %.
- **Heat transfer:** Nu lies above the correlations, but the gap shrinks with the grid: factor 1.8 at 48 and 1.5 at 96 cells. With the order from the test case, a Richardson estimate gives about 1.2–1.25. This is a rough estimate from two levels. The third level (144 cells, 62.5 million cells) gave 1.36 (gyroid) and 1.30 (diamond) and shows that the series is not yet asymptotic; extrapolated, Nu lands on Renon (Section 11.15).
- **Collision model:** BGK had overestimated Nu by about 25 % even without blowing up (194 instead of 154 at 48 cells, same source).
- **Designs:** Gyroid and diamond are level in area goodness (0.038 and 0.039 at 96 cells). Chevron plates are about 15 % lower: they reach similar Nu only with more friction. This supports the ranking of Section 11.13 (TPMS more compact than plates). The simulation shows no advantage of gyroid over diamond.
- **Limits:** uniform cell size (the grading was not simulated), a single Re, Pr = 1.5, isothermal wall without conduction in the steel, turbulence via an LES model at only moderate resolution.

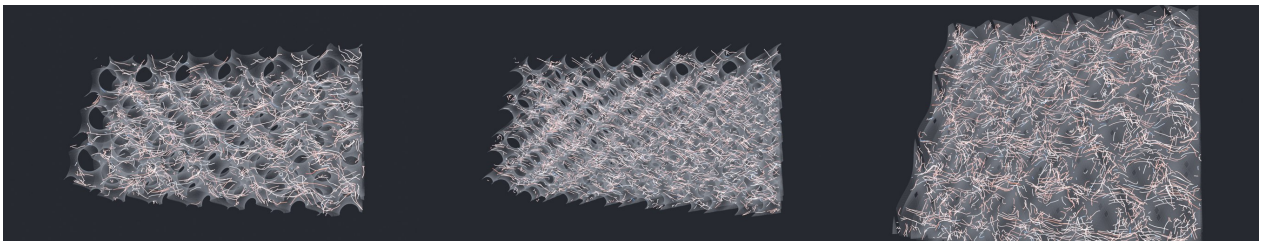

Figure 38: Particle paths in gyroid, diamond and chevron plates (from left), Re 3000, 48 cells per d_h. The wall is shown in X-ray view. A video was made from each set of paths. The colour of the paths comes from the runs with BGK collision and is for illustration only.

> *How to read the figure:* Stills of the paths of water particles (bright lines) through three designs: gyroid, diamond and chevron plates. The wall is shown semi-transparent. The more the paths swirl and cross, the better the water mixes and the more heat it picks up from the wall. The colour has no meaning here.

These and 30 further runs went into the surrogate model of Section 11.15.

11.15 Surrogate model from the flow simulations and the question of grading

Why a surrogate? For every design, the preheater calculation needs Nu and f on both sides. So far there were three ways to get them, each with a catch:

1. **Correlations from the literature** are fast but scatter widely. At Re 3000, Renon, Li 2020 and Li 2022 give $f = 0.54$, 0.15 and 0.81 for the same gyroid, more than a factor of 5, and Nu = 75, 63 and 79 [34, 35]. They also apply only to the cells and walls they were measured or computed for.

2. **Our own flow simulations** match our geometry exactly but cost 1–70 minutes per point. The segment model below evaluates about 440,000 combinations of approach, velocity, passes, cell and segment per variant. At two minutes each, that would take almost two years.

3. **Fitting our own correlation** (power law $\mathrm{Nu} = c\,\mathrm{Re}^m\,\mathrm{Pr}^n$) to the computed points would be fast, but it fixes the form in advance and does not say how reliable it is between the points.

A *surrogate* combines the advantages. It is trained once on a few expensive runs and afterwards delivers values for any input in microseconds, together with a measure of how reliable they are. The concrete question here is whether grading the cells along the flow path pays off. The literature only gives a blanket factor for that (Nu × 1.28 [36]). To check it ourselves, we need to know how Nu and f depend on cell size, and that is exactly what the surrogate learns from our own runs.

How a Gaussian process works. The surrogate is a Gaussian process (kriging). It assumes no formula, only that similar inputs give similar results. A kernel describes this similarity:

$$k(x, x') = \sigma^2 \exp\Big(- \sum_i \frac{(x_i - x'_i)^2}{2\ell_i^2} \Big) + \sigma_n^2\, \delta_{xx'}.$$

The length scale ℓ_i says over what distance in input i the result changes noticeably. A large ℓ_i means input i hardly matters. σ_n^2 absorbs the noise of the runs, such as the time fluctuation of the mixed-mean temperature. All parameters are determined from the data (maximum marginal likelihood). At a new point the process returns a mean, a weighted blend of the known points, and a standard deviation. The latter is small near computed points and grows in gaps, so one can see where the model is guessing. The length scales are a result in their own right: they show which input really matters.

Why multi-level? Fine grids are accurate but expensive (n144: 30–70 minutes, 8 GB of graphics memory); coarse grids are cheap (n48: 1–2 minutes) but systematically too high. A multi-level surrogate (co-kriging [60]) uses both. Many coarse runs describe the *shape* of the dependence on Re, Pr and cell. A few fine runs determine the *shift* caused by the grid. This works well if the grid error varies slowly and smoothly across the input space, and that is checked below.

Design of experiments. For each of gyroid and diamond, 14 points were chosen by Latin hypercube [61]: Re 500–10,000 (logarithmic), Pr 1.2–2.0, cell size $a = 0.75$–1.6 times the standard cell, wall fixed at 0.5 mm. These points ran on the coarse grid (48 cells per d_h). Three control points ran on the medium grid (96), and one run per type on the fine grid (144 cells per d_h, 62.5 million cells). Together with the runs of Section 11.14 that makes 40 runs. The series ran unattended from a job queue that could be paused at any time. A second AI agent watched memory, disk and divergence. No run failed.

Structure. The data are referred to the correlation of Renon [34], so that the Gaussian process only has to learn the deviation:

$$y_{\mathrm{Nu}} = \ln \frac{\mathrm{Nu}}{\mathrm{Nu}_{\mathrm{Renon}}(\mathrm{Re}, \mathrm{Pr})}, \qquad y_f = \ln \frac{f}{f_{\mathrm{Renon}}(\mathrm{Re})}, \qquad x = (\ln \mathrm{Re},\ \mathrm{Pr},\ a).$$

1. **Coarse level:** per type, a Gaussian process with an anisotropic RBF kernel and a noise term over the 15 points on 48 cells. Leave-one-out gives a mean error of 2.8 % (gyroid) and 0.9 % (diamond) for Nu. For f it is 9.5 % and 1.1 %: the gyroid friction scatters noticeably.

2. **Grid correction:** at each control point, $d_{96} = y_{96} - y_{48}(x)$. Over Re this is a straight line in $\ln \mathrm{Re}$ (residual 0.03 in ln). This is co-kriging after Kennedy and O'Hagan [60] with $\rho = 1$ in the logarithmic quantity.

3. **Finest level:** the series 48/96/144 at Re 3000 is *not* in the asymptotic range. We get $d_{96}/d_{144} = 0.58$, while a grid error $\propto n^{-p}$ needs more than $\ln 2/\ln 3 = 0.63$; no positive order fits. There are therefore two upper levels. *n144* is computed: $d_{96}(x)$ is scaled by $d_{144}/d_{96} \approx 1.7$. *Grid* $\to \infty$ is estimated, with Richardson over the last step only and the order from the plate test: $d_\infty = d_{144} + (d_{144} - d_{96})/(1.5^{1.33} - 1)$. Half the distance between the two levels serves as the uncertainty.

		$\mathrm{Nu}/\mathrm{Nu}_{\mathrm{Renon}}$		f/f_{Renon}	
Type	Re	n144	grid $\to \infty$	n144	grid $\to \infty$
Gyroid	1000	1.20	1.03 ± 11 %	0.94 ± 15 %	(uncertain)
Gyroid	3000	1.29	1.03 ± 14 %	1.06 ± 15 %	(uncertain)
Gyroid	8000	1.49	1.14 ± 16 %	1.35 ± 15 %	(uncertain)
Diamond	1000	1.14	0.97 ± 12 %	0.98 ± 6 %	(uncertain)
Diamond	3000	1.23	1.00 ± 14 %	1.01 ± 6 %	(uncertain)
Diamond	8000	1.40	1.08 ± 16 %	1.06 ± 6 %	(uncertain)

Table 17: Surrogate at Pr 1.5 and standard cell. The extrapolated friction cannot be used: it would drop below Renon, because f does not converge monotonically between the grids.

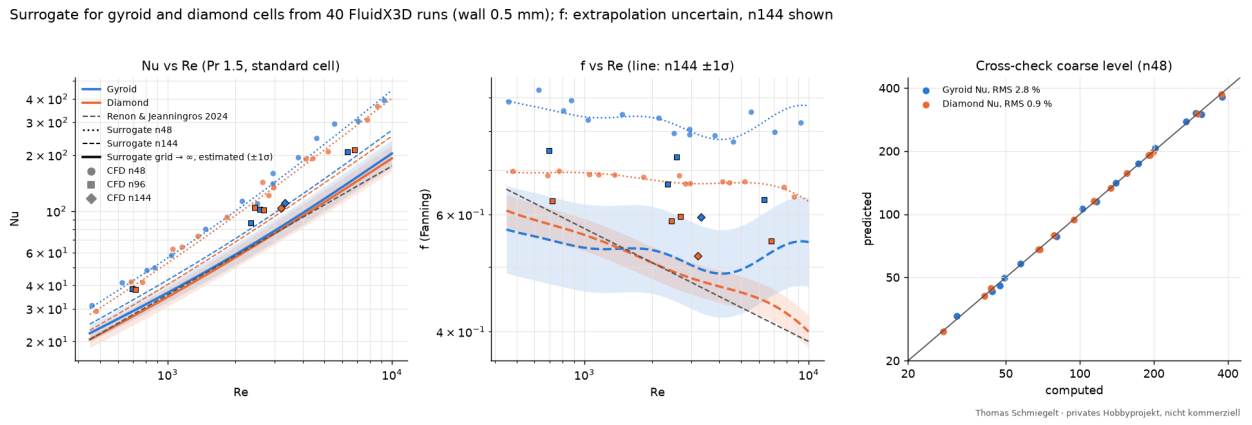

Figure 39: Surrogate for gyroid (blue) and diamond (orange). Left, Nu: dotted the coarse level, dashed n144, solid with band grid $\to \infty$, grey dashed Renon. Centre, f (n144 with band). Right, the cross-check of the coarse level.

How to read the figure: A *surrogate* is a fast stand-in model. It learns from 40 expensive flow calculations and then gives an instant estimate, with its uncertainty, for any input. Left: the Nusselt number (heat transfer) versus the *Reynolds number*, a measure of how fast and turbulent the flow is; both axes are

logarithmic. Dots are calculations (circles coarse, squares fine, diamonds finest), lines are the surrogate, the light band its uncertainty, and the grey dashed line the literature. Centre: the same for the friction factor f, which sets the pressure drop. Right: the cross-check. Each calculation was left out once and predicted from the others. Dots on the diagonal mean a correct prediction; the deviation is only 1 to 3 %.

What the surrogate says.

- Extrapolated to an infinitely fine grid, Nu lands on the Renon correlation (0.97–1.14). The excess of 1.5 at 96 cells was therefore grid error, not physics. The estimate of 1.2 from only two levels was too high.
- At equal d_h, Nu and f hardly depend on the cell size: for the gyroid, the kernel length scale for a sits at its upper bound. A smaller cell gives a smaller d_h but hardly changes the dimensionless numbers.
- Gyroid and diamond are level for heat transfer. For pressure drop, the gyroid is higher at high Re.

Application: does grading pay off? The preheater of Section 11.13 was set up again as a segment model (`stellarator/vorwaermer_stufung.py`):

- Counterflow, two-fluid TPMS; both channel networks share the cell, so d_h is the same on both sides.
- 8 segments of equal heat duty, each with local temperatures, properties, Pr and LMTD.
- Nu and f from the surrogate (level n144; Nu additionally at grid $\to \infty$ as a band).
- *Uniform:* one cell for all segments. *Graded:* each segment chooses its own cell so that $P_i + \lambda V_i$ is minimal, with λ sampled.
- Also free: approach, condensate velocity, and number of passes on the warm side.

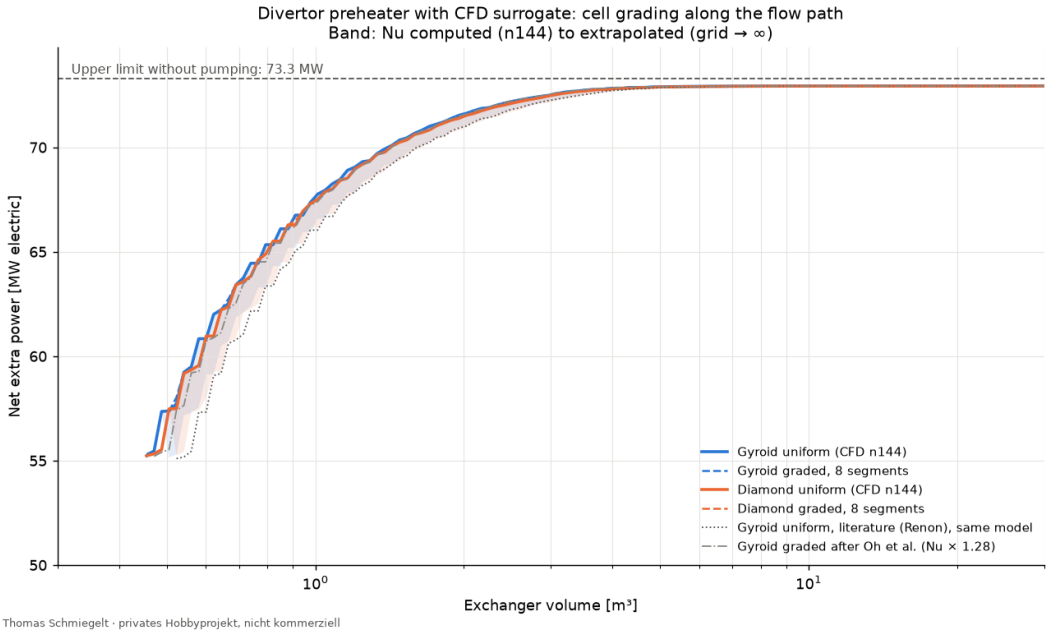

Figure 40: Net extra power over exchanger volume. Graded (dashed) and uniform coincide. For comparison, Renon and Renon with Nu × 1.28 (grading after Oh et al.) in the same model.

How to read the figure: Same layout as the net-gain figure above, now with the surrogate. The dashed lines are exchangers whose cell size changes in eight steps along the flow path (*graded*). The solid lines are exchangers with the same cells everywhere. They lie practically on top of each other. The band spans from the finely computed to the extrapolated heat transfer. Grading brings no benefit; the smallest cell is best everywhere.

Result.

- **Grading brings nothing.** In every segment the smallest allowed cell wins ($a = 0.75\,a_{\mathrm{std}}$, $d_h = 1.33\,\mathrm{mm}$). The graded design is therefore identical to the uniform one. The reason is that pumping power is small here (below 1 MW). Power is lost mainly through the approach at the warm end, and the cheapest way to reduce that is more area per volume, i.e. fine cells everywhere.
- **Volume:** the preheater reaches 95 % of the 73.3 MW upper limit with $1.3\,\mathrm{m}^3$ (n144) to $1.5\,\mathrm{m}^3$ (grid $\to \infty$). At $1\,\mathrm{m}^3$, CFD and Renon give 66.0–67.6 MW, a small difference.
- The literature factor for grading (Nu $\times$ 1.28 [36]) here matches almost exactly the gap between Renon and the n144 surrogate. The improvement reported by Oh et al. cannot be reproduced at the cell level. It must come from effects this model does not contain, such as the distribution in the headers.
- **Limits:** the best cell lies at the edge of the surrogate range; finer cells have not been computed. Whether the 0.5 mm wall carries the 3 MPa pressure difference has not been checked. Manufacturing such fine cells is an open question. Grading may pay off where pumping power matters, for example on the steam side of the condenser.

11.16 Collaboration

From the user came: the thinner wall with a larger outlet (f12), pairwise comparison as a tool for decisions and optimisation loops together with the judgements, the point that the walls of the pressure space must withstand the pressure, the point about the small context of the local model, and the decision for an existing turbine. The assistant did the calculations, built the tools, checked the research and named the consequences.

12 Plasma: where does the heat come from, and how is it produced?

So far the calculation started with the plasma as a source of 3053 MW (Section 11). This section asks how that power arises and by which routes it leaves the plasma. All numbers refer to the HTS design from stage 0: $R_0 = 10.4\,\mathrm{m}$, $a = 2.31\,\mathrm{m}$, $B_0 = 6.5\,\mathrm{T}$, plasma volume $1096\,\mathrm{m}^3$, central temperature $T_0 = 14\,\mathrm{keV}$. The script `stellarator/plasma_waerme.py` recomputes every step.

12.1 The reaction: mass becomes energy

In the plasma, deuterium (one proton, one neutron) and tritium (one proton, two neutrons) fuse:

$$\mathrm{D} + \mathrm{T} \;\rightarrow\; {}^4\mathrm{He}\;(3.5\,\mathrm{MeV}) + \mathrm{n}\;(14.1\,\mathrm{MeV}).$$

The products are lighter than the original nuclei. With the atomic masses [63],

$$\Delta m = m_\mathrm{D} + m_\mathrm{T} - m_\mathrm{He} - m_\mathrm{n} = 0.018\,88\,\mathrm{u} \quad\Rightarrow\quad Q = \Delta m\,c^2 = 0.01888 \cdot 931.494\,\mathrm{MeV} = 17.59\,\mathrm{MeV}.$$

That is 0.38 % of the mass involved. Momentum conservation decides the split: both products fly apart with equal momentum, so the lighter particle receives the larger share,

$$E_\alpha = Q\,\frac{m_\mathrm{n}}{m_\mathrm{n} + m_\alpha} \approx 3.54\,\mathrm{MeV}, \qquad E_\mathrm{n} \approx 14.05\,\mathrm{MeV}.$$

The usual rounded values are 3.5 and 14.1 MeV [66]; the balance uses 3.52 and 14.07 MeV. The 0.6 % difference is irrelevant for the balance. What matters is the ratio: **the neutron carries four fifths of the energy, the helium nucleus (alpha particle) one fifth.** Everything else follows from this.

12.2 Why it has to be so hot

Both nuclei are positively charged and repel each other. For the nuclear force to act they must come as close as the sum of their radii ($r \approx 1.2\,\mathrm{fm} \cdot A^{1/3}$, 3.2 fm together). There the repulsion is

$$E_C = \frac{e^2}{4\pi\varepsilon_0\, r} \approx 444\,\mathrm{keV}.$$

The plasma centre is at 14 keV, i.e. 162 million kelvin, but that is 32 times less than the barrier. Fusion still happens because of two effects acting together:

- **Maxwell distribution:** some particles are much faster than average; their share falls like $e^{-E/kT}$.
- **Tunnelling:** quantum mechanics lets a nucleus pass through the barrier without climbing over it. The probability rises with energy like $e^{-\sqrt{E_G/E}}$, with the Gamow energy $E_G = 1182\,\mathrm{keV}$ for D-T [62].

The product of the two curves is the *Gamow window*. At 14 keV it lies at $E_0 = (\sqrt{E_G}\,kT/2)^{2/3} \approx 39\,\mathrm{keV}$, about 2.8 times kT (Figure 41, top right). So almost only the fast particles in the tail of the distribution fuse.

The reactivity $\langle\sigma v\rangle$ sums up the reaction rate (formula of Bosch and Hale [62]). It peaks at about 67 keV. For a power plant, however, a different question counts: can heating by the alpha particles cover the losses on its own (ignition)? With the energy confinement time τ_E the condition is (Lawson, without radiation)

$$n\,T\,\tau_E \;\geq\; \frac{12\,T^2}{E_\alpha\,\langle\sigma v\rangle},$$

and the right-hand side is smallest at 13.6 keV: $2.8\times10^{21}\,\mathrm{keV\,s\,m^{-3}}$. **That is why the design temperature is** 14 keV **and not at the reactivity maximum.** In the centre, the design reaches $n_0 T_0 \tau_E = 8.0 \times 10^{21}\,\mathrm{keV\,s\,m^{-3}}$. It is above the limit because density and temperature fall off outwards and bremsstrahlung adds to the losses.

12.3 Where in the plasma the heat is produced

The power density is

$$p_\mathrm{fus}(\rho) = n_\mathrm{D}\,n_\mathrm{T}\,\langle\sigma v\rangle(T)\,Q, \qquad n_\mathrm{D} = n_\mathrm{T} = 0.425\,n_e,$$

with the stage-0 profiles $n_e = n_0(1-\rho^2)^{0.5}$, $T = T_0(1-\rho^2)$, $n_0 = 3.1 \times 10^{20}\,\mathrm{m^{-3}}$ (the remaining ions are helium ash and impurities). Integrated over the volume this gives 3051 MW; stage 0 gave 3053 MW.

- The centre produces $12\,\mathrm{MW/m^3}$, the mean is $2.8\,\mathrm{MW/m^3}$.
- Half the power is produced inside $\rho = 0.38$, i.e. in only 14 % of the volume, and 90 % inside $\rho = 0.62$. The heat comes from the hot core; the edge contributes almost nothing.
- There are $1.08 \cdot 10^{21}$ reactions per second. They consume 0.47 kg of tritium and 0.31 kg of deuterium per day, of which 2.9 g of mass is converted into energy.

12.4 Two routes out of the plasma

The neutron leaves at once. It is uncharged, so the magnetic field does not act on it. At 17 % of the speed of light it crosses the minor radius in about 45 ns without heating the plasma. It deposits its energy only in the wall, mostly in the blanket: by collisions with nuclei that slow it down, and by capture. The most important capture reaction also produces new tritium:

$$^6\mathrm{Li} + \mathrm{n} \;\rightarrow\; ^4\mathrm{He} + \mathrm{T} + 4.8\,\mathrm{MeV} \quad [66].$$

Such reactions release additional energy. The factor $M = 1.15$ (assumption as in stage 0) turns 2442 MW of neutron power into 2808 MW of heat in the blanket. **So more than four fifths of the plant's heat is not produced in the plasma at all, but in the volume of the blanket.**

The alpha particle stays and heats. It carries a double positive charge and gyrates around the field lines with a Larmor radius of

$$r_L = \frac{m_\alpha v_\alpha}{2e\,B_0} = 4.2\,\mathrm{cm} \ll a = 2.31\,\mathrm{m}.$$

It thus stays confined and hands its 3.5 MeV to the plasma through Coulomb collisions. The slowing-down time on the electrons is $\tau_s \approx 0.2\,\mathrm{s}$ [64]. Above the critical energy $E_c \approx 14.8\,A_\alpha T_e\,(\sum n_i Z_i^2/n_e A_i)^{2/3} \approx 415\,\mathrm{keV}$ mainly the electrons slow it down, below it the ions. Over the whole path, about 80 % therefore goes to the electrons and 20 % to the ions. This *self-heating* of 611 MW keeps the plasma hot. The design is ignited (no external heating in operation), but only reaches this with 1.29 times the confinement of the ISS04 scaling [65].

Equilibrium: what comes in must go out. In steady state,

$$P_\alpha = \frac{W}{\tau_E} + P_{\mathrm{brems}}: \qquad 611\,\mathrm{MW} = \frac{877\,\mathrm{MJ}}{1.82\,\mathrm{s}} + 129\,\mathrm{MW} = 482\,\mathrm{MW} + 129\,\mathrm{MW}.$$

- **Bremsstrahlung** (129 MW): electrons emit X-rays in collisions with ions. This hits the whole first wall evenly.
- **Transport** (482 MW): heat and particles slowly migrate outwards across the field, reach the scrape-off layer and flow along the field lines onto the divertor.

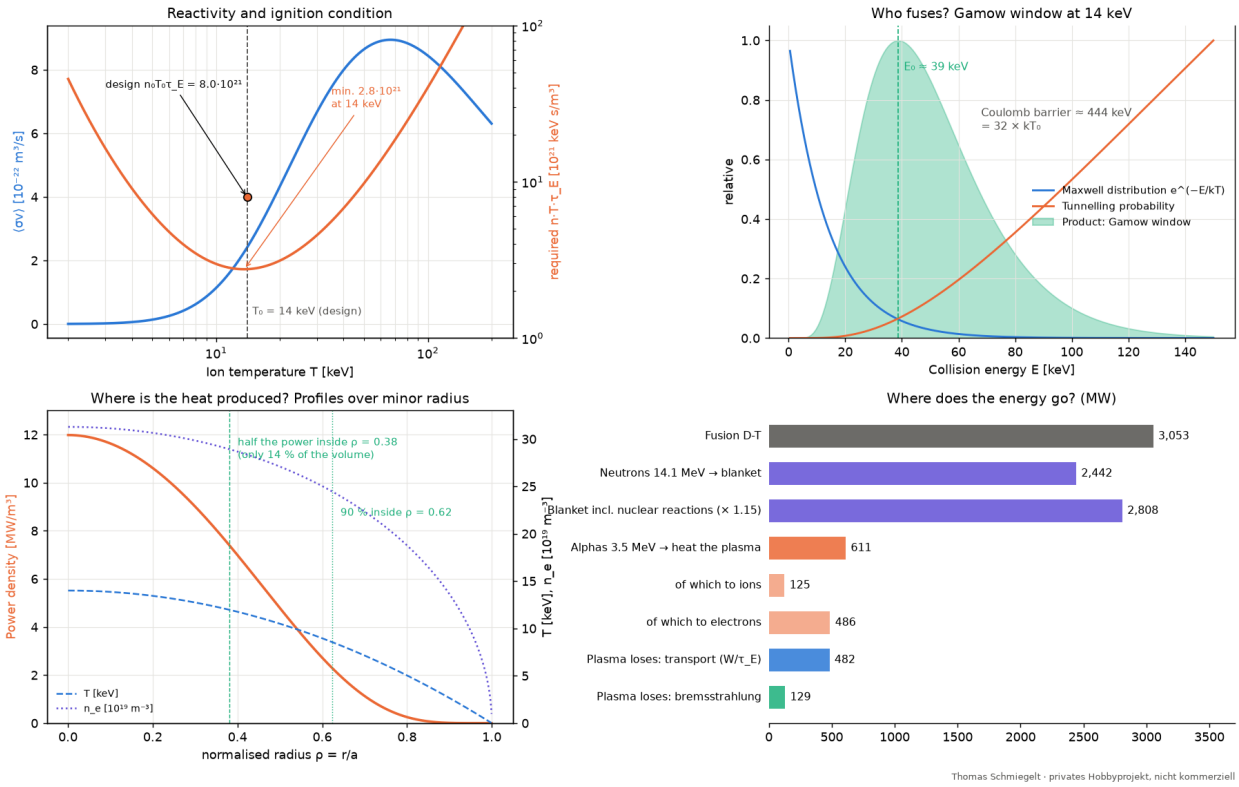

Figure 41: Where the heat comes from. Top left: reactivity and the $nT\tau_E$ required for ignition, with the design point. Top right: Gamow window at 14 keV. Bottom left: profiles and power density over the minor radius. Bottom right: routes of the energy in MW.

How to read the figure: Top left, in blue: how often deuterium and tritium fuse at a given temperature (*reactivity*). In orange: how well the heat must be confined for the plasma to heat itself (*ignition*). Temperature is in keV; 1 keV is about 11.6 million degrees. The orange curve has its minimum near 14 keV, where the design point sits. Top right: the *Gamow window*. Blue is the share of particles with a given energy, which falls steeply. Orange is the chance of overcoming the electric repulsion, which rises steeply. Their product (green) shows that most fusions happen near 39 keV, among the few fast particles rather than at the average. Bottom left, across the radius (0 centre, 1 edge): temperature, density and, in red, the power produced per volume. Half of it comes from the innermost seventh of the volume. Bottom right: where the energy goes, in MW.

12.5 What this means for the plant

1. **The 80/20 split is nuclear physics, not design.** The blanket has to take 82 % of the heat, and in its volume rather than at its surface. This fits the thick, water-cooled WCLL-type blanket (Section 11).

2. **The heat balance has probably set the divertor too favourably.** It splits the alpha power half and half between first wall and divertor ($f_{\text{div}} = 0.5$). Without further measures, however, 79 % goes to the divertor as transport, i.e. 482 MW instead of 305 MW. At 10 MW/m² that needs at least 48 m² of divertor area instead of 31 m². The 50/50 split assumes that about 180 MW extra is radiated at the edge, for example by deliberately injected impurity gas. That is a task for the divertor design.

3. **Start-up needs external heating.** Until alpha heating takes over, the plasma has to be brought to temperature from outside. W7-X uses microwaves (electron cyclotron resonance heating, up to 15 MW), neutral beam injection (up to 8 MW) and ion cyclotron heating [67].

4. **Limits of this calculation:** 0-D balance with fixed profiles, bremsstrahlung only (no line or cyclotron radiation), Lawson limit without profiles, slowing down with $\ln\Lambda = 17$ and central values. A transport calculation, for example with a 1-D code, would be the next step.

13 Liquid metal in a magnetic field: MHD in FluidX3D

In the section on liquid lead (Section 10) the pressure drop in the magnetic field was only estimated with a rule of thumb: $\mathrm{d}p/\mathrm{d}x = \sigma\, v\, B^2 k_p$, with $k_p \approx 1/\mathrm{Ha}$ for insulating walls. The user asked whether the flow in the magnetic field could also be computed dynamically, as a kind of CFD simulation, and whether this could be coupled with FluidX3D. FluidX3D has no magnetic fields of its own. This section describes how the required force was added in two stages and checked against cases with known solutions.

What happens physically? When an electrically conducting metal moves across a magnetic field, a voltage arises in it, as in a dynamo. It drives currents $\mathbf{j}$, and these currents in turn feel a force in the field, the *Lorentz force* $\mathbf{j} \times \mathbf{B}$. It brakes the motion across the field. At a small magnetic Reynolds number, as for lead in a coil, the currents hardly change the external field (*inductionless* approximation). Then Ohm's law for moving conductors and charge conservation apply:

$$\mathbf{j} = \sigma\,(-\nabla\varphi + \mathbf{u} \times \mathbf{B})\,, \qquad \nabla \cdot \mathbf{j} = 0 \quad \Rightarrow \quad \nabla^2\varphi = \nabla \cdot (\mathbf{u} \times \mathbf{B}), \qquad \mathbf{f} = \frac{\mathbf{j} \times \mathbf{B}}{\rho}.$$

σ is the electrical conductivity, φ the electric potential. How strongly the field acts compared with viscosity is measured by the *Hartmann number* $\mathrm{Ha} = B\,a\sqrt{\sigma/(\rho\nu)}$ (a half the duct width along the field). Thin boundary layers of thickness a/Ha form at the walls perpendicular to the field (*Hartmann walls*); thicker layers of thickness $a/\sqrt{\mathrm{Ha}}$ form at the walls parallel to the field (*side walls*) [20]. Whether the currents close through the walls or only through these layers decides the pressure drop.

Terms used in this section.

Hartmann walls the two duct walls that the magnetic field meets at right angles. If the field points upwards, they are the floor and the ceiling of the duct. Named after the Danish physicist Julius Hartmann, who studied this flow in the 1930s.

side walls the other two walls, along which the field runs (left and right).

insulating wall a wall through which no electric current can flow, for example one with a ceramic layer. The currents generated in the metal must then close inside the metal, in the thin layers at the wall.

conducting wall a wall that conducts current well, for example copper or thick steel. The currents can then close through the wall as through a wire; the circuit has little resistance and the braking force becomes large. *Perfectly conducting* means without any resistance, a limiting case for the calculation.

conducting Hartmann walls floor and ceiling conduct, the side walls insulate. This is Hunt's case (1965), for which an exact solution exists.

Hartmann number Ha a measure of how strongly the magnetic field acts compared with the viscosity of the metal. $\mathrm{Ha} = 0$ means no field; at $\mathrm{Ha} = 100$ the field almost completely dominates the flow.

Hartmann layer the thin layer at the Hartmann walls in which the velocity rises from zero to the value in the middle. Its thickness is half the duct width divided by Ha, i.e. one hundredth at $\mathrm{Ha} = 100$.

side layer the corresponding layer at the side walls. It is thicker: half the duct width divided by $\sqrt{\mathrm{Ha}}$.

potential φ the electric voltage at each point in the metal, measured against a fixed point. Its gradient steers the currents so that they close.

Jacobi step a simple method for computing φ: each cell repeatedly takes the average of its neighbours, corrected by the source, until nothing changes any more.

pressure-drop coefficient k_p the fraction of the largest possible magnetic braking force that actually acts (1 = full braking, small = the currents find only a difficult return path).

Stage 1: Lorentz force without potential

Between two infinitely wide plates $\nabla\varphi$ vanishes, leaving $\mathbf{f} = -K\,\mathbf{u}_\perp$ with $K = \sigma B^2/\rho$. This force was built into the main kernel of the project's own FluidX3D copy, semi-implicitly with the Guo forcing: $\mathbf{u}_\perp = (\mathbf{u}^*_\perp + \mathbf{F}_\perp/2\rho)/(1 + K/2)$. This keeps the calculation stable even for large K. The changes are published as diffs in the folder `fluidx3d/` (FluidX3D licence [17]). The check is the Hartmann flow, which has an exact solution: $u(y) = f_x/K\,(1 - \cosh(\mathrm{Ha}\,y/a)/\cosh\mathrm{Ha})$.

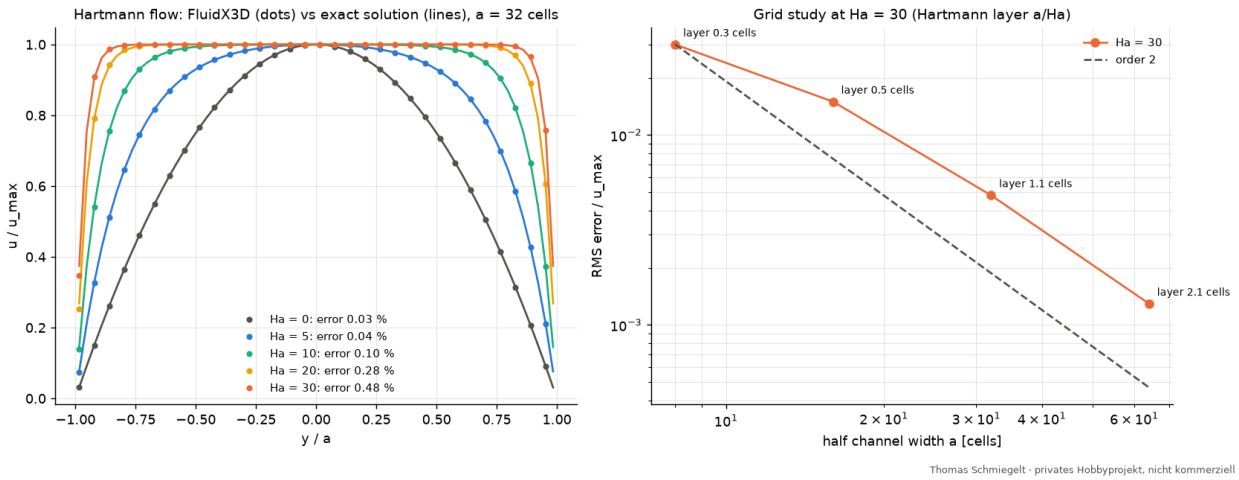

Figure 42: Stage 1: Hartmann flow between two plates. Left: FluidX3D (dots) against the exact solution (lines) for Ha = 0 to 30. Right: the error at Ha = 30 versus grid resolution.

> *How to read the figure:* On the left each colour is one field strength. Horizontally is the position between the two plates (−1 and +1 are the walls), vertically the velocity relative to its maximum. Without a field (grey) the familiar parabola appears. The stronger the field, the flatter the profile in the middle and the steeper its drop at the walls: the field brakes the fast middle, and only thin layers at the wall keep a gradient. The dots (FluidX3D) lie on the lines (exact solution). On the right you see the error shrink as the grid is refined. The labels give the number of cells inside the thin wall layer. Even with a single cell the error stays below 1 %; on finer grids it falls roughly like the dashed line (second order).

Without a field the deviation is 0.03 %, at Ha = 5 it is 0.04 %, and at Ha = 30 with 32 cells (just over one cell in the Hartmann layer) 0.5 %. On finer grids the error falls roughly with second order (3.0 / 1.5 / 0.48 / 0.13 % for a = 8 / 16 / 32 / 64 cells).

Correction. The values obtained first were much worse (4.3 % at Ha = 30 with 32 cells, 68 % with 8 cells) and already appeared in an earlier version. They were an error in the evaluation, not in the calculation. Without the switch `UPDATE_FIELDS`, FluidX3D recomputes the output velocity in a separate kernel, and that kernel lacked half of the Lorentz force. The larger K, the larger the error. This only came to light when the Hartmann test was repeated with the stage 2 code, in which `UPDATE_FIELDS` is on. The rule "at least two cells in the Hartmann layer" was therefore too strict; about one cell is enough for 1 % accuracy.

Stage 2: ducts with side walls

In a real duct the currents flow in loops. In the middle they run across the flow; to return they must go through the walls or through the thin layers at the walls. This return path is described by the potential φ. Without φ, stage 1 would overestimate the braking force in a duct with insulating walls several times over.

Implementation. A new switch `MHD_POT` allocates two buffers for φ. Before every time step the GPU performs some Jacobi steps for $\nabla \cdot \mathbf{j} = 0$, here 20 double steps. The equation is written as a balance of currents over the six faces of each cell, so that charge is also conserved discretely. On an insulating wall the current through the wall face is zero. A perfectly conducting wall holds $\varphi = 0$ on its surface, which lies half a cell in front of the wall node. In the main kernel the force $K\rho\,(\mathbf{E} \times \hat{\mathbf{B}})$ with $\mathbf{E} = -\nabla\varphi$ is added from the same face values.

Validation. Comparing the code only with itself would be worthless. So a separate 2D solver for fully developed duct flow was written that takes a different route. It does not use φ but the induced magnetic field b (Shercliff [18], Hunt [19]):

$$\nabla^2 u + \mathrm{Ha}\,\frac{\partial b}{\partial y} = -1, \qquad \nabla^2 b + \mathrm{Ha}\,\frac{\partial u}{\partial y} = 0.$$

It reproduces three known limits: the square duct without field ($\bar{u} = 0.1406$, i.e. $f\,Re = 14.23$), the exact Hartmann solution for a wide duct with conducting walls, and the core velocity 1/Ha for a wide duct with insulating walls. Along the way a bug in its own boundary treatment was found and fixed.

Two classic cases in the square duct were checked. In *Shercliff*'s case all walls are insulating. In *Hunt*'s case the Hartmann walls are perfectly conducting and the side walls insulating; the current returns through the wall, and fast jets form at the side walls.

A bug that only the validation revealed. In the first run the error with insulating walls grew strongly with Ha: 12 % at Ha = 15, 36 % at Ha = 30. With conducting walls everything was fine. The reason: in the core of an insulated duct two large forces almost cancel, the braking force $-K\mathbf{u}_\perp$ and the force from the potential. Only a fraction of about 1/Ha remains, so any small inconsistency between the two parts is amplified by Ha. In the kernel the potential force was added only after the semi-implicit part, so that part missed some of the force. After swapping the order, and using the same stored velocity for $\mathbf{u} \times \mathbf{B}$ as in the Jacobi step, the error at Ha = 15 dropped from 12 % to 0.45 %.

Case	Ha	a [cells]	profile error	flow-rate error
no field	0	32	0.01 %	−0.03 %
Shercliff (insulating)	5 / 10 / 15	32	0.09 / 0.23 / 0.45 %	−0.15 / −0.30 / −0.37 %
Shercliff	30	64	0.49 %	−0.19 %
Hunt (conducting Hartmann walls)	5 / 10 / 15	32	0.08 / 0.42 / 0.86 %	−0.09 / −0.30 / −0.68 %
Hunt	30	64	0.79 %	−0.78 %
grid study Shercliff	10	16 / 32 / 64	0.75 / 0.23 / 0.09 %	

Table 18: Stage 2: FluidX3D against the independent 2D solver (RMS error of the profile relative to the maximum; deviation of the mean flow rate).

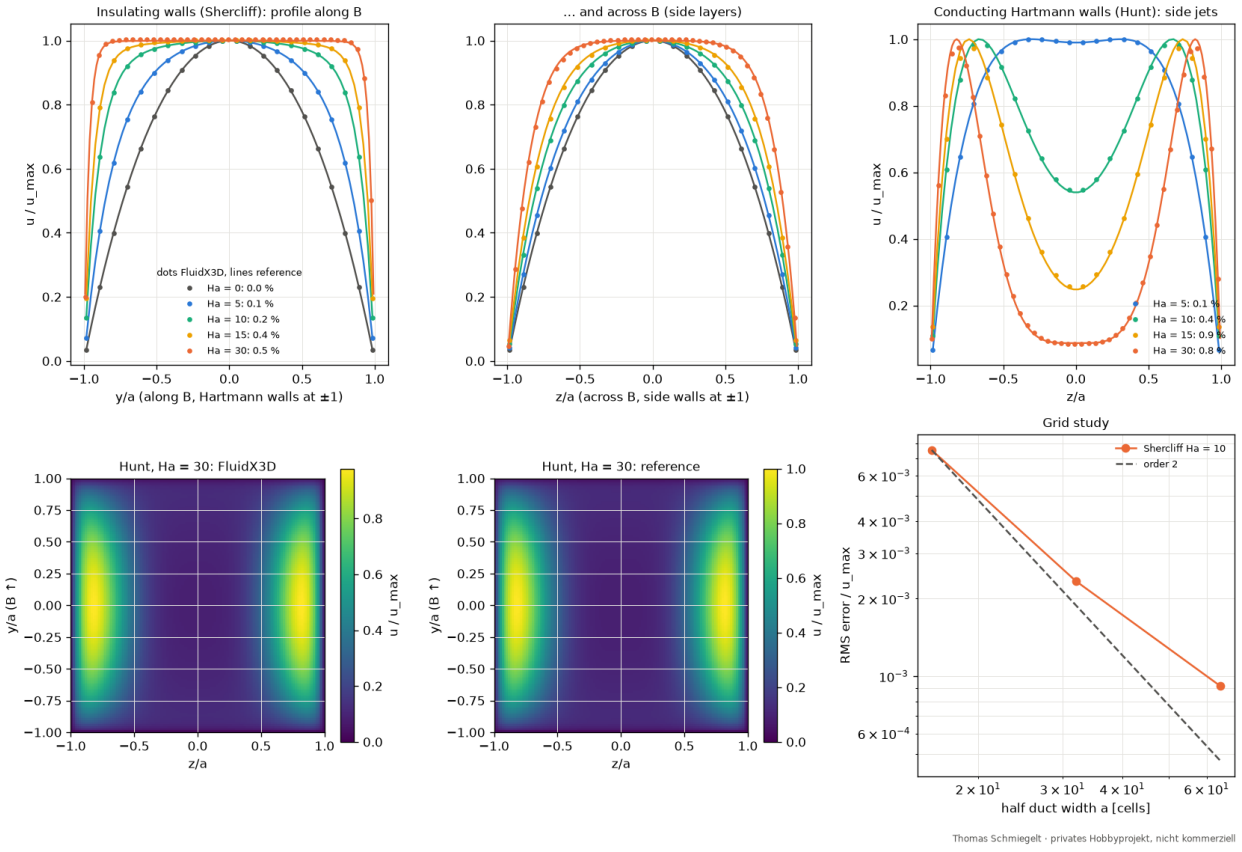

Figure 43: Stage 2: square duct, B pointing straight up. Top left and centre: insulating walls (Shercliff); top right: conducting Hartmann walls (Hunt). Bottom left and centre: the whole cross-section for Hunt, Ha = 30, FluidX3D and reference. Bottom right: the grid study.

How to read the figure: You are looking into a square duct; the metal flows towards you and the magnetic field points up. Top left: the velocity profile from bottom to top, i.e. along the field; top centre: from left to right, across the field. Dots are FluidX3D, lines the independent reference solver; the percentages are the deviations. With insulating walls a stronger field makes the profile flat and box-like along the

field (thin layers at top and bottom). Across the field it stays rounder because the side layers are thicker. Top right: the top and bottom walls conduct. Now the field strongly brakes the middle, and the metal escapes into two fast jets along the side walls. The two colour maps at the bottom show exactly this over the whole cross-section (yellow fast, dark slow); left FluidX3D, centre the reference. They are hard to tell apart. Bottom right: the error falls as the grid is refined.

All twelve cases agree with the reference solver to better than 1 %. The grid study shows an order of 1.4 to 1.7, somewhat worse than in stage 1. The likely reason is the treatment of the wall faces; this has not been investigated yet.

Checking the earlier rule of thumb. The reference solver gives the pressure-drop coefficient $k_p = \mathrm{d}p/\mathrm{d}x \,/\, (\sigma \bar{u} B^2) = 1/(\mathrm{Ha}^2\, \bar{U})$ in the square duct:

Ha	1/Ha	k_p insulating	k_p conducting Hartmann walls
5	0.200	0.407	0.925
10	0.100	0.153	0.634
20	0.050	0.065	0.475
50	0.020	0.023	0.335
100	0.010	0.011	0.256

Table 19: Pressure-drop coefficient from the 2D solver (grid with at least three cells per Hartmann layer).

With insulating walls k_p does approach 1/Ha; at Ha = 100 it is still 10 % above. The rule of thumb was therefore right for large Ha and too low for small Ha. If the Hartmann walls conduct, k_p falls only slowly, and at Ha = 100 the pressure drop is about 25 times larger than with insulating walls.

Limits and next steps.

- **Real Hartmann numbers are much larger.** Lead–lithium in the 10 mm channels at 6 T gives roughly Ha ≈ 600, in blanket ducts of a few centimetres several thousand (material data not yet checked). With one cell per Hartmann layer, Ha = 600 alone would need about 600 cells per half duct width. Memory suffices, but the simple Jacobi solver converges too slowly on such fine grids. A multigrid method or a grid refined towards the wall would be needed; until then computed values hold up to Ha ≈ 30, and the validated scaling laws beyond.
- **Finitely conducting walls** (steel, wall conductance ratio c) are still missing; so far there are only perfectly insulating and perfectly conducting walls. For steel ducts the case in between is the most important one.
- **Single GPU only**, and the field is uniform so far. The coil field ($\mathbf{B}(\mathbf{x})$ from Biot–Savart) would be the step towards coupling with the stellarator.

14 Alpha particles in the coil field: do they stay in the plasma long enough?

In the plasma section (Section 12) the alpha particles heat the plasma with 611 MW. This assumes that every alpha gives off its energy inside the plasma before it reaches the edge. Whether that holds depends on the magnetic field. The user asked for a dynamic simulation with the magnetic field; this is the first part of it (variant A): the orbits of 1000 alphas in the field of the project's own coils.

Terms used in this section.

alpha particle the helium nucleus created in every deuterium–tritium fusion, with 3.5 MeV of kinetic energy. It is electrically charged and therefore follows the magnetic field lines.

guiding centre a charged particle circles rapidly around a field line (about 4 cm radius for an alpha). Instead of computing every circle, one follows only the centre of this circle, the guiding centre. It moves along the field line and drifts slowly across it.

passing and trapped an alpha flying mainly along the field keeps going around the whole ring (*passing*). One flying mainly across the field is reflected at places of stronger field, as by a mirror, and bounces back and forth (*trapped*). Trapped particles are especially sensitive to small field errors.

$v_\parallel/v$ the share of the velocity along the field: $+1$ or -1 means exactly along it, 0 exactly across.

slowing-down time the time in which an alpha gives its energy to the plasma through collisions with electrons and ions, here about 0.2 s. An alpha that stays in the plasma that long has done its job.

last closed surface the outermost of the nested flux surfaces, the edge of the confined plasma. An alpha that crosses it counts as lost here.

quasisymmetry a special form of stellarator field: although the machine is twisted, the field strength along the orbits looks as in a symmetric ring. Trapped particles then do not drift outwards on average. Our plasma shape is designed to be quasi-axisymmetric (QA).

prompt loss an alpha whose orbit already grazes the edge on its first transit; it is lost within fractions of a millisecond. This mainly affects alphas born close to the edge.

Method. The orbits are computed with simsopt [7] as collisionless guiding centres in the vacuum field of the 16 coils from stage 2. Coils and shape are available at unit size ($R_0 = 1.03\,\mathrm{m}$). Collisionless orbits, however, depend only on the ratio of gyration radius to plasma radius ($\rho_L/a = 0.018$). So instead of enlarging the coils, the velocity is reduced accordingly, and times and positions are scaled back to the plant. The 1000 starting points follow the distribution of fusion power, and the directions are uniformly distributed. The calculation runs to 0.2 s, the slowing-down time. For the energy balance, each lost alpha is assigned the energy it would still have at the time of loss according to classical slowing down [64]:

$$u(t) = \big(u_0 + 1\big)\, e^{-3t/\tau_s} - 1, \qquad u = (E/E_c)^{3/2}, \qquad \tau_s = 0.195\,\mathrm{s},\ E_c = 415\,\mathrm{keV}.$$

The loss positions are assigned to a grid on the edge surface (64×32 points, combined into 16×8 tiles of about $10\,\mathrm{m}^2$ each). This gives the heat flux.

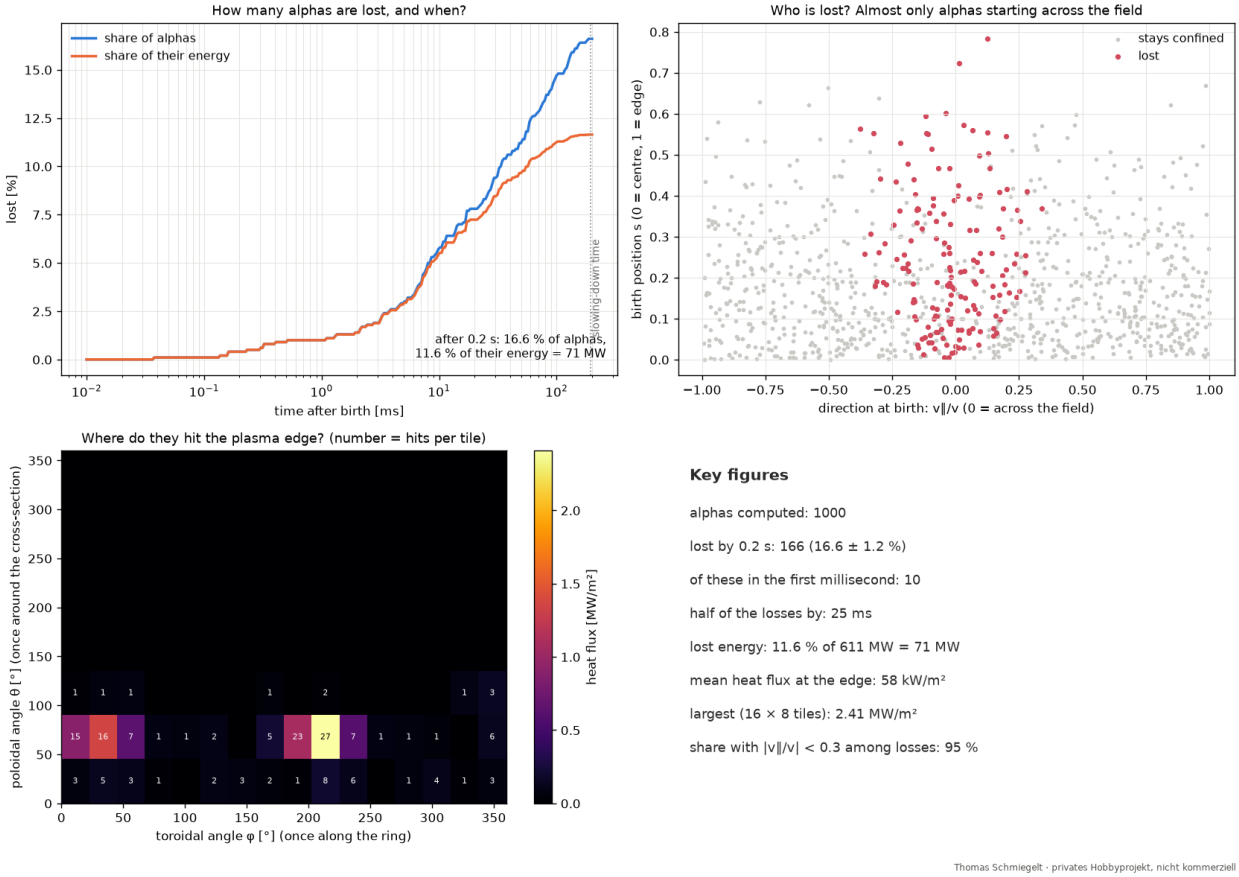

Figure 44: Alpha particle losses: time history, who is lost, where they hit the edge, key figures (1000 alphas, collisionless, field of the project's own coils).

How to read the figure: Top left: horizontally the time since the alphas were born, logarithmic from a hundredth of a millisecond to 0.2 s. Vertically, the percentage lost. Blue counts particles, orange their energy. The orange curve stays lower because alphas lost late have already given off much of their energy. Hardly anything is lost in the first millisecond; most losses happen between 5 ms and 100 ms. Top right: each dot is one alpha, horizontally its direction at birth (0 = across the field), vertically its birth position (0 = plasma centre, 1 = edge). The red, lost alphas lie almost all in a narrow strip around 0: they are trapped particles. Bottom left: the unrolled edge surface of the plasma, horizontally once along the ring, vertically once around the cross-section. The colour is the heat flux, the number the count of hits. The hits are bunched in one band, and twice over, because the machine repeats itself twice. Bottom right: the key figures.

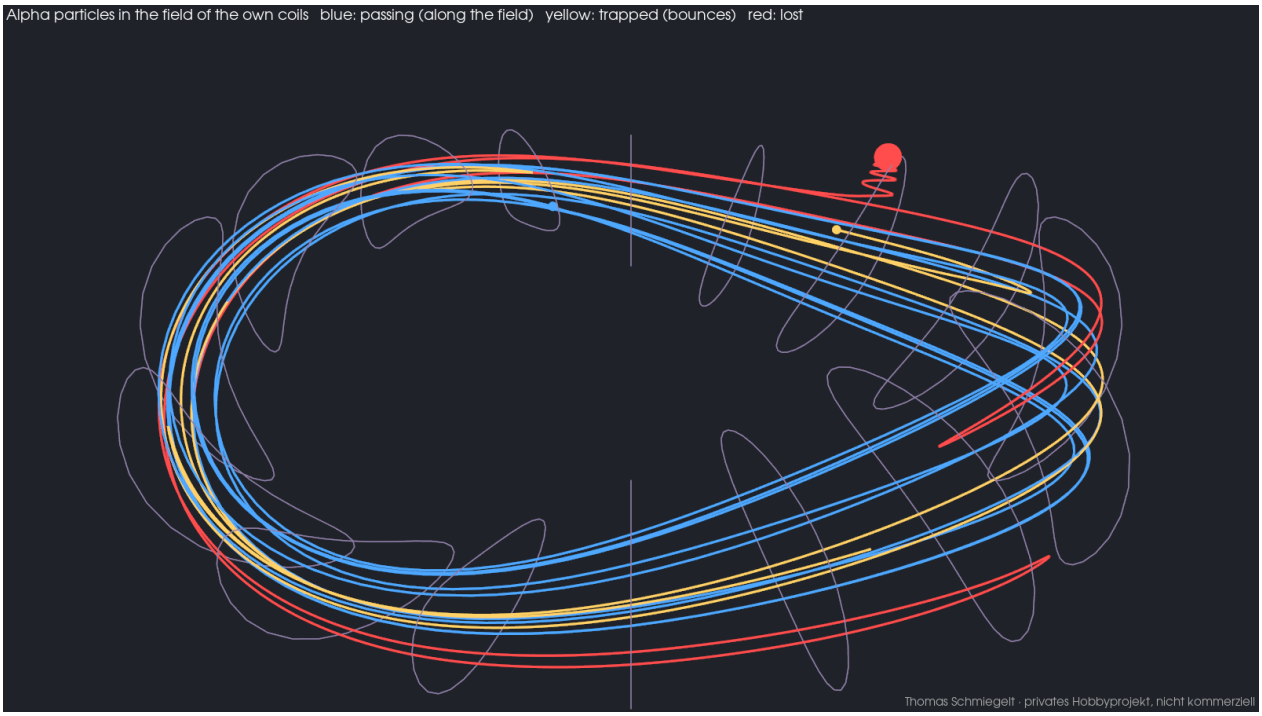

Figure 45: Three alpha orbits in the field of the project's own coils (still from the video `docs/filme/alpha_bahnen.mp4`): passing (blue), trapped (yellow), lost early (red).

> *How to read the figure:* You are looking at the ring obliquely from above. The thin grey loops are cross-sections of the plasma edge at 16 places; they show how the shape twists around the ring. Each coloured line is the path of an alpha's guiding centre. The blue alpha flies along the field and keeps going around the whole ring; it stays safely inside. The yellow one flies almost across the field and turns back at places of strong field; its path is bent back at the ends. The red one is also a trapped alpha. Its orbit slowly moves outwards until it reaches the edge after 0.13 ms (red dot at the top). Orbits like this cause the losses in the figure above, though mostly much more slowly.

Results.

- **16.6 % of the alphas are lost** (± 1.2 % statistical uncertainty with 1000 particles), only 10 of them in the first millisecond. Half of the losses have occurred after 25 ms.
- **They are almost all trapped alphas**: 95 % of the lost ones had $|v_\parallel/v| < 0.3$. On average they were born somewhat further out ($s = 0.24$ versus 0.18).
- **Lost energy: 11.6 % of the alpha power, i.e.** 71 MW. The plasma misses this part of its self-heating; instead of 611 MW about 540 MW remain.
- **The losses arrive bunched.** Averaged over the whole edge it is only 58 kW/m^2, but almost all hits lie in one band in the upper half of the cross-section, in two spots. In the hottest tile of about 10 m^2 it is around 2.4 MW/m^2. The first wall of this project is designed for 0.5 MW/m^2 (cooling section). Because of the coarse tiles and only 27 hits in the hottest tile, the peak value is uncertain; the true peak is more likely higher.

Assessment. The losses do not come as a prompt loss in the first transits but slowly, over milliseconds up to the slowing-down time. This is the typical picture of trapped particles drifting outwards in a field that is not quite quasisymmetric. The plasma shape itself is very precisely quasisymmetric (quasisymmetry objective in DESC $2.4 \cdot 10^{-5}$, smaller than the reference *precise QA* with $5.8 \cdot 10^{-5}$). The coils, however, only approximate this shape: the field is inclined to the edge surface by 0.27 % on average and 0.96 % at most ($|B_n|/|B|$). Whether most of the losses come from these coil errors or are already inherent in the shape is settled by the following comparison. A loss of more than 10 % of the alpha energy would be too much for a power plant, above all because of the bunched wall load.

Before and after: coils or shape? The same calculation was done a second time, now in the field of the plasma shape alone, i.e. as if the coils reproduced the shape perfectly. For this, the equilibrium of the

shape is converted with `booz_xform`[1] into *Boozer coordinates*, a coordinate system in which the field lines are straight and the field strength is a simple Fourier series. simsopt then computes the orbits directly in it, with the same distribution of birth positions and directions as before (mean $s = 0.18$, 31 % trapped starting directions in both runs).

Expectation beforehand. Before the result was available, the event log recorded: the shape is more precisely quasisymmetric than *precise QA*, for which Landreman and Paul found no alpha losses within 0.2 s [21]. Less than 1 % loss was therefore expected. The interpretation was also fixed in advance: below 2 % means the coils are the main cause; between 2 and 10 % both contribute; above 10 % it is the shape itself or a calculation error.

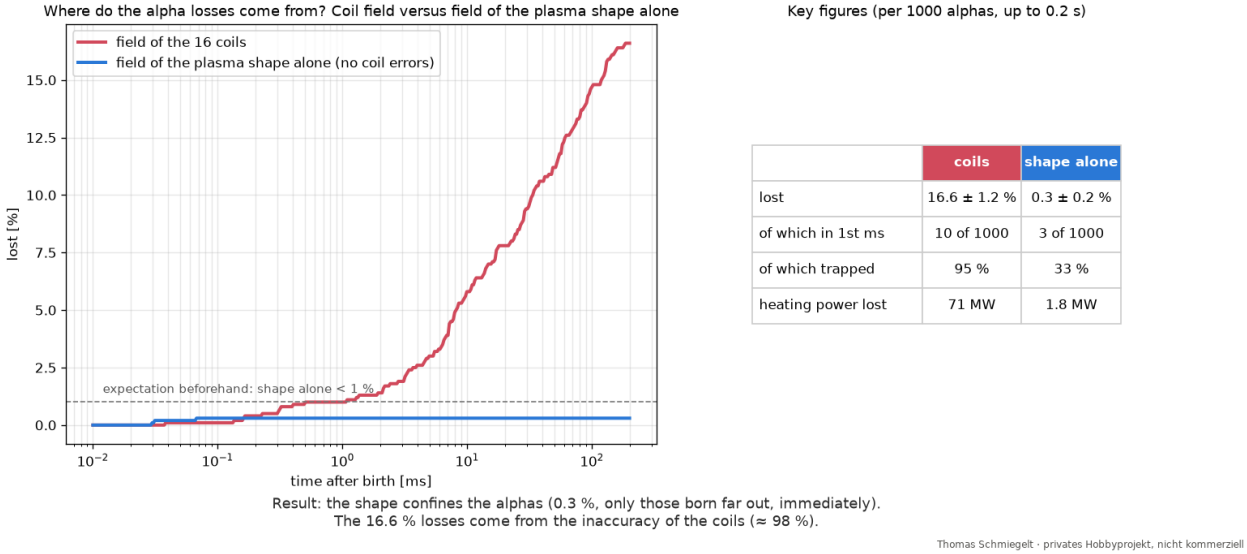

Figure 46: Before and after: alpha losses in the field of the 16 coils (red) and in the field of the plasma shape alone (blue), 1000 alphas each up to 0.2 s.

> *How to read the figure:* Left: horizontally the time since the alphas were born (logarithmic), vertically the fraction lost. The red curve (coils) rises steadily from about one millisecond to 16.6 %. The blue curve (shape alone) jumps to 0.3 % right at the start and then stays flat: after that not a single alpha is lost. The dashed line is the expectation of 1 % fixed beforehand; the blue curve stays below it. Right: the key figures of both runs.

Result. In the field of the shape alone only 3 of 1000 alphas are lost (0.3 ± 0.2 %), all born far out ($s = 0.60$ to 0.72) and within the first 0.1 ms. The slow losses of trapped alphas, which make up 95 % of the losses in the coil field, are completely absent. The lost heating power drops from 71 MW to 1.8 MW. Because this run was much faster than the coil run, a spot check verified that the orbits are really computed up to 0.2 s: six alphas that were not lost (four trapped, two passing) all reach 0.2000 s; the trapped ones bounce ($v_{\parallel}$ changes sign again and again) and stay inside. The expectation is confirmed: about 98 % of the losses come from the inaccuracy of the coils, not from the plasma shape. This fits the still image above: the red orbit gets caught in a small field well, which only a not quite quasisymmetric field forms. This comparison is the baseline for the next step: re-optimise the coils and repeat the same calculation.

Limits.

- **Collisionless**: in reality the orbits change while the alpha slows down and changes direction through collisions. Collisions can turn trapped particles into passing ones and vice versa; this can shift the losses either way.
- **Vacuum field**: currents in the plasma (finite pressure) are not included.
- **The loss location is the plasma edge**, not the wall. The wall lies further out, and the hits spread somewhat differently there.
- **Statistics**: 1000 alphas are enough for the loss fraction, but not for a fine wall-load map.

[1] `https://github.com/hiddenSymmetries/booz_xform`

- **Field of the shape**: the conversion to Boozer coordinates uses 24 Fourier modes in each direction; finer components are missing. This could slightly underestimate the losses of the shape, but does not change the order of magnitude (0.3 % versus 16.6 %).

Next steps. (1) Done: the causes are separated, it is the coils. (2) Fit the coils more closely to the shape, with a target for alpha losses, and repeat the same comparison ("after"). (3) More particles and collisions for a reliable wall-load map.

15 Strength: where the force comes from and how a support structure takes it up

The first structural calculation of the 1-GW plant (fusion surrogate with 79 samples, hex model and Elmer) gave stresses in the coil casing of about 2000 MPa, four times more than steel at 4 K can bear in the long term. The user noted on this: *"Wall, coils and strength are one topic, they belong together. The plasma belongs to it too. The coil shape determines the plasma, and the coil also has to hold, and that also affects the temperature." (translated)* Later the user added the package (everything between plasma and coil) and a 3D model as the goal, and when the numbers came: *"We have to concentrate on strength. 2000 MPa is a killer, even 500 MPa is a lot. Maybe we need more coils, or we have to understand how the force arises." (translated)* This section describes how, step by step, a design emerged that keeps within the limits, under which conditions it holds, and why it works. The design study is complete and evaluated; a separate variant report presents and compares all designs computed, this chapter shows the best ones and those close to the limit.

Terms used in this section.

Lorentz force the force that a magnetic field exerts on a conductor carrying current. Per metre of conductor it is $\vec{f} = I\,\vec{t} \times \vec{B}$ (current I, direction of the conductor $\vec{t}$, field $\vec{B}$). In our coils 22 MA flows; that gives up to 140 MN per metre, the weight of 14 000 t on every metre.

self-field the field that a coil produces at its own conductor. It pushes the coil apart like an inflated tube.

equivalent stress (von Mises) a number that condenses the multiaxial stress state in the material into a value that can be compared with the strength from the tensile test.

99-% stress σ_{99} (formerly called "p99") the stress that is not exceeded in 99 % of the steel volume. Single peaks at edges of the coarse calculation mesh are thus not overrated.

rope tension A curved rope can carry a load pressing transversely on it by tension alone, if the load matches the curvature (like a ring under internal pressure). A straight piece cannot do this: it has to carry the load by *bending*, and bending produces much higher stresses than tension.

bearing, bucking cylinder the places where the coils are held. In the tokamak each coil presses with its straight inner leg against a bucking cylinder in the centre of the machine.

connector steel blocks between neighbouring coil casings that pass forces from coil to coil.

strain the relative change in length. The superconducting tape (REBCO) tolerates only a few tenths of a percent of tension in the long term before it is damaged.

package all layers between plasma and coil: first wall, blanket (where tritium and heat are produced), shielding against neutrons and vacuum vessel.

design study (DOE) many designs whose control variables are spread over the whole range by plan (here Latin hypercube [61]), so that one can see which quantity has which effect.

Division of labour. The user set the direction and decided at every fork. The AI (Claude Opus) planned, set expectations, wrote the coupling and evaluated the results. Subtasks went to smaller models: Claude Sonnet researched the limit values (every source checked via Crossref, sample re-checked by Opus) and extended the coil optimisation by a force target; Claude Haiku explored how a new coil set runs through the structural model. The calculations were done with simsopt [7] (coils) and Elmer [10] (strength).

Expectation beforehand

Before each calculation, the event log recorded what was expected and how the result was to be interpreted:

1. A force target in the coil optimisation lowers the peak force by about 30 % at double the field error.
2. Reducing force alone does not give the required factor 4; what is needed is a support structure that passes the forces on in a closed way.
3. The force per metre lies at 100–150 MN/m; the required rope tension varies along the coil by more than a factor of 3, because the curvature changes strongly. Bending, not tension, is the cause of the high stress.

4. The net force on each coil points to the centre of the machine.
5. More coils lower the force per metre roughly in inverse proportion to the number of coils; the steel needed for the same stress stays about the same.
6. The package is tight at the present spacing.

Step 1: Force target in the coil optimisation

The coil optimisation (stage 2) received an additional term that penalises high force per metre (simsopt `LpCurveForce`). Without this term exactly the previous coil set results (deviation of the field error 0.01 %).

Weight of force target	Field error $B \cdot n$ (mean)	Peak force	Mean force
0 (previous)	0.27 %	100 %	100 %
0.01	0.30 %	−9 %	−3 %
1	1.56 %	−25 %	−6 %

Result: The peaks can be capped, the mean force hardly. It is fixed by field strength and the size of the machine. Expectation 1 was too optimistic, expectation 2 is confirmed: the coil shape alone does not solve the problem. The force weight is only used further in the range 0 to 0.03.

Step 2: A coupled calculation chain

So that plasma, coils, package, strength and temperature can be evaluated together, a chain was built (`stellarator/kopplung.py`). A design runs through it in 1–3 minutes:

1. **Coils** (simsopt): number of coils, distance to the plasma, coil length, force target → field error as a measure of plasma quality.
2. **Calibration**: current such that $B_0 = 6.5\,\mathrm{T}$ prevails on the magnetic axis. Winding pack square, edge length such that the current density stays as with 16 coils.
3. **Strength** (hex mesh, Elmer): casing with four wall thicknesses, potting, connectors, support; load $\vec{f} = I\vec{t} \times \vec{B}$ from all coils.
4. **Package**: smallest distance from the inside of the casing to the plasma edge, all round; from this the shield thickness, nuclear heating and neutron dose in the winding pack.
5. **Temperature**: field at the edge of the winding pack → utilisation of the superconductor and temperature margin.

The limits and assumptions:

Quantity	Limit	Origin (status)
Casing steel 1.4429/316LN, σ_{99}	500 MPa	2/3 of the yield strength at 4 K, ITER grade C3; best grade C1 allows 667 MPa (ITER slides, unverified)
Strain winding pack	0.3 %	REBCO irreversible from 0.45–0.72 % [69]; ARC kept < 0.2 % [68]
Nuclear heating winding pack	2000 W/m^3	ARIES-CS (unverified); strict 50 W/m^3 for EU-DEMO [72]
Neutron dose REBCO	$3 \cdot 10^{18}$ n/cm^2	lower bound after ARC [68]
Utilisation of superconductor	70 %	assumption; $J_c(B,T)$ after [70, 71]
Field error $B \cdot n$	0.3 %	previous coil set as benchmark
Package at least	1.0 m	ARIES-CS design [73]
Neutron attenuation	e-fold per 0.1 m	assumption; tungsten alone 0.061 m [74]

While building the chain an old error came to light: the previous structural model had scaled the coils 3 % larger than the plasma and alpha calculation. The chain now uses the same scale everywhere. The first test point (previous coils) confirmed the expectations with one exception: the field at the conductor came out at 6.4 T instead of about 15 T. Cause: it had been measured on the centreline, where the self-field cancels. At the edge of the winding pack it is 18.7 T. The test point showed: package (1.77 m free), temperature (utilisation 34 %) and field accuracy fit. It fails only on strength (σ_{99} = 2129 MPa) and on strain in the winding pack (0.76 %). Expectation 6 (package tight) was wrong: ARIES-CS manages with 1.3 m from plasma to coil centre [73], we have 2.5 m.

Step 3: Where the force comes from

For each coil the force per metre along the conductor was calculated: the field of the other coils by Biot–Savart, the self-field with the regularised formula for a rectangular cross-section (simsopt). It is split into a part pointing outwards (the coil widens) and a sideways part.

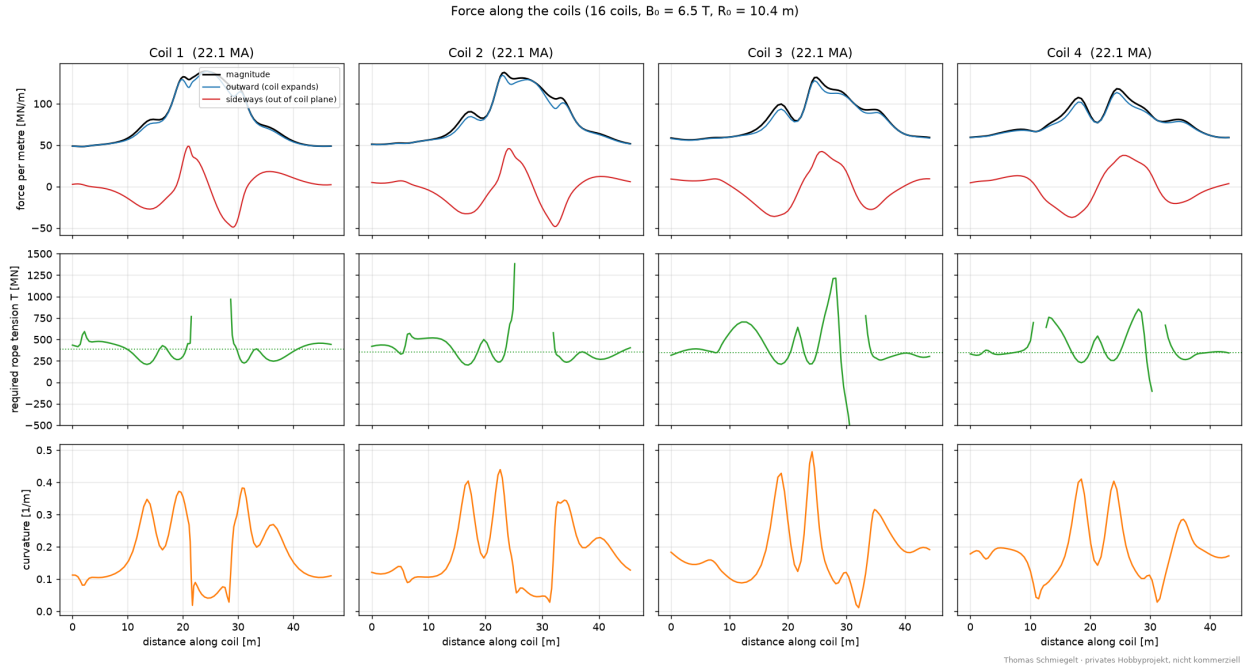

Figure 47: Force along the four basic coils (16 coils in total, the others are mirror images): force per metre, required rope tension, curvature.

> *How to read the figure:* Each column is one coil, horizontally the path along the coil (one coil is about 45 m long). Top: black is the total force per metre, blue the part that widens the coil, red the sideways part. The force is everywhere 50 MN/m or more and reaches 140 MN/m in the middle of the path: this is the inner leg of the coil on the inside of the ring, where the coils stand close together and the field is strongest. Middle: the rope tension with which the coil could carry this load as pure tension. Where the curve is missing, the coil is almost straight; there this would only work with infinite rope tension. Bottom: the curvature. Peaks are tight bends (radius about 2.5 m), valleys are almost straight pieces. The picture shows the core problem: the largest force sits exactly on the straight piece.

Derivation: why straight pieces are so costly. For a rope with rope tension T, curvature κ and transverse load f_N, equilibrium gives

$$\frac{dT}{ds}\,\vec{t} + T\kappa\,\vec{N} + \vec{f} = 0 \quad \Rightarrow \quad T = \frac{f_N}{\kappa}.$$

With $f_N \approx 100\,\mathrm{MN/m}$ and $\kappa \approx 0.25\,\mathrm{m}^{-1}$ one needs $T \approx 400\,\mathrm{MN}$. With about $1.3\,\mathrm{m}^2$ of steel cross-section in the casing, this is already about 300 MPa. This is the share that cannot be avoided if field and size are fixed. On the inner leg the coil is almost straight over about $l = 7\,\mathrm{m}$ ($\kappa < 0.08\,\mathrm{m}^{-1}$), the load there is $q \approx 130\,\mathrm{MN/m}$. As a beam on two supports this gives

$$M = \frac{q\,l^2}{8} \approx 800\,\mathrm{MNm}, \qquad W = \frac{H^4 - h^4}{6H} \approx 0.38\,\mathrm{m}^3 \;(H = 1.36\,\mathrm{m}\ \text{casing},\ h = 0.755\,\mathrm{m}\ \text{winding pack}), \qquad \sigma_B = \frac{M}{W} \approx 2100$$

This back-of-the-envelope estimate matches the FEM value of 2129 MPa remarkably well. The high stresses therefore come from the bending of the straight inner leg.

How tokamaks solve this. A tokamak coil has the shape of a "D": the arc is shaped so that it carries the load as pure tension, and the straight leg on the inside lies flat against a bucking cylinder or against the neighbouring coils (wedging). It is therefore not bent but compressed. Our model, in contrast, held each coil only at its outermost point. The net force per coil, however, points inwards with up to 1.1 GN, to the centre of the machine (expectation 4 confirmed). The coil therefore hung like a beam supported on the wrong side.

More coils? The same analysis for 20 and 24 coils:

Coils	Current per coil	Force per metre (mean)	Sum of all coils	Field error $B \cdot n$
16	22.1 MA	85 MN/m	58 GN	0.27 %
20	17.7 MA	65 MN/m	57 GN	0.074 %
24	14.7 MA	54 MN/m	58 GN	0.028 %

The force per metre falls, the sum stays (expectation 5 confirmed): with the same steel share the same stress follows. The gain lies elsewhere: with more coils the field becomes ten times more accurate. This margin can be traded for shorter, smoother coils.

Step 4: Support-structure trials with the same 16 coils

Step 3 directly showed what has to change: support where the force points, and couple the coils into a composite that passes transverse loads from coil to coil. For this, the structural model received a bearing at the innermost point of each coil (face towards the centre of the machine, like a bucking cylinder), and optionally inside and outside.

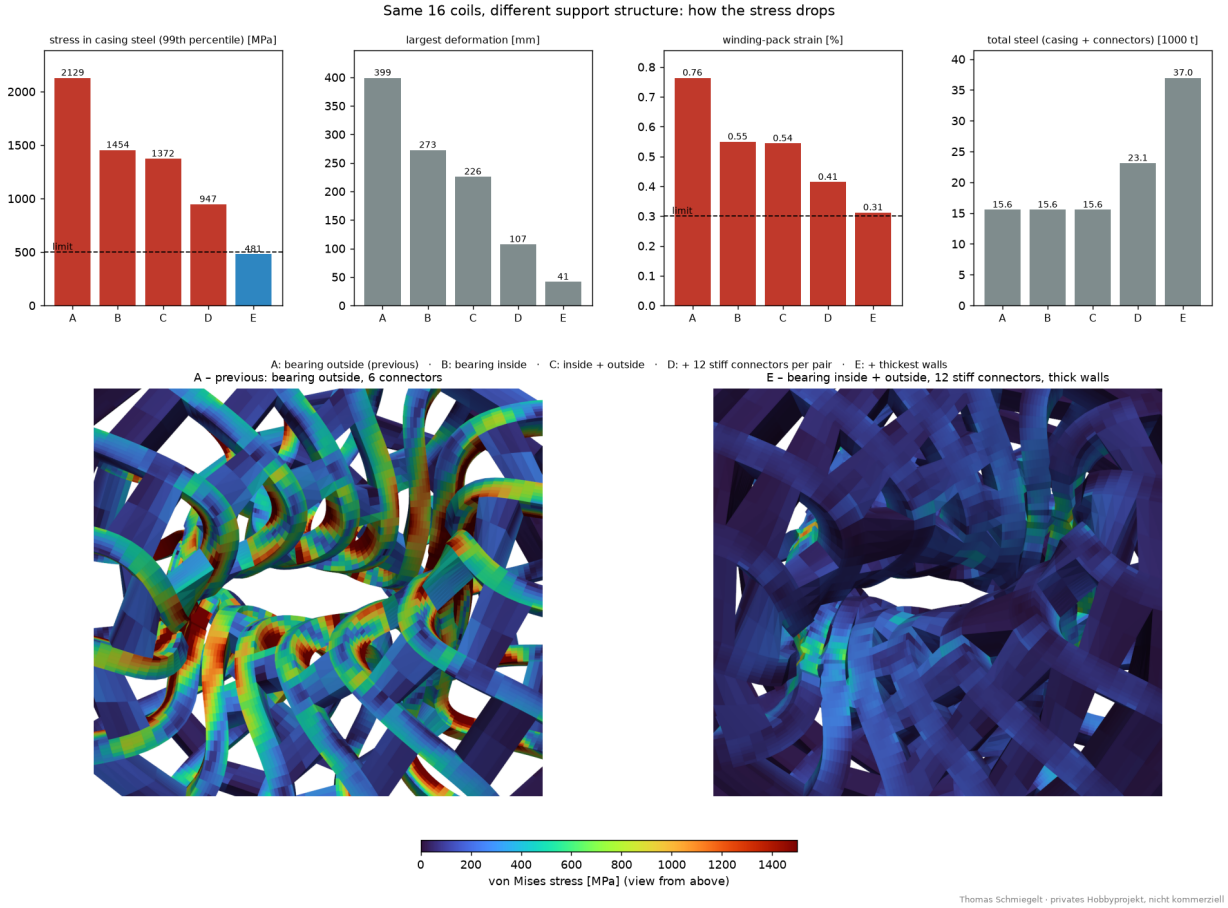

Figure 48: Same 16 coils, different support structure: stress, deformation, strain and steel mass for five variants (A–E), including the stress seen from above for A and E.

How to read the figure: Top: four bar charts for the five variants A to E (key in the line below). Red bars lie above the limit (dashed), blue ones below it, grey quantities have no fixed limit. From A to E the stress in the steel falls from 2129 MPa to 481 MPa and the deformation from 399 mm to 41 mm; the steel mass rises from 15 600 to 37 000 t in exchange. Bottom left: variant A seen from above; red marks places above 1200 MPa, mainly at the inner legs around the hole in the middle. Bottom right: variant E; almost everything is blue (below 400 MPa); the coils have become thicker and, through connectors, a composite.

	Variant	σ_{99} steel	Deformation	Strain winding pack	Total steel
A	bearing outside (previous), 6 connectors per pair	2129 MPa	399 mm	0.76 %	15.6 kt
B	bearing inside	1454 MPa	273 mm	0.55 %	15.6 kt
C	inside and outside	1372 MPa	226 mm	0.54 %	15.6 kt
D	+ 12 connectors per pair, steel (200 GPa)	947 MPa	107 mm	0.41 %	23.1 kt
E	+ thickest walls of the range	481 MPa	41 mm	0.31 %	37.0 kt

Why this works. B: the net force goes into the bearing over a short path, instead of bending the whole coil (−32 %, expectation 30–50 % confirmed). D: more and stiffer connectors pass the transverse load of the straight leg on to the neighbours; the free length l of the beam becomes shorter, and $M \propto l^2$. E: a larger section modulus W lowers the bending stress further. With this, for the first time a design lies below 500 MPa. The price is mass. The problem has shifted: from "does not hold" to "too heavy".

Step 5: Design study

To find the most robust admissible design, 200 designs were computed (Latin hypercube). Eleven quantities were varied: number of coils (16, 20, 24), coil–plasma distance, length per coil, force weight, four wall thicknesses, size of the bearing areas, number (6–14 per pair) and stiffness (20–200 GPa) of the connectors. Bearings inside and outside are fixed. During the study the user changed the goal: *"The weight is not so important. It is a building, after all." (translated)* Since then the aim is not the lightest but the most robust design, i.e. the one with the largest smallest reserve over all limits. One design failed in meshing, 199 were computed completely.

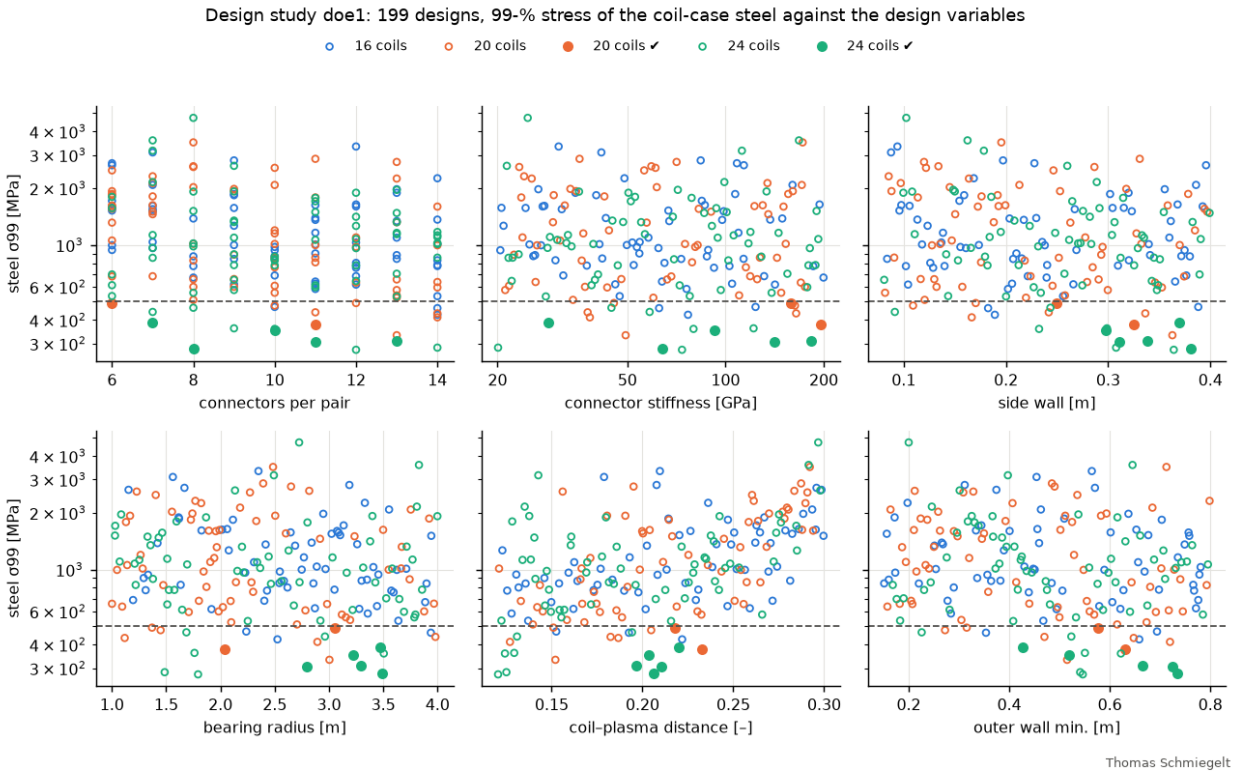

Figure 49: Design study: 99-% stress in the case steel against six design variables, all 199 designs.

How to read the figure: Each circle is a computed design; the colour shows the number of coils (blue 16, orange 20, green 24), filled circles meet all eight limits. Vertically the stress (logarithmic), dashed the limit of 500 MPa. No single variable orders the points; the clearest trend is the rise of stress with the coil–plasma distance (bottom centre).

A clean-up was necessary. Eleven designs did not fit into memory and were computed with a coarser mesh. Only the mesh study revealed that their connectors became one third longer, because a connector always spans four stations. These designs were therefore too favourable. Recomputed with the standard mesh, two of the originally nine admissible designs drop out (0063: 369 →595 MPa, 0136: 479 →577 MPa). **Seven admissible designs remain**, all with 20 or 24 coils.

List 1: the best and those close to the limit. The table shows the admissible designs ranked by their smallest reserve and the twelve that fail most narrowly. All other designs and a catalogue of all geometrically representable ones are in the variant report.

No.	coils	σ_{99} MPa	max. MPa	strain %	$B \cdot n$ ‰	clearance m	steel kt	smallest reserve (bottleneck)
admissible								
0061	24	351	725	0.17	1.49	1.62	19	30 % (steel σ99)
0169*	24	308	717	0.15	2.13	1.39	28	29 % (package space)
0093	24	283	1190	0.16	1.45	1.37	28	27 % (package space)
0008	24	305	1695	0.17	2.20	1.48	26	27 % (field error B·n)
0014	20	377	1419	0.21	2.21	1.65	28	25 % (steel σ99)
0017	24	389	1721	0.20	1.66	1.30	23	20 % (package space)
0157	20	492	1658	0.22	2.37	1.35	18	2 % (steel σ99)
narrowly missed								
0173	24	533	1827	0.24	1.85	1.28	27	−7 % (steel σ99)
0103	20	440	738	0.19	2.66	1.25	21	−8 % (neutron fluence)
0044	20	540	1583	0.23	2.76	1.35	26	−8 % (steel σ99)
0004	20	558	1085	0.26	1.81	1.54	19	−12 % (steel σ99)
0136*	20	577	2238	0.30	1.62	1.84	26	−15 % (steel σ99)
0022	20	600	2183	0.26	0.70	1.57	24	−20 % (steel σ99)
0137	24	612	2562	0.29	0.94	1.57	16	−22 % (steel σ99)
0063*	20	595	4265	0.37	0.75	1.53	41	−24 % (winding-pack strain)
0042	20	632	3962	0.34	2.66	1.61	21	−26 % (steel σ99)
0043	24	653	1208	0.29	1.79	1.30	16	−31 % (steel σ99)
0140	20	656	4169	0.34	1.46	1.76	22	−31 % (steel σ99)
0085	20	608	2067	0.28	2.00	1.23	30	−31 % (neutron fluence)

Table 20: Ranking (cleaned). * = recomputed with the standard mesh.

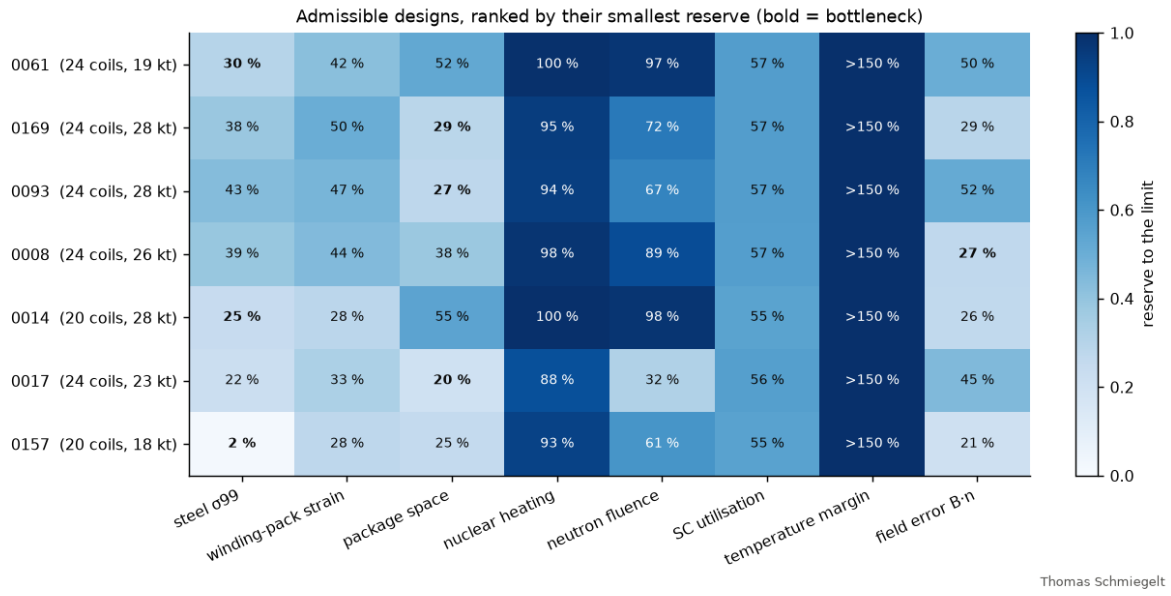

Figure 50: Reserves of the admissible designs per limit.

How to read the figure: Rows are designs, columns limits. The number is the distance to the limit in per cent, darker means more reserve; bold is the bottleneck. For the best designs, steel stress, package space and field error alternate as the bottleneck; superconductor and nuclear heating have large reserves everywhere.

Step 6: What determines the stress

For six quantities a surrogate model (Gaussian process [75]) was trained and checked by five-fold cross-validation. Package space and nuclear heating are predicted almost perfectly ($R^2 = 0.98$), steel mass and field error well (0.95 and 0.90), the steel stress poorly ($R^2 = 0.46$). Its most important variable is the coil–plasma distance, then the coil length. Number and stiffness of the connectors matter less than expected. Interpretation: the stress depends on the coil shape – coils further away must be more strongly twisted and carry more current – and on the random position of the connectors, which sets the local peaks.

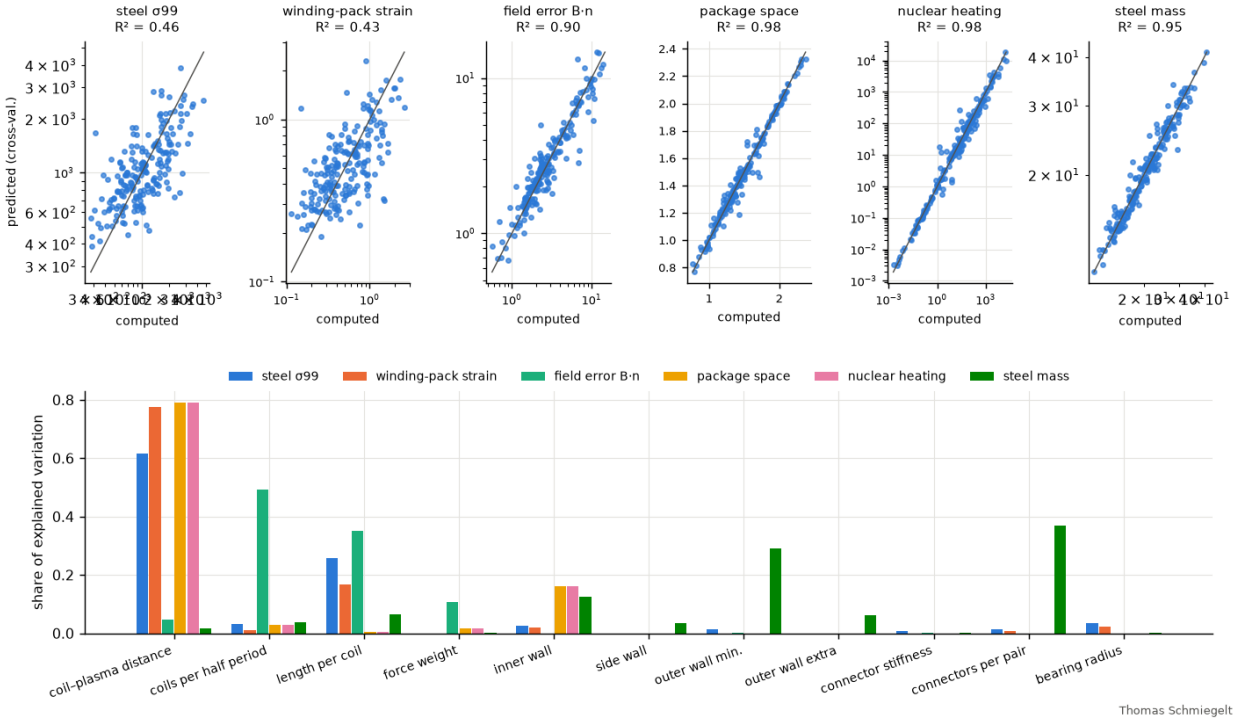

Figure 51: Surrogate models: prediction against computation (top) and influence of the variables (bottom).

> *How to read the figure:* Top: each point is a design, horizontally computed, vertically predicted by a model that had not seen it; on the diagonal the prediction would be perfect. Bottom: how strongly each variable moves each quantity. The field error depends on the number of coils, package and heating on the distance, the mass on connector count and wall thickness.

Putting it to the test. The surrogate proposed four new designs that maximise the smallest reserve minus one standard deviation. When computed, three were admissible and one failed clearly (σ_{99} 851 MPa). The best proposal V3 reached a robustness of 0.35 on the standard mesh, more than 0169, but lost on the fine mesh (σ_{99} 272 → 394 MPa) and dropped to 0.21. The surrogate is therefore useful for pre-selection, but stress still has to be computed directly; 0169 remains the most robust design.

Step 7: Mesh study – how certain is "admissible"?

The five best designs were recomputed with the same coils and a finer mesh: 1.5 times as many stations along the coil (same connector length) or three instead of two cells across the wall.

Design	doe1		standard		1.5× stations		wall 3 cells	
	σ_{99}	max.	σ_{99}	max.	σ_{99}	max.	σ_{99}	max.
0008	305	1695	–	–	407	3250	306	1931
0014	377	1419	–	–	549	3814	375	1639
0018	811	2650	996	4566	–	–	–	–
0055	366	753	413	942	–	–	–	–
0060	662	1313	704	1499	–	–	–	–
0061	351	725	–	–	556	6553	346	861
0063	369	2016	595	4265	–	–	–	–
0084	395	911	462	1018	–	–	–	–
0093	283	1190	–	–	410	2343	280	1432
0121	264	520	286	749	–	–	–	–
0136	479	995	577	2238	–	–	–	–
0145	362	1838	426	2490	–	–	–	–
0149	837	1541	636	1423	–	–	–	–
0169	219	421	308	717	319	1262	308	794
0176	245	615	434	2674	–	–	–	–

Table 21: Mesh study: σ_{99} and largest element value in the steel (MPa).

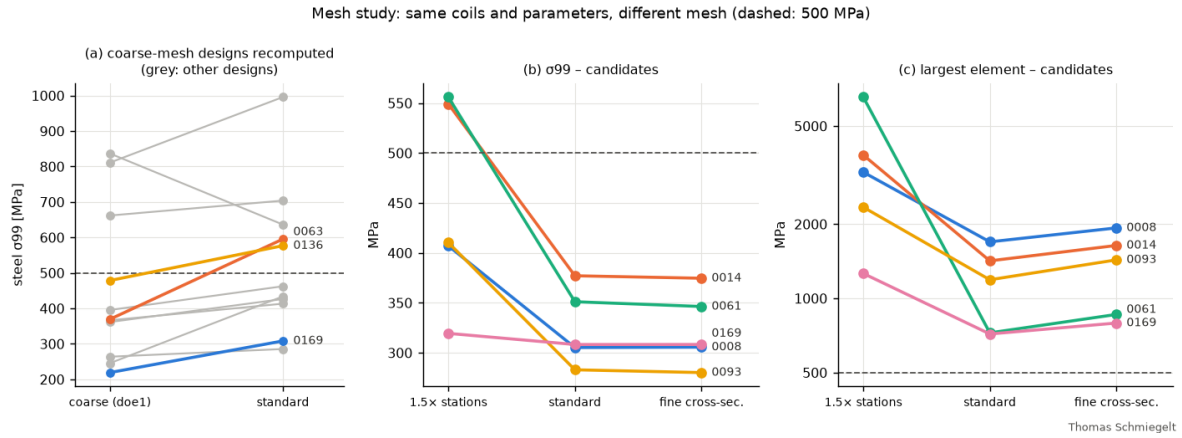

Figure 52: Mesh study of the best designs.

How to read the figure: Left σ_{99}, right the largest element value, finer from left to right. A flat line means the result does not depend on the mesh. Finer walls change almost nothing; more stations along the coil raise σ_{99} by 34–58 % for four designs. Only 0169 stays nearly the same (308 →319 MPa). The maxima grow with every refinement: the sign of a sharp edge.

Of the 200 most highly stressed steel elements, 77–100 % lie less than 1 m from a connector joint. The connectors are sharp-edged steel blocks; at their ends along the coil the stiffness jumps, as at the end of a lap joint. **Only design 0169 is robust against the mesh** (24 coils, 13 stiff connectors per pair, 28 kt steel); it is therefore the best design of this study. The others are "admissible" only on the standard mesh.

Step 8: Which steel, which limit?

The user asked: *"I want to know exactly which steel you are using. I don't know p99 steel." (translated)* The case steel is **1.4429** / **X2CrNiMoN17-13-3** (AISI 316LN): austenitic, about 17 % chromium, 13 % nickel, 2.5 % molybdenum, low carbon, strengthened with nitrogen; ITER builds its coil cases from it. "p99" was not a material but the quantity σ_{99} (see terms).

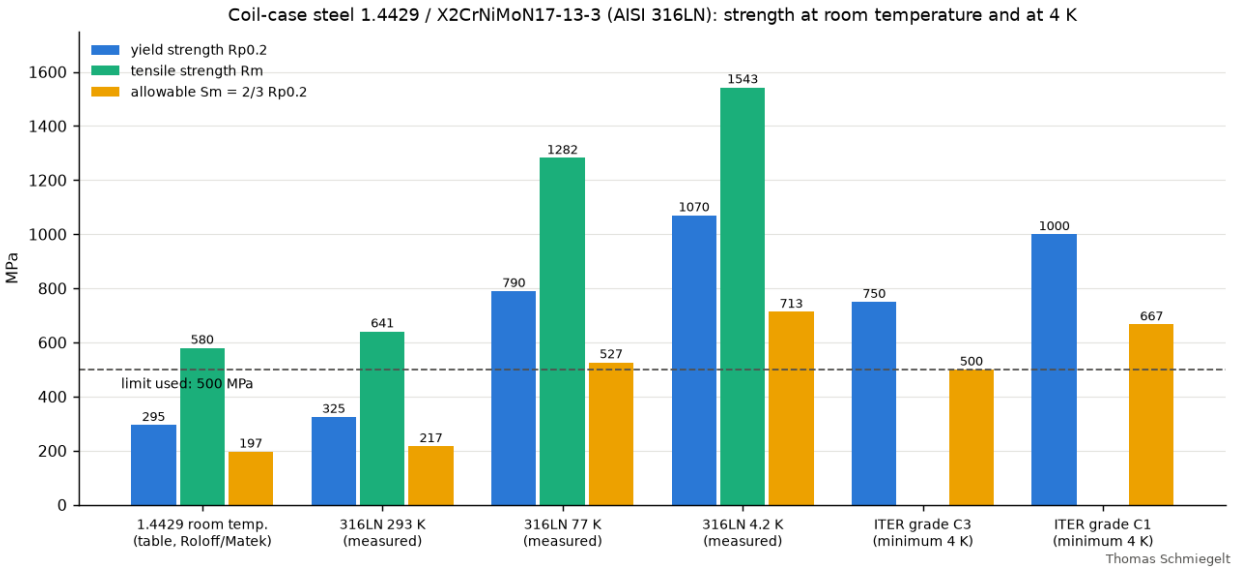

Figure 53: 1.4429 / 316LN at room temperature and at 4 K.

How to read the figure: Per group blue is the yield strength, green the tensile strength, yellow two thirds of the yield strength. On the left the handbook value for room temperature (295 MPa [76]), next to it measurements: at 4.2 K 316LN reaches 1070 MPa yield and 1543 MPa tensile strength [77], with high nitrogen above 1000 MPa even in thick plates [78]. On the right the guaranteed ITER minimum values.

Derivation: The ITER magnet criteria allow $S_m = \frac{2}{3} R_{p0.2}$ at operating temperature. With the weakest ITER grade C3 ($R_{p0.2} \geq 750$ MPa at 4 K) this gives 500 MPa; the ASME form $\min(\frac{2}{3} R_{p0.2}, \frac{1}{3} R_m)$ with the

measured 4-K values gives 514 MPa. At room temperature only about 200 MPa would be allowed; this does not matter, because the magnetic force acts only in cold operation.

Stress categories and maxima. In addition to σ_{99}, the categories according to ITER/ASME were determined: for each section across the wall the mean (membrane stress P_m, allowable S_m) and the linear part (bending P_b, allowable $P_m + P_b \leq 1.5\,S_m$).

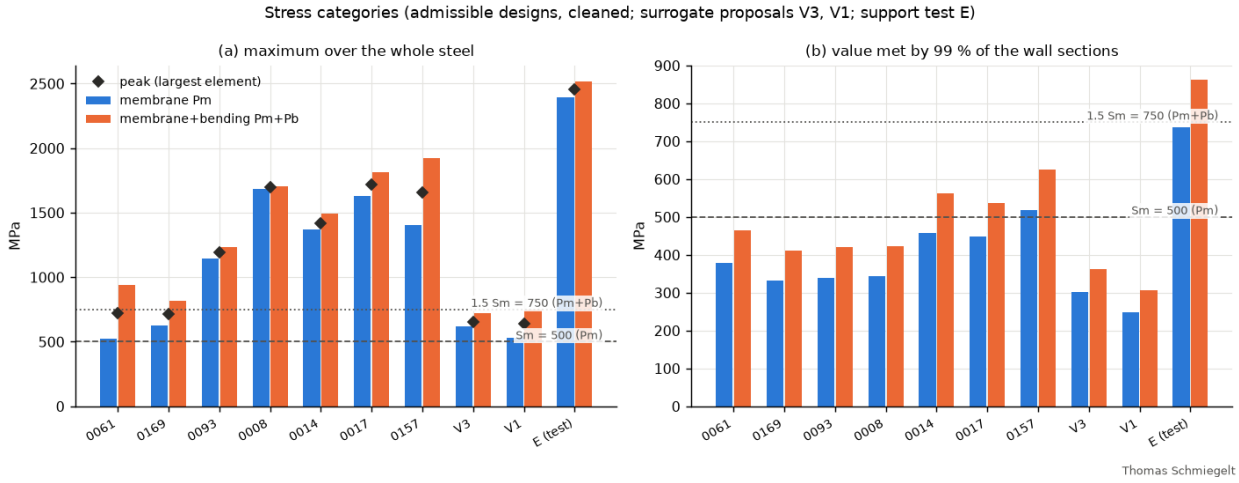

Figure 54: Stress categories and maxima.

How to read the figure: Two bars per design: membrane stress P_m (blue) and membrane+bending (orange); dashed the limit 500 MPa for P_m, dotted 750 MPa for $P_m + P_b$. Left the maximum over the whole steel; the diamond marks the largest single element: maximum and peak lie together, at the end of a connector – a local discontinuity that mostly does not even meet the limit $1.5\,S_m$. Right the value met by 99 % of all wall sections: for the best designs and the proposals V3 and V1 both bars stay below their limits, so the walls themselves can carry the load.

Result: in 99 % of the wall sections P_m stays below 400 MPa for the best designs; the walls can carry the load. The connector joints cannot in the model; there the design needs a shape (fillet, tapered thickness, longer contact area).

Step 9: Weakening selectively – first attempt

The user's idea (*"It can also help to weaken certain areas so that the load is distributed more evenly." (translated)*) was first tested in its simplest form: all connectors of 0169 softer.

No.	E connectors	stations	σ_{99} MPa	max. MPa	deformation mm	strain %	$B \cdot n$ ‰	all limits
0105	50 GPa	64	358	743	27	0.17	2.13	yes
0110	100 GPa	64	331	734	22	0.16	2.13	yes
0114	140 GPa	64	318	725	21	0.15	2.13	yes
0205	50 GPa	96	372	1541	29	0.19	2.13	yes
0210	100 GPa	96	343	1398	25	0.18	2.13	yes
0214	140 GPa	96	330	1323	23	0.17	2.13	yes

Table 22: 0169 with softer connectors; numbers 1xx standard mesh, 2xx fine mesh (original: 183 GPa).

It was expected (F-139) that the peaks at the connector ends would drop and σ_{99} would fall by 5–15 %. The opposite happened: with 50 GPa instead of 183 GPa, σ_{99} in the case rises from 308 MPa to 358 MPa (+16 %) and the deformation from 19 mm to 27 mm; only the connectors themselves are relieved (192 →127 MPa). The reason: here the connectors are the relief path. Making them soft overall sends the load back into bending of the case walls. The idea remains right but must be applied *locally*: soft ends (where the peak sits) and a stiff core that keeps carrying the load. This needs connectors with graded stiffness in the model – the next step.

Step 10: The 3D model, entirely in PicoGK

The user asked: *"Can you please try to build a 3D model entirely in PicoGK?" (translated)* PicoGK [29] describes bodies with voxels and distance fields; union, subtraction and offset always succeed, even for the twisted cases where Boolean operations in FreeCAD often fail. Python passes only geometric data (cross-sections per station, connector surfaces, plasma boundary, bearing locations). In PicoGK the winding packs, cases (outer contour minus potting) and potting (minus winding pack) are built as Boolean differences, the package by offsetting the plasma (first wall and blanket to 0.55 m, shield to 0.88 m, vacuum vessel to 1.0 m), the bearings as rods to an inner support cylinder and an outer ring. Three test bodies show overlaps.

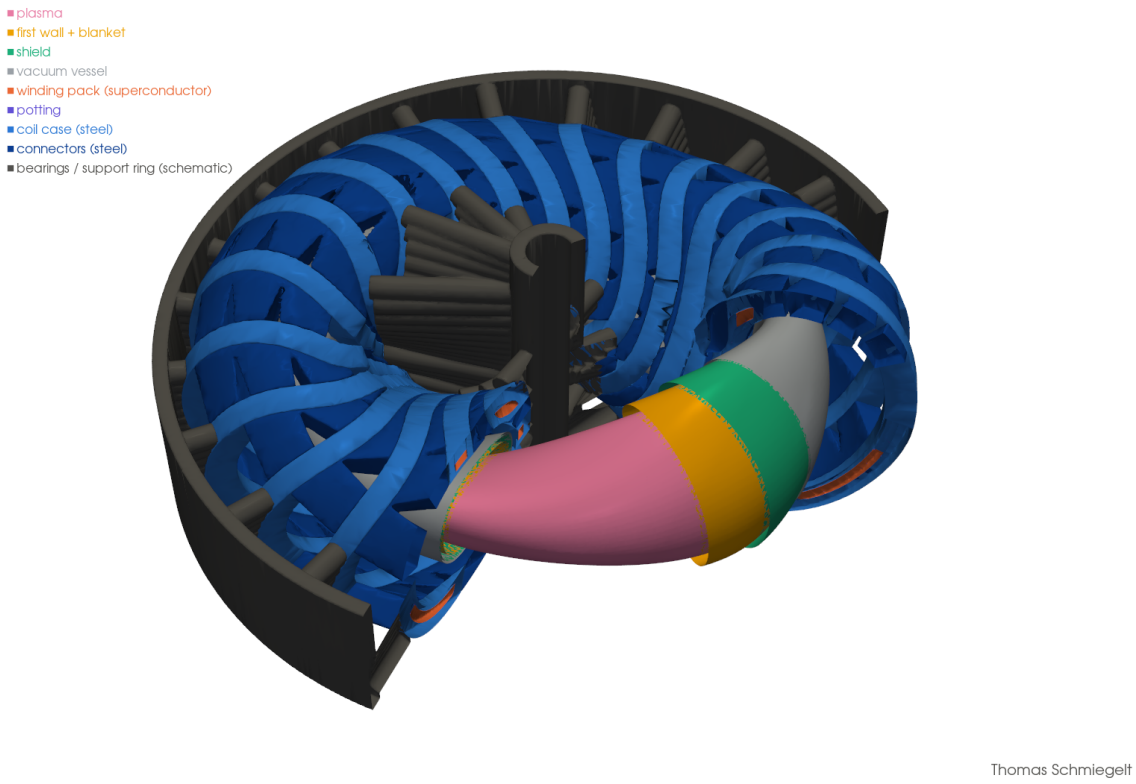

Figure 55: Design 0169 as PicoGK model in a stepped cut.

How to read the figure: Each layer is cut open at a different angle. From inside: plasma (pink), first wall and blanket (yellow), shield (green), vacuum vessel (grey). Around it the 24 coil cases (blue) with their winding pack (orange, visible in the cut), connectors in between (dark blue). Dark grey the schematic bearings: rods from the coils to a support cylinder in the centre and a ring outside.

Winding pack and case agree with the computational model to 0.5–1 %, the connectors mostly to 0–3 % (expected ±3 %). For 0169, 13 % of the connector volume is missing: neighbouring connectors partly interpenetrate there, and the computational model counts the shared space twice. Connector mass and stiffness of 0169 are therefore somewhat too high in the computational model; according to the study with softer connectors this changes σ_{99} by a few percent, but the connectors should lie without overlap in a recomputation. Only the 3D model revealed this error. PicoGK finds no overlap of package and coils and none between the cases – an independent confirmation of the clearance and penetration checks. Only the 2 cm thin potting cannot be represented with 40 mm voxels. One model takes 2–5 minutes. Models of all designs of list 1 and of all geometrically representable designs are in the variant report.

Comparison with the expectation

Expectation	Result
1. Force target: −30 % at double field error	more expensive: −9 % at +12 %, −25 % at ×5.7
2. Force reduction alone is not enough	confirmed
3. 100–150 MN/m, rope force varies > ×3, bending is the cause	confirmed; hand calculation matches FEM
4. Net force towards the plant centre	confirmed, up to 1.1 GN per coil
5. More coils: same sum, same stress	confirmed; the gain is field accuracy
6. Package tight	refuted: 1.3–1.8 m free, ARIES-CS needs about 1 m
Bearing inside: −30 to −50 %	confirmed: −32 %
DOE: 10–25 % below 500 MPa	confirmed: 11 %
DOE: strain second bottleneck (< 10 % pass)	refuted: 24 % pass
DOE: connectors and side wall set the stress	refuted: coil–plasma distance and coil length
Surrogate for stress $R^2 \geq 0.8$	refuted: 0.46 (package, heating 0.98)
Finer mesh: +5 to +25 %	refuted: +34 to +58 % along the coil; only 0169 +4 %
Softer connectors: σ_{99} −5 to −15 %	refuted: rises (see step 9)
PicoGK: volume ±3 %, no overlap	confirmed

Limitations

- **Rigid bearings.** The bearing areas are clamped as if the support cylinder were infinitely stiff. This is optimistic; the support cylinder itself must be modelled.
- **Sharp-edged connector joints.** Mesh-dependent peaks arise there; the assessment applies to the walls, not to the joints.
- **Primary load only.** Linear elastic, magnetic force only. Cool-down, dead weight, fatigue and fracture mechanics (often governing at 4 K) are missing.
- **Package and temperature coarse.** Neutron attenuation and tape fraction are assumptions.
- **Plasma only via the field error.** Alpha losses for 0169 are still to be computed in full.

Next steps

Shape the connector joints and make them compliant locally (not globally): soft ends, stiff core, iteration by stress ratio; elastic support cylinder instead of rigid bearings; alpha orbits in the coil field of 0169; fatigue and fracture mechanics at 4 K.

Code and data

Software: ElmerFEM, gmsh, FreeCAD, simsopt, DESC, PyVista, ParaView, Blender, FluidX3D 3.8 [17], PicoGK [29]. `https://github.com/ThomasSchmiegelt/stellarator-design` (MIT licence): pipeline code, findings log, figures.

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